

THE ELEMENTS OF ACCOUNTANCY

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*Being a Plain Account of Recording, Classifying,
Summarising and Reporting*

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*The question asks for a reader,
a branch and a rule for each
party.*

*Reasoning begins with the
bank and the publishers, who
are outsiders with money at
stake.*

*Reasoning continues with the
tax authority and the staff.*

*Nurul needs both kinds of
accounting.*

*The conclusion states the
match and the check confirms
it is complete.*

*The chapter's main ideas in a
few lines.*

*Bookkeeping and accounting
are different things.*

Question. For each reader, which question does the reader ask, which branch of accounting serves the reader, and which rule makes the answer trustworthy?

Reasoning. Take the bank first. It is an external user, so financial accounting serves it. It asks whether the loan of Tk 600,000 can be repaid. The answer is trustworthy because accounting standards make the shop report in the same way as other borrowers. Take the publishers next. They are external users. They ask whether the shop will pay the Tk 1,150,000 owed to them. The business entity rule matters here, because the debt belongs to the shop and not to Nurul's household.

The tax authority is an external user. It asks whether the profit return is honest, and standards give it a common method to test. The staff, treated here as external users, ask whether the wages of Tk 2,940,000 are safe and whether the shop will go on. The going concern assumption is the rule that matters to them, because it states that the shop is expected to continue.

Nurul is the internal user. Nurul reads the profit of Tk 750,000, which comes from financial accounting. Nurul also needs management reports, such as the weekly report on fast-selling books, to decide what to order. Historical cost supports the profit figure, because every book is recorded at the price paid.

Conclusion. Five readers, five questions, one set of records. Four readers are served by financial accounting, and Nurul is served by both branches. **Check.** Count the readers: five in the case file, five in the answer. Each answer names one reader, one question and at least one rule. The wages figure, the loan figure and the profit figure all agree with the case file.

* * *

SUMMARY

Accounting is the process of recording, sorting and reporting the money effects of a business's events to people who need them. Bookkeeping is its recording part. Financial accounting serves outsiders and follows written standards; management accounting serves managers. Internal users run the business; external users do not. Four ideas keep reports trustworthy: accounting standards, the business entity rule, the going concern assumption and historical cost. Financial statements carry the result to the readers.

THINK IT THROUGH

2.T1 Nurul pays a household bill from the shop's till. Using the business entity rule and the point of view of the bank, explain why a shop's money and a home's money must be kept apart.

WHERE PEOPLE GO WRONG

Many students think that bookkeeping and accounting are the same. They are not. Bookkeeping is the recording part. Accounting

*The book I need most and miss most,
dedicated to the person I need most and miss
most. Naima Islam Oyshee*

An accounting standard is a written rule that makes businesses report alike.

Accounting standard. A written rule about how to record and report items, so that different businesses report in the same way.

Standard. Word in use. In daily life, a level of quality, as in a high standard of service. In accounting, a written rule about how to record and report items.

The first basic idea keeps the shop's money apart from the owner's own money.

GO SLOWLY HERE.

The business entity rule treats the business as separate from its owner.

Business entity. The rule that a business is treated as separate from its owner when records are kept.

The business entity rule is an accounting rule, which may differ from the law.

A short review of the first basic idea.

The second basic idea assumes that the business will continue.

Going concern. The assumption that a business will continue to operate for the foreseeable future.

An **accounting standard** is a written rule about how to record and report items, so that different businesses report in the same way. For example, a standard may say how the cost of goods held for sale must be measured. In daily speech, the word *standard* means a level of quality, as in "a high standard of service". Here it means a rule, not a level of quality. Which standards apply depends on the place. Many countries, including Bangladesh, use the International Financial Reporting Standards, while others use national standards of their own.¹

Begin with a household example. A father who runs a small shop takes money from the till to pay the family electricity bill. If the shop's records show that payment as a shop cost, the shop will look less profitable than it is. The bank, reading the report, will form a false picture of the shop. The remedy is to keep two sets of money, the family's and the shop's, and to write only the shop's events in the shop's records.

This remedy is the rule of the **business entity**. The business entity rule says that a business is treated as separate from its owner when records are kept. Nurul's household spending does not appear in the records of Meghna and Sons. The rule also applies between two businesses of one owner. Meghna Binding Service, a book-binding workshop that began on 1 January 2026, keeps its own separate set of books, even though Nurul owns both businesses.

The rule has an exception that causes confusion. In law, a sole trader, meaning one person who owns a business alone, is often not a separate person from the business. A company, in contrast, is a separate person in law. The accounting rule does not wait for the law: it treats every business as separate from its owner, whether or not the law agrees. The rule serves the readers, who ask about the business and not about the family.

PAUSE AND CHECK 2.P1 Nurul takes a book from the shop shelf as a gift for a relative. Using the business entity rule, explain why the shop's records should show this event.

Before moving on. The business entity rule keeps the shop's records free of the owner's private spending, and it applies between two businesses of one owner. You should now be able to say: the shop's records describe the shop, not the family.

Consider a farmer who plants fruit trees. The farmer expects to pick fruit for many years, so the trees are valued by what they will bear, not by the timber they would give if cut down today. Accounting makes a similar choice. The **going concern** assumption is that a business will continue to operate for the foreseeable future. Under this assumption, the shelves and counter of the bookshop are treated as items that the shop will use for years, not as items that must be sold tomorrow.

¹ The International Financial Reporting Standards are published by the International Accounting Standards Board, which replaced the committee described in the history section on 1 April 2001.

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Management accounting serves managers, and its reports follow no outside rule.

Management accounting. Accounting that prepares reports inside the business to help managers plan and control.

A short review of the ideas in this section.

A sentence the reader can test against the section.

One question sorts the readers into an inside group and an outside group.

Each external user wants to know something different about the business.

GO SLOWLY HERE.

Staff are a borderline group, and this book treats them as external users.

A table lays out each reader beside the question that reader asks.

Management accounting is accounting that prepares reports inside the business to help managers plan and control. Nurul's weekly report on which books sell fastest is management accounting. No outside rule governs such a report; the manager designs it to answer whatever question needs an answer. The chapters that follow concentrate on financial accounting. Management accounting is named here so that you know it exists.

PAUSE AND CHECK 2.P2 A school employs a clerk who writes each fee payment into the fee book. Is the clerk's work bookkeeping or accounting? Give one reason.

Before moving on. Accounting is the process of recording, sorting and reporting the money effects of events to people who need them. Bookkeeping is one part of that process, the recording part. Financial accounting serves outsiders by written rules, and management accounting serves managers without outside rules.

You should now be able to say: accounting is the whole journey from an event to a report, and bookkeeping is only its first stage.

* * *

2.2 WHO USES ACCOUNTING INFORMATION

Return to the five readers of the shop's records. Is there a way to sort them? One question gives a useful sorting: does this reader help to run the business from day to day? Nurul does. The bank, the publishers and the tax authority do not. Readers who help to run the business are called internal users, and those who do not are called external users. Internal users are people inside the business, such as managers and owner-managers, who use information to plan and control. External users are people outside it who have a claim on it or an interest in it.

Take the external users one at a time, and imagine each person's point of view. A bank that has lent money asks whether the shop can repay the loan on the agreed dates. A publisher that has delivered books not yet paid for asks whether the shop will pay what it owes. A tax authority asks whether the profit has been reported honestly, because the tax due depends on it. An owner who does not manage the business asks whether the profit is worth the money invested. A customer rarely reads accounting reports at all, but cares whether the shop will still be open next year to order a special book.

Staff are a harder case, and many students find them confusing, because staff work inside the shop yet do not run it. In this book we treat them as external users. They read finished reports, not the working records, and they cannot change what the reports say. Their question is whether the wages of Tk 2,940,000 recorded for them are safe and whether the shop will continue to employ them. Other books may place employees among the internal users, so the grouping is a convention, not a law of nature.

The table that follows gathers the five readers of the case. Read each row from left to right: the reader, the group, the question that

TO THE READER

This book is written for readers who are meeting the subject for the first time. It assumes no earlier knowledge. Every new word is explained when it first appears, and the book moves slowly wherever an idea is new or difficult.

Beside most paragraphs you will find a note in the narrow column at the edge of the page. A note in italics gives the main point of the paragraph in one line. Read these notes first to see the shape of a chapter, and again when you review.

A word in **bold type** is a new term. It is shown in bold once, where it is first explained, and its meaning is repeated in the narrow column. After that the word appears in ordinary type. Italics are used for emphasis, for foreign words and for the titles of books.¹

Where an everyday word has a special meaning here, a note marked "Word in use" says so. Where an idea is both new and hard, a note reads "Go slowly here", and the text slows down with it.

HOW A CHAPTER RUNS

A chapter opens with a puzzle, then builds its ideas one section at a time. Each section ends with a short recap. The chapter closes with a summary, a problem to think through, the common mistakes people make, a true account from history, and review questions. Questions are numbered, such as 7.1, and their answers are collected at the back of the book.

When a chapter points to another part of the book, it names the chapter and the section, for example Chapter 3, section 3.2. Use the Contents to find it. At the back you will find a glossary of every term, the answers to the questions, a list of further reading, and an index.

This book teaches The Elements of Accountancy one idea at a time, from the very beginning.

The narrow column beside the text holds short notes.

Key term. A new word, shown in bold where it first appears, with its meaning in one line.

A bold word is a new term, and its meaning waits beside it.

Some everyday words have a second, special meaning in this subject.

Each chapter has the same shape, so you always know where you are.

Chapters refer to one another by name, never by page.

YOU WILL NEED

Chapter 1 'Business Activity and Companies', section 1.1 'What Businesses Do'; Chapter 1 'Business Activity and Companies', section 1.3 'Why Records Are Needed'.

A family notebook can answer several questions for several readers.

The shop in the running case has five readers and one set of records.

Five readers ask five different questions, which creates a puzzle for one set of records.

Think of a family that writes every payment in one notebook: rent, food, school fees, bus fares. At the end of the month, the mother opens the notebook and asks one question: did we save anything? The same notebook can answer other questions. The landlord may want proof that the rent was paid. A bank, before lending money for a repair, may want to know whether the family can repay it. There is one notebook, but there are three readers and three questions. The notebook is useful only if every reader believes what it says.

Case file, illustrative. Meghna and Sons is a family-run bookshop in Dhanmondi, Dhaka, owned by Nurul Meghna. Five parties read its records for 2025. Nurul reads the profit of Tk 750,000. The bank reads the loan of Tk 600,000. The publishers, who supply the books, read the Tk 1,150,000 owed to them. The tax authority reads the profit return. The staff read the wages of Tk 2,940,000.

Notice that the five readers do not ask the same question. Nurul asks whether the shop earned a fair reward for the work put into it. The bank asks whether the loan will be repaid. The publishers ask whether they will be paid for their books. The tax authority asks whether the profit has been reported honestly. The staff ask whether the shop is secure enough to keep them employed. If each reader demanded a private set of records, the sets would disagree, and nobody would trust any of them. How can one business keep one set of records that serves all five readers? And why should any of them believe it? This chapter answers both questions. It also states the purpose of the whole book, because every later chapter teaches one step of the process described here.

* * *

2.1 THE IDEA OF ACCOUNTING

Picture a woman who sells vegetables at a market stall. Each morning she buys sacks of produce. During the day she sells by the kilogram. In the evening she counts the money in her tin and writes three numbers in a notebook: what she spent, what she received, and what is left. She has done three things without naming them. She has noticed events, written them down, and drawn a conclusion. Accounting is these same actions, performed in an orderly way for many readers.

The name for the whole process is **accounting**. Accounting is the process of recording, sorting and reporting the money effects of a business's events to people who need them. An event is anything that changes the money position of a business, such as buying books, selling a novel or paying a wage. Recording means writing

A market seller already performs the three actions of accounting.

GO SLOWLY HERE.

Accounting means recording, sorting and reporting the money effects of events.

Accounting. The process of recording, sorting and reporting the money effects of a business's events to

¹ A footnote gives a side point that would interrupt the main argument. Footnotes are numbered within each chapter and appear at the foot of the page.



CHAPTER I

Business Activity and Companies

Who does the work, who owns the shop, and who needs to know?



This chapter shows what businesses do: some sell work, some sell goods bought from others, and some make what they sell. It then explains who may own a business, from one person to many shareholders, and why every owner needs written records.

books to schools for a fee. When you meet a new business, ask these three questions in turn. The answers tell you what the business must count.

Why does business activity exist? Nobody can grow all their food, make all their clothes and print all their books. So people specialise: one person does a task well and exchanges the result for what others do well. Profit is the reason a business continues. Profit is the money left after the costs of the business are paid.² The tea seller earns profit only when the money from customers is larger than the cost of leaves, milk, fuel and her own time.

Where does business activity fail? It fails when the costs are larger than the money received. The difference is then a loss, not a profit. A loss is not a disgrace, but a business that suffers losses for long cannot continue. This is why the owner must learn the result, and why the result must be learned from facts and not from feeling.

Different people see the same business differently. The owner asks whether the business rewards the money and effort put into it. A manager, who directs the daily work for the owner, asks whether the work is well organised. An employee, who works for a wage, asks whether the wage will be paid. A customer asks whether the goods are good and the price fair. Every one of these questions depends on facts about the business.

Businesses differ in what the customer receives. We use three groups, and the first test is simple: does the customer receive work, goods that the business bought, or goods that the business made? Each group needs different records, so each group has its own name. We shall meet the groups one at a time, starting with the one that is easiest to see.

A **service business** earns money by doing work for customers, not by selling goods. A hairdresser, a tailor who stitches the customer's own cloth, and a teacher are examples. In this book, Meghna Binding Service is a service business: it repairs and binds books for schools. The customer leaves with a repaired book, but the business was paid for the work.

A **merchandising business** buys finished goods from others and sells them again at a higher price. A bookshop is the clearest example: it buys books from publishers and sells the same books to readers. It does not change the books. It earns its profit from the gap between the price it pays and the price it receives.

PAUSE AND CHECK 1.P1 A shop buys sacks of rice and sells the rice by the kilogram. Which kind of business is it, and why?

A **manufacturing business** makes goods from materials and then sells them. A bakery turns flour, sugar and fuel into bread. A factory turns timber into chairs. The business holds three things that the other two kinds do not: materials waiting to be used, goods half finished, and goods ready to sell. Later chapters will give these three names.

² Profit has a precise meaning in accounting, and later chapters teach it. For now, the plain meaning is enough.

Businesses exist because no one can make everything that he or she needs.

Business activity can fail, and failure is called a loss.

Owners, managers, employees and customers each look at a business differently.

Businesses differ in what the customer finally receives.

A service business is paid for work, not for goods.

Service business. A business that earns money by doing work for customers rather than by selling goods.

A merchandising business buys goods and sells the same goods again.

Merchandising business. A business that buys finished goods and sells them again at a higher price, such as a bookshop.

A manufacturing business changes materials into goods and sells them.

Manufacturing business. A business that makes goods from materials and then sells them.

person. Written records let owners and outsiders check the facts and make decisions.

THINK IT THROUGH

Meghna Binding Service repairs and binds books for schools. The four problems below move from a familiar case, to the same case reshaped, to a new setting, and then to a case that combines earlier ideas.

1.T1 The workshop earns money from binding work. Why does it keep no records of goods bought to sell again?

1.T2 A hairdresser begins to sell bottles of shampoo bought from a wholesaler. Which kinds of business does the salon now combine, and which kind brings in most of its money if shampoo is only a small part of its income?

1.T3 A village club collects a monthly fee from its members and spends it on a reading room. Is the club a business in the sense used in this chapter? Give a reason.

1.T4 A bakery owned by three friends bakes bread and also sells tea that it buys from a wholesaler. State the kinds of business it combines, the shape of its ownership, and one person who would need its records.

WHERE PEOPLE GO WRONG

A mistaken belief: "The cash in the drawer is the profit." *Correction:* cash is only money held. Some of it may be owed to suppliers or may belong to customers who paid in advance. Profit is what remains after all the costs of the period are paid.

A mistaken belief: "A service business has nothing to record." *Correction:* it holds no goods for resale, but it earns fees, pays wages and rent, and owes money. These all need written notes.

A mistaken belief: "A company and its owner are the same." *Correction:* a company is a separate legal person. It can own goods and owe money in its own name, and its shareholders remain separate people.

A mistaken belief: "A business with more owners is stronger." *Correction:* more owners bring more money and more opinions, and more people to inform. Strength depends on the work the business does and the care with which it is run.

A mistaken belief: "Every firm belongs to exactly one kind." *Correction:* many firms mix kinds. We name the kind that earns most of the money and treat the others as additions.

IN HISTORY

On 31 December 1600, Queen Elizabeth I granted a royal charter to a group of London merchants trading to the East Indies. The charter gave the group the sole right to trade there for fifteen years, and its aim was to compete with the Dutch merchants who then controlled much of the spice trade. Thomas Smythe became

Four short problems test whether the kinds of business and ownership are secure.

Cash in the drawer is not the same as profit.

A service business still needs records, though it holds no goods for sale.

A company and its owners are different persons in law.

More owners do not by themselves make a business better.

The three kinds of business are tools for thinking, not strict boxes.

A royal charter of 1600 created an early company that served its investors.

* * *

1.2 WHO OWNS THE BUSINESS

Every business belongs to someone. The owner provides the money and effort to begin, and receives the profit or bears the loss. Owners come in three main shapes: one person, a few people, and many people. Each shape gives the business different duties, because other people lend money, sell goods or buy shares according to who stands behind it.

The first shape is the sole proprietor, a single person who owns the business and answers for it. The tea seller beside the bus stop is a sole proprietor. So is the owner of Meghna and Sons, Nurul Meghna, who has traded in Dhanmondi since 4 April 1996. The business is simple to begin and simple to direct. Its weakness is that all the money, work and risk rest on one person.

The second shape is the partnership: a business owned by two or more people who agree to run it together and share its profit. For example, two friends might open a bookshop and divide the profit between them. Partnerships bring more money and more skills than one person can supply. They also bring a new question, which is who receives how much. Chapter 18, 'Partnership Accounts', teaches how partners' records answer it.

The third shape is the **company**. A company is a business that the law treats as a person separate from its owners, with its ownership divided into equal parts called shares. A person can buy a share, so a company can have hundreds of owners. The idea is strange at first. The company can own goods and owe money in its own name, even though the people who own it remain separate.³

A **shareholder** is a person who owns one or more shares of a company. The proposal for Meghna and Sons Limited illustrates this: it names 410,000 shares of Tk 10 each. Suppose a person held 41,000 of those shares. That person would own one tenth of the company, because 41,000 is one tenth of 410,000. More shares give a larger part of the ownership.

PAUSE AND CHECK 1.P2 A company has 100 shares, and one person holds 25 of them. What part of the company does this person own?

Consider the three shapes from the viewpoint of an outsider who thinks of lending money. For a sole proprietor, the lender deals with one person. For a partnership, the lender deals with several people who agreed together. For a company, the lender deals with the company itself, and the shareholders may be many strangers. As the number of owners grows, more people need clear, written facts, because they cannot watch the business every day.

The company shape has one more feature that matters for records. Because a company can issue shares, it can collect small sums from many people and gather a large total. This is how very

³ The rules for forming a company differ from country to country. In Bangladesh, companies are formed under the Companies Act 1994; other countries have their own laws.

Every business belongs to someone, and the kind of owner changes the records.

A sole proprietor is the one person who owns and answers for the business.

A partnership shares ownership among two or more people by agreement.

A company is a business that the law treats as a person of its own.

Company. A business that the law treats as a person separate from its owners, with ownership divided into shares.

A shareholder is an owner of a company through its shares.

Shareholder. A person who owns one or more shares of a company.

The three shapes differ in who owns, who decides and who must be told.

A company's shares let many people own a small part of one business.

A sole proprietor keeps control, which is why many small shops remain one.

Advanced note: a shareholder's risk is limited, and a beginner may defer it.

The recap gathers the section into three sentences.

A record is a written note that lets anyone check what happened.

Record. Word in use. In daily life, it can mean the best result ever achieved, as in a sports record. In accounting, it is a written note of an event in the business, such as a sale or a payment.

Memory is weak, and other people cannot see inside it.

Many decisions depend on records, and each decision has its own reader.

A useful record is dated, complete and open to checking.

is needed, how many people share the work, and how many outsiders must be told.

Why would an owner choose to remain a sole proprietor? Control is the main reason. One person decides what to buy, what to charge and when to open, and no agreement or vote slows the decision. The profit also belongs to one person. For a small family shop with modest needs, these benefits often outweigh the extra money that more owners might bring.

Advanced note, which you may defer. One more feature explains why companies are popular. In most countries, a shareholder who has fully paid for his or her shares cannot be made to pay the company's debts from personal wealth. This protection is called limited liability. It makes people willing to buy shares, because the most they can lose is the amount they paid. The rules differ by country, so check the law of the place where the company is formed. Chapter 19, 'Company Accounts: Shares', returns to shares.

Before moving on. A sole proprietor is one owner, a partnership is two or more owners by agreement, and a company is a separate legal person owned through shares. A shareholder owns one or more shares. You should now be able to say who owns each of three businesses, and why a company with many shareholders needs written facts.

* * *

1.3 WHY RECORDS ARE NEEDED

Return to the shop owner and the pile of notes. What would turn the pile into an answer? The owner needs a written note of each sale, each purchase and each payment, made when the event happened. Such written notes are the records of a business. The word has two meanings, so notice that this book uses the second. Records are the memory of the business, written down so that it does not depend on any one person.

Why not trust memory? A busy owner forgets a small payment, remembers a large one wrongly, and cannot show either to anyone else. A lender, a tax officer or a partner will not accept "I believe so". A written record can be checked, copied, filed and examined years later. It also protects honest people, because it shows that an account was kept fairly.

Several decisions depend on records. The owner decides whether to hire another worker, buy more books or begin a new workshop. A lender decides whether a business can repay a loan. The tax authority decides how much tax is due. A person thinking of buying shares decides whether the company is sound. For Meghna and Sons, the proposal to form a company in 2026 will need records, because strangers will be asked to trust it. Chapter 2, 'What Accounting Is', describes these readers more fully.

What makes a record useful? Three qualities matter. It is dated, so that anyone can place the event in time. It is complete, so that a sale or payment is never left out because it seemed small. It is open

Begin with your own life. You may own a bicycle, a mobile telephone and Tk 2,000 in cash. Each has value to you, and you control each one. A business owns things of the same kind. An **asset** is something of value that a business owns or controls, such as cash, goods held for sale or equipment. The shelving that the workshop buys on 2 January 2026 is an asset, and so is the cash in its drawer. Goods held for sale have a name of their own, inventory, which Chapter 13, 'Inventory: Meaning and Measurement', section 13.1 'What Inventory Is' teaches in full. We use that word from here onward.

Now suppose you borrow Tk 1,500 from a friend to buy a second-hand textbook. The book is yours, but you owe your friend. A business owes money in the same way. A **liability** is an amount a business owes to others, such as a loan or an unpaid supplier bill. On 3 January 2026 the workshop buys books from a publisher for Tk 200,000 and promises to pay later. Until it pays, it has a liability. The amount owed to one supplier is called an account payable; Chapter 4, 'Accounts, Debits and Credits', section 4.1 'The Account' explains that name more fully. Some books say creditor for the same idea. This book says account payable.

Return to the bicycle. Suppose it is worth Tk 12,000, and Tk 4,000 of its price is still owed to your uncle. How much of the bicycle is really yours? Tk 12,000 minus Tk 4,000 leaves Tk 8,000. A business works in the same way. **Owner's equity** is the owner's claim on the business: what is left of the assets after the liabilities are paid. Other books call it capital, net worth or proprietorship. This book uses only owner's equity, so that one idea always has one name. At 31 January 2026 the workshop holds assets of Tk 565,000 and owes Tk 120,000, so its owner's equity is Tk 445,000.

Go back to Rahim's stall. It holds Tk 100,000. His uncle's claim is Tk 30,000 and his own is Tk 70,000. In words: what the business holds equals what it owes plus what the owner can claim. The **accounting equation** is the rule that assets always equal liabilities plus owner's equity. In symbols, $A = L + E$, where A stands for assets, L for liabilities and E for owner's equity. For the stall, $100,000 = 30,000 + 70,000$. The equation is true because each asset was paid for by someone: a lender, who gains a liability claim, or the owner, who gains an equity claim. No third source exists.

Start from your own belongings; a business owns things of the same kind.

GO SLOWLY HERE.

Asset. Word in use. In daily life, a useful quality or a helpful person, as in "she is an asset to the team". In accounting, an item of value that a business owns.

Asset. Something of value that a business owns or controls, such as cash, goods held for sale or equipment.

Money owed to others is the second building block.

Liability. An amount a business owes to others, such as a loan or an unpaid supplier bill.

The owner's share is what remains after the debts are taken away.

GO SLOWLY HERE.

Capital. Word in use. In daily life, a city where a government sits, or any money used to start something. In accounting, it often means the owner's money in the business; this book says owner's equity.

Owner's equity. The owner's claim on the business: what is left of the assets after the liabilities are paid.

Both sides of a business must agree, and this agreement has a name.

Accounting equation. The rule that assets always equal liabilities plus owner's equity.

3.2 REVENUE, EXPENSES AND DRAWINGS

On the first day the owner's claim is the money put in. Then the business starts to work. How can the owner's claim grow beyond what the owner invested? Think of a household. Your wealth rises when you receive a salary, falls when you pay the rent, and falls again when you give money to a relative. Business owners see the same three movements, and each has a name.

A business earns by selling goods or doing work. **Revenue** is money earned or becoming due from selling goods or doing services. On 10 January 2026 the workshop binds books for a school and receives Tk 40,000 in cash. That is revenue of Tk 40,000. Notice the words "or becoming due". A customer may take the work today and pay next month. The revenue is earned when the work is done, not when the cash arrives. Chapter 7, 'Adjusting the Accounts', section 7.1 'Why Adjust' returns to this timing.

Earning revenue costs something. An **expense** is a cost used up in earning revenue, such as rent or wages. On 15 January 2026 the workshop pays rent of Tk 60,000, and on 31 January it pays wages of Tk 35,000. Both are used up already: the rented room has served its month, and the worker has done the work. Two payments are not expenses, and many students confuse them. Buying shelving only changes one asset into another, and paying a supplier only reduces a liability.

The owner may also withdraw money. **Drawings** means money or goods the owner withdraws from the business for personal use. If Rahim takes Tk 500 from the tin to buy his family's dinner, that is drawings. It reduces the assets and the owner's claim, yet it did not help to earn any revenue, so it is not an expense. The workshop has no drawings in January 2026.

The four movements can now be joined. The owner's equity at any date equals the money the owner has put in (with any equity carried from earlier periods), plus revenue, minus expenses, minus drawings. In symbols, $E = I + R - X - D$, where I is the owner's investment, R is revenue, X is expenses and D is drawings. Put this into the equation and the expanded form appears: $A = L + (I + R - X - D)$. It tells you where the owner's claim came from and where it went.

Given. Karim's umbrella stall has an owner's investment of Tk 30,000, revenue of Tk 18,000, expenses of Tk 11,500, drawings of Tk 4,000 and liabilities of Tk 9,500. *Find.* The owner's equity and the total assets. *Formula.* Owner's equity equals investment plus revenue minus expenses minus drawings: $E = I + R - X - D$. Then $A = L + E$. *Steps.* $E = 30,000 + 18,000 - 11,500 - 4,000 = 32,500$. $A = 9,500 + 32,500 = 42,000$. *Answer.* Owner's equity is Tk 32,500 and total assets are Tk 42,000. *Check.* Change the order: $30,000 - 4,000 = 26,000$; $26,000 + 18,000 = 44,000$; $44,000 - 11,500 = 32,500$. The result is the same.

Case file, illustrative. Meghna Binding Service, January 2026: owner's investment Tk 500,000; revenue Tk 40,000; expenses Tk 60,000 (rent) plus Tk 35,000 (wages) = Tk 95,000; drawings none. Owner's equity = $500,000 + 40,000 - 95,000 - 0 = \text{Tk } 445,000$.

The owner's claim changes after the first day, and three new ideas explain how.

Revenue is what a business earns by selling or by doing work.

Revenue. Money earned or becoming due from selling goods or doing services.

An expense is a cost used up while earning revenue.

Expense. A cost used up in earning revenue, such as rent or wages.

Drawings are the owner's own withdrawals, and they are not a cost of earning.

Drawings. Money or goods the owner withdraws from the business for personal use.

The owner's claim at any date is money put in, plus revenue, minus expenses and drawings.

Worked example: a messier Stall with all four movements and a liability.

The workshop's January figures give the same owner's equity by a second route.

THE BALANCE OF EVERY LEDGER ACCOUNT AT 31 JANUARY 2026

ACCOUNT	DEBIT TOTAL (TK)	CREDIT TOTAL (TK)	BALANCE (TK)
Cash	540,000	295,000	245,000 debit
Shelving	120,000		120,000 debit
Inventory	200,000		200,000 debit
Publisher Payable	80,000	200,000	120,000 credit
Owner's Equity		500,000	500,000 credit
Service Revenue		40,000	40,000 credit
Rent	60,000		60,000 debit
Wages	35,000		35,000 debit

The answer and the check agree with the case file.

Answer. The Cash account closes with a debit balance of Tk 245,000, and the other balances are those of the table above. *Check.* The debit balances total $245,000 + 120,000 + 200,000 + 60,000 + 35,000 = \text{Tk } 660,000$. The credit balances total $120,000 + 500,000 + 40,000 = \text{Tk } 660,000$. The two totals are equal, which is what correct journalising and posting must produce. Chapter 6 shows how this test is used.

Common slips are easy to name. A student may post a debit to the credit side of the account, which reverses the amount. A student may forget to enter the folio, so the line seems unposted and is posted twice. A student may total the wrong side or subtract in the wrong order and obtain a negative balance, a sure sign of a slip. Check the side, the folio and the subtraction, in that order.

* * *

SUMMARY

A transaction starts with a source document, is recorded in the journal in date order with debits and credits, and is then posted to the ledger, which holds one account for each kind of item. These are steps 1 to 3 of the eight-step accounting cycle, which repeats each period. A narration explains each entry, and a compound entry has more than one debit or credit. A ledger account is balanced by subtracting the smaller side total from the larger.

THINK IT THROUGH

5.T1 A clerk posts the 20 January entry to the Cash account but forgets to post it to Publisher Payable. What balance will Publisher Payable then show, and by how much will the credit balances exceed the debit balances?

WHERE PEOPLE GO WRONG

Many students think the journal and the ledger hold different information. They hold the same transactions. The journal arranges

The journal and the ledger arrange the same facts in

WORKED EXAMPLE: SEVEN TRANSACTIONS OF JANUARY

Case file, illustrative. *Given.* Meghna Binding Service, January 2026: 1 January, the owner invests cash of Tk 500,000; 2 January, buys shelving for cash, Tk 120,000; 3 January, buys books (inventory) on credit from a publisher, Tk 200,000; 10 January, earns Tk 40,000 in cash for binding work; 15 January, pays rent, Tk 60,000; 20 January, pays the publisher Tk 80,000; 31 January, pays wages, Tk 35,000.

Find. Total assets, liabilities and owner's equity at 31 January 2026, and a check that they balance. *Formula.* $A = L + E$. Each row of the table below shows the effect of one event on A, L and E, in taka. *Steps.* The table lists each event with its effect. Assets rise or fall by the cash and other items affected, and equity changes only when revenue or an expense arises or the owner invests.

EFFECTS OF THE SEVEN EVENTS ON THE EQUATION (TK)

DATE	EVENT	ASSETS	LIABILITIES	OWNER'S EQUITY
1 Jan	Owner invests cash	+500,000	0	+500,000
2 Jan	Shelving bought for cash	+120,000 and -120,000	0	0
3 Jan	Inventory bought on credit	+200,000	+200,000	0
10 Jan	Binding fee earned (revenue)	+40,000	0	+40,000
15 Jan	Rent paid (expense)	-60,000	0	-60,000
20 Jan	Publisher paid	-80,000	-80,000	0
31 Jan	Wages paid (expense)	-35,000	0	-35,000
Total		565,000	120,000	445,000

Event	Assets	Liabilities	Owner's equity
1 Jan owner invests	500,000	0	500,000
2 Jan shelving (cash)	500,000	0	500,000
3 Jan inventory (credit)	700,000	200,000	500,000
10 Jan binding fee	740,000	200,000	540,000
15 Jan rent paid	680,000	200,000	480,000
20 Jan publisher paid	600,000	120,000	480,000
31 Jan wages paid	565,000	120,000	445,000

Check at 31 January: $120,000 + 445,000 = 565,000$

FIG. 3.2. Totals of assets, liabilities and owner's equity after each of the seven events in taka

shelving for cash Tk 120,000. 3 January: buys book inventory on credit from a publisher Tk 200,000. 10 January: earns Tk 40,000 cash for binding work. 15 January: pays rent Tk 60,000. 20 January: pays Tk 80,000 to the publisher. 31 January: pays wages Tk 35,000.

Find. The journal entry for each transaction, with narrations; the postings to the ledger accounts Cash, Shelving, Inventory, Publisher Payable, Owner's Equity, Service Revenue, Rent and Wages; and the balance of Cash.

Formula. For each transaction: debit the account that gains (assets, expenses) or loses (liabilities, owner's equity, revenue) value, credit the other, and make the totals equal. For the Cash balance: $B = L - S$, where L is the larger side total and S the smaller.

Steps. Step 1: journalise. The first table covers 1 to 10 January.

JOURNAL ENTRIES FOR 1 TO 10 JANUARY 2026

DATE	PARTICULARS	FOLIO	DEBIT (TK)	CREDIT (TK)
1 January 2026	Cash	L1	500,000	
	Owner's Equity	L5		500,000
	Narration: Owner invested cash to begin the workshop.			
2 January 2026	Shelving	L2	120,000	
	Cash	L1		120,000
	Narration: Bought shelving for cash.			
3 January 2026	Inventory	L3	200,000	
	Publisher Payable	L4		200,000
	Narration: Bought book inventory on credit from a publisher.			
10 January 2026	Cash	L1	40,000	
	Service Revenue	L6		40,000
	Narration: Received cash for binding work.			

The second table covers 15 to 31 January. In every entry the debit total equals the credit total.

The seven events of the month are the data, and the task is to find the final position.

Find, formula and steps: classify each event, then total the effects.

The task is three jobs: journalise, post, and balance Cash.

The method reuses the debit rules and the balance formula.

Steps 1 and 2 journalise the first four transactions.

The second table covers the last three transactions.

the records would be wrong. Chapter 6, 'The Trial Balance', shows how such mistakes are sought.

IN HISTORY

In 1397 Giovanni di Bicci de' Medici (1360–1429), a banker of Florence in Italy, founded the Medici Bank. It grew into one of the most powerful banking houses of the fifteenth century, with branches in cities such as Rome, Venice, Geneva, Lyon, Bruges and London. Each branch was a separate company with its own name, owner's equity, managers and account books, under a head office in Florence.

A branch of such a bank held coins, loans owed to it and goods, and it owed deposits and shares of profit to its partners. These are the two lists that the accounting equation says must match. A set of confidential ledgers, discovered in 1950, allowed the historian Raymond de Roover to reconstruct how the bank was organised and run. The bank collapsed in 1494, when the Medici were expelled from Florence. Its books remain a reminder that every business, old or new, must show what it holds and who has a claim on it.

A bank of the fifteenth century needed the same two lists that the equation joins.

Surviving ledgers let a modern historian reconstruct what the branches held and owed.

REVIEW QUESTIONS

3.1 Classify each item as an asset, a liability or owner's equity: (a) cash in the drawer, Tk 30,000; (b) a bill of Tk 12,000 owed to a paper supplier; (c) a binding press owned by the workshop; (d) a bank loan of Tk 90,000; (e) the owner's claim after all debts are paid.

3.2 A business has assets of Tk 240,000 and owner's equity of Tk 150,000. Find the liabilities.

3.3 At the start of a year a workshop's owner's equity was Tk 300,000. During the year the owner invested no more money. Revenue was Tk 410,000, expenses Tk 365,000 and drawings Tk 50,000. Find the owner's equity at the end of the year.

3.4 A workshop reports, in thousands of taka, assets of 1,250 and liabilities of 480. State its owner's equity in taka.

3.5 On 1 March a workshop starts with Tk 200,000 in cash from the owner. It then (a) buys equipment on credit for Tk 90,000; (b) earns Tk 30,000 in cash; (c) pays wages of Tk 18,000; (d) the owner withdraws Tk 10,000; (e) pays Tk 40,000 of the equipment debt. Find assets, liabilities and owner's equity after the fifth event, and check that they balance.

reader who was not present: an auditor, a new bookkeeper or the owner a year later.

PAUSE AND CHECK 5.P1 A workshop pays Tk 10,000 in cash for electricity. Which account is debited, which is credited, and what narration would you write?

Every journal entry obeys one rule: the total of the debit amounts equals the total of the credit amounts. In the entry above, Tk 500,000 is debited and Tk 500,000 is credited. If the totals differ, the entry is wrong, and the error must be found before anything is posted. This rule is the safeguard that makes the records dependable.

Most entries touch two accounts, but some touch three or more. A **compound entry** is a journal entry that has more than one debit or more than one credit. Take a different workshop, not part of the case, that buys a binding press for Tk 30,000. It pays Tk 10,000 in cash and owes Tk 20,000 to the dealer. The press is debited for Tk 30,000. Cash is credited for Tk 10,000 and the dealer's payable account is credited for Tk 20,000. The debit total, Tk 30,000, equals the credit total, Tk 30,000, even though three accounts appear.

A COMPOUND ENTRY FOR A DIFFERENT WORKSHOP

DATE	PARTICULARS	FOLIO	DEBIT (TK)	CREDIT (TK)
5 March 2026	Equipment		30,000	
	Cash			10,000
	Equipment Dealer Payable			20,000
	Narration: Bought a binding press, part paid in cash, balance owed.			

A short review of the journal.

Before moving on. The journal is the book of first entry. Each entry has a date, a debit line, a credit line, and a narration. Debits equal credits in every entry. A compound entry has more than one debit or more than one credit, and it still balances.

* * *

5.3 THE LEDGER AND POSTING

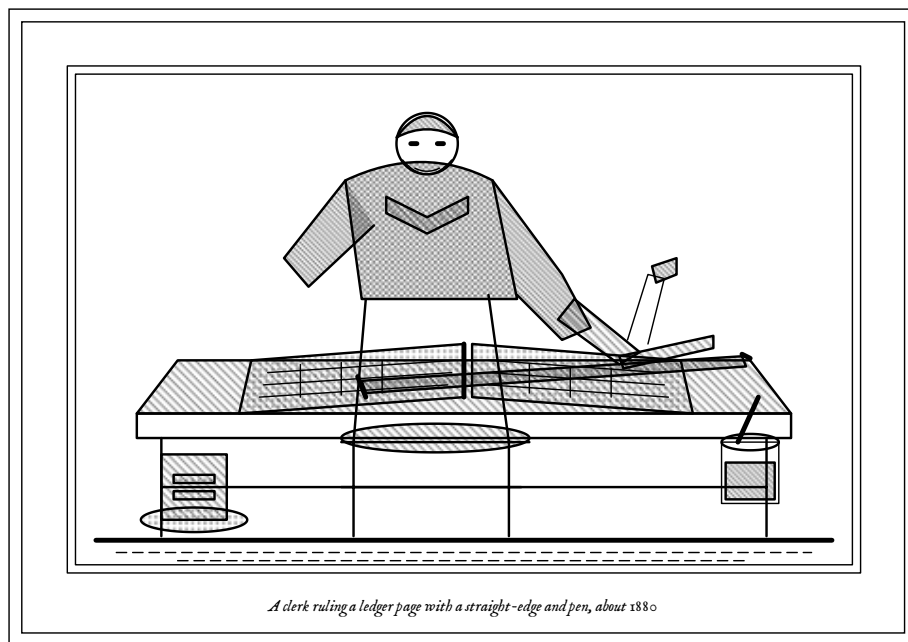
Return to the shopkeeper's question: how much was spent on rent? The journal lists events by date, so the rent payments lie scattered among all the other entries. A second book is needed, in which all the rent entries sit together on one page. Another page holds all the Cash entries, another all the Owner's Equity entries, and so on. Each page is one account.

The **ledger** is the book containing all of a business's accounts, where the effects of journal entries are collected. The ledger of the workshop will hold eight accounts: Cash, Shelving, Inventory, Publisher Payable, Owner's Equity, Service Revenue, Rent and

The second book gathers every entry that touches one account.

The ledger holds every account, and posting fills it.

Ledger. The book containing all of a business's accounts, where the effects



CHAPTER IV

Accounts, Debits and Credits

How can one fact be written down so that it can be found, added and checked?



This chapter teaches the account, the record that gathers every change to one kind of item. It shows the six types of account and the rule of debit and credit, called double entry, and applies the rule to seven transactions of a small business.

The ring shows all eight steps and the return to the beginning.

A short review of the cycle and of source documents.

One transaction on 1 January shows what the first book must hold.

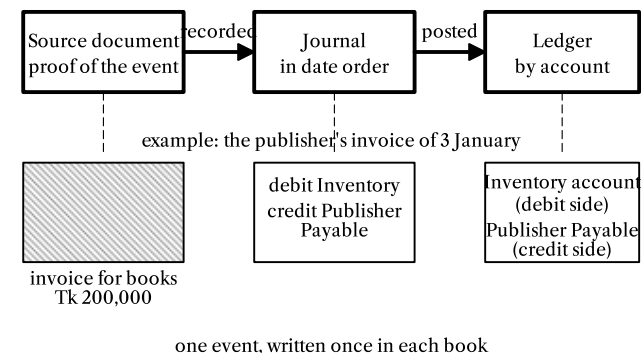


FIG. 5.1. The path from source document to journal to ledger

The second figure draws the whole cycle as a ring. Begin at the box marked 1 at the top and follow the arrows clockwise. Each box is one step. After step 8 the ring returns to step 1, which is why it is drawn as a circle and not as a line. The boxes with heavy outlines are the three steps taught in this chapter.

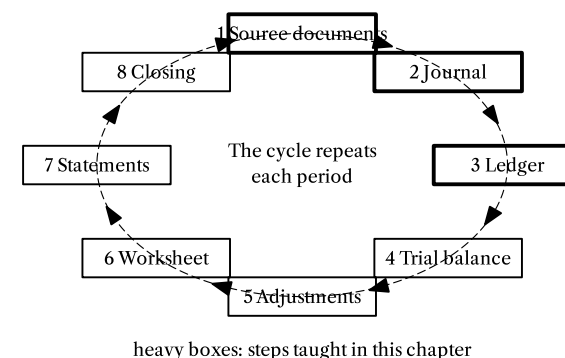


FIG. 5.2. The accounting cycle as a ring of steps

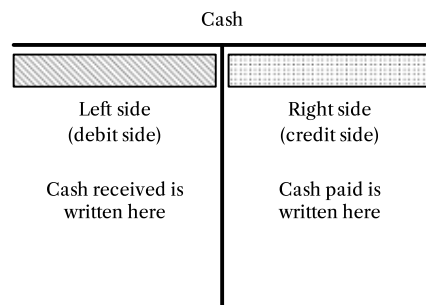
Before moving on. The accounting cycle is the series of steps repeated each period. Steps 1 to 3 move a transaction from a source document to the journal and then to the ledger. A source document is the proof on which the whole record rests.

* * *

5.2 THE JOURNAL

On 1 January 2026, Nurul invests Tk 500,000 in cash in the workshop. What must be written down? Two accounts are touched. The account Cash rises, because the workshop now holds more cash.

figure is one entry. Later in this chapter, the two sides receive their proper names.



The line under the name and the line down the middle make the letter T.

FIG. 4.1. A T-account for Cash, with its two sides labelled

Before moving on. An account collects all the changes to one kind of item, the chart of accounts lists every account in use, and a T-account is a quick drawing with a left side and a right side. You should now be able to say what the Cash account would show, and why a business keeps a separate account for each kind of item.

* * *

4.2 SIX TYPES OF ACCOUNT

Chapter 3 sorted the items of a business into groups when it built the accounting equation.¹ Accounts follow the same groups, and there are six types. Why six? Because each type moves in its own way when something happens, and the rules of the next section depend on knowing the type. The table below lists them. Read each row from left to right: the type, what its accounts record, and examples from Meghna Binding Service.

THE SIX TYPES OF ACCOUNT

TYPE	WHAT ITS ACCOUNTS RECORD	EXAMPLES
Asset	Things of value the business owns or controls	Cash, Shelving, Inventory
Liability	Amounts the business owes to others	Accounts Payable
Owner's capital	Value the owner has put into the business	Owner's Capital
Drawings	Value the owner takes out for personal use	Drawings
Revenue	Money earned or due from goods	Service Revenue

¹ Recall: asset, liability and equity, Chapter 3, 'The Accounting Equation', section 3.1 'Assets, Liabilities and Equity'; revenue, expense and drawings, section 3.2 'Revenue, Expenses and Drawings'.

The recap gathers the section into three sentences.

Every account belongs to one of six types taken from the equation.

YOU WILL NEED

Chapter 4 'Accounts, Debits and Credits', section 4.2 'Six Types of Account'; Chapter 4 'Accounts, Debits and Credits', section 4.3 'Debit and Credit'; Appendix A1 'Maths You Will Use'.

A drawer of loose slips holds every fact but answers no question quickly.

Two books, one arranged by date and one by account, answer both questions.

Imagine a shopkeeper who writes every payment and every receipt on a loose slip of paper and drops the slip into a drawer. After a month the drawer holds hundreds of slips. A question arrives: how much did the shop spend on rent? The shopkeeper must read every slip to find the answer. A second question follows: what happened on the tenth of the month? Again every slip must be read. The slips contain all the facts, yet they cannot answer either question without a long search.

The puzzle is how to arrange the facts so that both kinds of question can be answered at once. Accountants found the answer long ago: write each event twice, in two different books. The first book lists events in the order in which they happened. The second collects the same events under the name of the thing they affect, such as Cash or Rent. This chapter explains both books, the path that leads from the first to the second, and the repeating series of steps to which they belong.

* * *

5.1 THE ACCOUNTING CYCLE

A baker's daily routine is a cycle, and accounting has a cycle too.

The accounting cycle is the series of steps repeated each period.

Accounting cycle. The series of steps repeated each period, from recording transactions to preparing closing records.

This book divides the cycle into eight steps, and three belong to this chapter.

Think of a baker's working day. The baker buys flour, mixes dough, bakes loaves, sells them, counts the money and cleans the ovens. Tomorrow the baker performs the same steps in the same order. A routine that repeats in this way is a cycle. Accounting has a routine of this kind. In each period a business performs the same ordered steps to turn its events into reports, and then it begins again.

The name for this routine is the **accounting cycle**. The accounting cycle is the series of steps repeated each period, from recording transactions to preparing closing records. A period is the stretch of time covered by one set of reports, such as a month or a year. When one cycle ends, the next begins with the first transaction of the new period. Meghna Binding Service, the book-binding workshop that began on 1 January 2026 with its own separate set of books, runs one cycle for each period it chooses to report.

The eight steps appear in the table that follows. Read each row from left to right: the number of the step, what happens in it, and where this book teaches it. Steps 1 to 3 are the subject of this chapter. The later steps are previewed here in one line each and taught in full later. A trial balance, for example, is a list of the balances of all the accounts, used to test that the records agree; Chapter 6 teaches it. Some books group the steps differently, but the order never changes.

sides.² Assets stand on the left of the equals sign, and liabilities and owner's equity stand on the right. The rule copies this layout. An increase in an item from the left side of the equation is recorded on the left of its account. An increase in an item from the right side is recorded on the right. Decreases are recorded on the opposite side from increases.

Three types need a second thought: revenue, expenses and drawings. They are parts of owner's equity. Revenue raises equity, so it sits with the right side. Expenses and drawings lower equity, so their increases take the opposite side, the left. The expanded equation shows the result. In words: assets plus drawings plus expenses equal liabilities plus owner's capital plus revenue. In symbols: $A + D + E = L + C + R$, where A is assets, D is drawings, E is expenses, L is liabilities, C is owner's capital and R is revenue.

Now the rule can be read straight from the equation. The three types on the left, which are assets, drawings and expenses, increase on the left, so they increase by debit. The three types on the right, which are liabilities, owner's capital and revenue, increase on the right, so they increase by credit. Every decrease is on the other side. The table gathers the rule. Read each row from the left: the type, the side of an increase, the side of a decrease, and the normal balance, which the next paragraph explains.

THE SIDE OF INCREASE AND DECREASE FOR EACH TYPE

TYPE	INCREASE	DECREASE	NORMAL BALANCE
Asset	Debit	Credit	Debit
Drawings	Debit	Credit	Debit
Expense	Debit	Credit	Debit
Liability	Credit	Debit	Credit
Owner's capital	Credit	Debit	Credit
Revenue	Credit	Debit	Credit

An account usually grows more than it shrinks, so its balance usually ends on the side of its increases. The **normal balance** is the side of an account, debit or credit, on which increases are recorded. Cash has a debit normal balance, because cash increases by debit. Accounts Payable has a credit normal balance. When a balance falls on the other side, it is a signal to look closer: an asset that ends on the credit side may mean an error, or a bank overdraft.

PAUSE AND CHECK 4.P1 Rent is an expense. On which side is its normal balance, and why?

The figure shows the same rule as a picture. The left column lists the three types that increase by debit, and the right column lists the three that increase by credit. A short bar under each name marks its normal balance. You can rebuild the whole picture from one fact: assets, drawings and expenses stand on the left, and the rest stand on the right.

² Recall: the equation in expanded form, Chapter 3, 'The Accounting Equation', section 3.2 'Revenue, Expenses and Drawings'.

Drawings and expenses lower equity, so they behave like assets; revenue behaves like equity.

The three types on the left increase by debit; the three on the right increase by credit.

The normal balance is the side on which an account usually ends.

Normal balance. The side of an account, debit or credit, on which increases are recorded and where its balance usually falls.

The figure shows each type with its side of increase and its normal balance.

computis et scripturis, which describes how merchants of Venice kept their books.

Pacioli did not claim to have invented the method. He described what he called the method of Venice, which merchants had used for generations before the printing press made it widely available. His book was a textbook for students of mathematics and business. That is why it mattered: a method kept in merchants' habit became a method that anyone could read and teach.

REVIEW QUESTIONS

- 4.1 A small kiosk begins when its owner puts Tk 2,000 in cash into the business. It then buys a shelf for Tk 500 in cash and earns Tk 300 in cash from sales. Give the debit and the credit for each event, and state the balance of Cash.
- 4.2 A workshop has a Cash balance of Tk 50,000 on the debit side. It buys paper for Tk 18,000 on credit, pays Tk 7,500 to the supplier, earns Tk 26,400 in cash, pays wages of Tk 9,200 in cash, and the owner takes Tk 4,000 in cash for personal use. Give the entries, and find the balances of Cash and Accounts Payable.
- 4.3 A business borrows Tk 100,000 in cash from a bank, and later repays Tk 40,000 in cash. Give the entries for both events and the balance of the Bank Loan account.
- 4.4 A Cash account shows a balance of Tk 5,000 on the credit side. Say what this would normally mean, and name one way to find the cause.
- 4.5 For each account, state its type, the side that increases it, and its normal balance: Inventory, Accounts Payable, Drawings, Service Revenue, Wages.

Before moving on. A debit is the left side and a credit is the right side. Assets, drawings and expenses increase by debit, while liabilities, owner's capital and revenue increase by credit, and the normal balance of each type is the side of its increase. Under double entry every transaction has equal debits and credits. You should now be able to say, for any account, which side raises it and which side lowers it.

* * *

4.4 APPLYING THE RULES

Every transaction can be recorded by the same four questions, asked in order. First, which two accounts does the transaction affect? Second, what is the type of each? Third, does each account increase or decrease? Fourth, use the table to find the side: an increase in an asset is a debit, a decrease is a credit, and so on for each type. Finish by checking that the debit amount equals the credit amount. If you follow the questions in order, you do not need to memorise any transaction.

* * *

WORKED EXAMPLE: RECORDING THE JANUARY TRANSACTIONS

We apply the method twice. The first example uses small round numbers, so that the method is the only new thing. The second is the real case, with seven transactions. Each example follows the same order: Given, Find, Formula, Steps, Answer and Check.

Example 1. Given. A student begins a tea stall. (a) She puts Tk 1,000 of her own cash into the business. (b) She buys a stove for Tk 300 in cash. (c) She earns Tk 200 in cash by selling tea. *Find.* The debit and credit of each event, and the balance of each account.

Formula. In words: increases of assets, drawings and expenses are debits, increases of liabilities, owner's capital and revenue are credits, and each transaction has an equal debit and credit. In symbols, the rule comes from $A + D + E = L + C + R$.

Steps. (a) Cash is an asset and increases, so debit Cash 1,000. Owner's Capital increases, so credit Owner's Capital 1,000. (b) The stove is an asset and increases, so debit Equipment 300. Cash decreases, so credit Cash 300. (c) Cash increases, so debit Cash 200. Tea Revenue increases, so credit Tea Revenue 200. In the Cash T-account the left side holds 1,000 and 200, which total 1,200. The right side holds 300. The balance is $1,200 - 300 = 900$ on the left.

Answer. Cash has a debit balance of Tk 900. Equipment has a debit balance of Tk 300. Owner's Capital has a credit balance of Tk 1,000, and Tea Revenue has a credit balance of Tk 200.

Check. The debit balances total $900 + 300 = 1,200$, and the credit balances total $1,000 + 200 = 1,200$. They agree. In the equation, assets of Tk 1,200 equal owner's capital of Tk 1,000 plus revenue of

The recap gathers the section into three sentences.

Four questions turn any transaction into a debit and a credit.

Two examples follow: a small one to learn the method, then the full case.

The given facts of the small example are three events of a tiny stall.

The formula says which side raises each type and that the amounts are equal.

Each event gets one debit and one credit of equal amount.

The answer states each balance and its side.

The check compares the two sides of the equation.

The find and the formula are the same as in the first example.

The table shows the debit and credit of each transaction.

Adding each account's two sides gives its balance.

The answer lists the eight balances in one sentence.

Two checks confirm the result: equal totals and the equation.

15 January, it pays rent of Tk 60,000. 20 January, it pays Tk 80,000 to the publisher. 31 January, it pays wages of Tk 35,000.

Find. The debit and credit of each transaction, and the balance of each account. *Formula.* The same as in Example 1. Use the four questions of the last section for each line, and remember that Inventory is an asset and Accounts Payable is a liability.

Steps. The table below gives the result of the four questions for each transaction. Read each row from left to right: the date, the account debited, the account credited and the amount. Notice the sixth row. Paying the publisher lowers a liability, so Accounts Payable is debited, and it lowers an asset, so Cash is credited.

THE SEVEN TRANSACTIONS WITH THEIR DEBITS AND CREDITS

DATE	DEBIT	CREDIT	AMOUNT (TK)
1 January	Cash	Owner's Capital	500,000
2 January	Shelving	Cash	120,000
3 January	Inventory	Accounts Payable	200,000
10 January	Cash	Service Revenue	40,000
15 January	Rent	Cash	60,000
20 January	Accounts Payable	Cash	80,000
31 January	Wages	Cash	35,000

Now add the two sides of each account. Cash has debits of 500,000 and 40,000, which total 540,000, and credits of 120,000, 60,000, 80,000 and 35,000, which total 295,000. The difference is $540,000 - 295,000 = 245,000$ on the debit side. The other accounts are found in the same way, and the table below gives them all.

THE BALANCE OF EACH ACCOUNT AT 31 JANUARY 2026

ACCOUNT	DEBITS	CREDITS	BALANCE
Cash	540,000	295,000	245,000 debit
Shelving	120,000	0	120,000 debit
Inventory	200,000	0	200,000 debit
Accounts Payable	80,000	200,000	120,000 credit
Owner's Capital	0	500,000	500,000 credit
Service Revenue	0	40,000	40,000 credit
Rent	60,000	0	60,000 debit
Wages	35,000	0	35,000 debit

Answer. At 31 January 2026, Cash has a debit balance of Tk 245,000, Shelving Tk 120,000 and Inventory Tk 200,000. Rent has a debit balance of Tk 60,000 and Wages Tk 35,000. Accounts Payable has a credit balance of Tk 120,000, Owner's Capital Tk 500,000 and Service Revenue Tk 40,000.

Check. First, the debit balances total $245,000 + 120,000 + 200,000 + 60,000 + 35,000 = 660,000$. The credit balances total $120,000 + 500,000 + 40,000 = 660,000$. They agree. Second, the debit entries of the seven rows total 1,035,000, and so do the credit entries. Third, the assets are $245,000 + 120,000 + 200,000 =$

The list that carries out the test has a name. A **trial balance** is a list of all ledger account balances, with debits and credits totalled, to test that they are equal. The list has three parts: the name of each account, its balance in the Debit column if the balance is a debit, and its balance in the Credit column if the balance is a credit. At the foot, each column is added. The trial balance is taken at one date, so its heading reads "as at" that date.

The preparation has five steps. First, balance every account in the ledger, using the method of Chapter 5. Second, write the heading: the name of the business, the words "Trial Balance", and the date. Third, list each account with its balance in the correct column. Fourth, add each column. Fifth, compare the two totals. If they are equal, the test is passed. If they differ, the books contain a slip that must be found before the next step of the cycle.

The accounts are usually listed in the order of the ledger: assets first, then liabilities, owner's equity, revenue and expenses. An account with no balance, such as a suspense account that has been cleared, may be left off the list or shown with a dash. The order is a convention that makes lists easy to read and to compare from month to month. It does not affect the test.

The next figure shows the layout of a trial balance. Read it from the top. The first row holds the headings. The rows below hold one account each, with its balance in one of the two columns. A ruled line separates the accounts from the totals, and the totals sit in the last row, where the eye can compare them. A trial balance is step 4 of the accounting cycle.

Trial balance as at 31 January 2026		
Account	Debit	Credit
Cash	245,000	
Shelving	120,000	
Inventory	200,000	
Publisher Payable		120,000
Owner's Equity		500,000
Service Revenue		40,000
Rent	60,000	
Wages	35,000	
Totals	660,000	660,000
equal totals: the test is passed (amounts in taka)		

FIG. 6.1. A trial balance as a two-column list with totals

The January list is an **unadjusted trial balance**: a trial balance taken before the period-end adjustments have been recorded. An adjustment is a change that brings the accounts up to date at the end of a period; Chapter 7, 'Adjusting the Accounts', teaches them. Every trial balance in this chapter is unadjusted, because no adjustment has yet been made.

Case file, illustrative. Given: Meghna Binding Service, trial balance as at 31 January 2026. Debit balances: Cash 245,000; Shelving

A trial balance lists every ledger balance and compares the two column totals.

Trial balance. A list of all ledger account balances, with debits and credits totalled, to test that they are equal.

Preparing the list takes five steps, always in the same order.

Accounts are listed in a fixed order, and accounts with no balance are left out.

The figure that follows shows the layout, row by row.

Recall: accounting cycle, Chapter 5, section 5.1.

The trial balance in this chapter is taken before any period-end adjustment.

Unadjusted trial balance. A trial balance taken before the period-end adjustments have been recorded.

The January ledger is balanced first, then each balance is listed in its

Misunderstanding: Drawings belong to the income statement columns.

Misunderstanding: net income is entered once, on the same side everywhere.

Misunderstanding: equal totals prove that the worksheet is correct.

In 1979 two Americans built an electronic version of the accountant's columnar worksheet.

The modern Spreadsheet keeps the logic of the paper worksheet.

Some students extend Drawings to the income statement columns, because it looks like a cost. Drawings is money taken out by the owner. It is not an expense, and it reduces owner's equity. It belongs in the balance sheet debit column.

Some students enter net income on the debit side of both pairs of statement columns. The income columns need it on the debit side to balance, but the balance sheet columns need it on the credit side, because it increases owner's equity. A net loss reverses both sides.

Some students believe that equal totals prove a worksheet correct. They prove only that debits equal credits. A cost entered in the wrong account, or an amount left out of both sides, leaves the totals equal. The totals are a necessary test, not a sufficient one.

IN HISTORY

In 1979, Dan Bricklin and Bob Frankston wrote VisiCalc, the first electronic spreadsheet for the Apple II. Bricklin was then a student at Harvard Business School, in Massachusetts, in the United States. Frankston was a programmer. The Apple II was a small personal computer, sold to homes and offices. Their program was built from the accountant's columnar worksheet. It gave the computer the same columns and rows that an accountant draws on a large ruled sheet, and it recalculated the dependent figures whenever one figure changed.²

Today, the worksheet of this chapter often lives on a computer screen. A clerk who changes one adjustment sees every total change at once. Yet the logic has not changed. Figures still move from left to right through the same pairs of columns, and the same equal-total checks still apply. A machine calculates faster than a clerk, but it cannot tell whether an amount sits in the right account. That judgement remains with you.

REVIEW QUESTIONS

8.1 Clean numbers. A business has these unadjusted balances: Cash Tk 3,000, Prepaid Insurance Tk 1,200 and Wages Tk 3,000 (debits); Capital Tk 3,200 and Fees Revenue Tk 4,000 (credits). Insurance of Tk 300 has expired. Complete the ten-column worksheet and state the net income.

8.2 Realistic data. The unadjusted balances are, in taka: Cash 24,500; Accounts Receivable 18,300; Supplies 4,200; Equipment 48,000; Drawings 6,000; Wages 17,200; Rent 4,800; Utilities 2,000 (debits); Accounts Payable 9,600; Capital 54,000; Fees Revenue 61,400 (credits). Supplies on hand at the end are Tk 1,500; wages owed are Tk 1,300; depreciation of equipment is Tk 4,800. Complete the worksheet and state the net income.

8.3 Changed sign and unit. For one quarter, the income columns total 21,500 (debit) and 18,000 (credit), and the balance sheet

² Software Arts was the company that Bricklin and Frankston formed to develop the program. A firm called Personal Software sold it.

6.2 WHAT THE TRIAL BALANCE DOES NOT SHOW

Suppose that the rent payment of 15 January had never been written in the journal. Rent would show nothing, and Cash would show Tk 305,000 instead of Tk 245,000. The debit total would be $305,000 + 120,000 + 200,000 + 35,000 = \text{Tk } 660,000$. The credit total would still be Tk 660,000. The test is passed, yet the books are wrong. This example is the central lesson of the section: the test checks equality, not truth.

A framework helps. Divide errors into two groups. The first group changes only one side of the books, so the totals differ and the trial balance finds the error. Examples are a debit posted without its credit, an amount posted wrongly on one side only, a column added wrongly, and a balance listed in the wrong column. The second group changes both sides by the same amount, so the totals stay equal and the test says nothing.

The table that follows lists six kinds of error. Read a row from left to right: the kind of error, whether the totals differ, and an example from the January case. The example of the wrong account is a payment of wages debited to Rent. Rent would show Tk 95,000 and Wages nothing, but the debit total would remain Tk 660,000.

SIX KINDS OF ERROR AND THE TRIAL BALANCE

KIND OF ERROR	TOTALS DIFFER?	EXAMPLE FROM THE JANUARY CASE
One side of an entry not posted	Yes	The Rent debit of 60,000 is never posted; the debit total becomes 600,000
Wrong amount on one side only	Yes	Wages posted as 53,000 on the debit side only
Whole transaction left out	No	The 15 January rent entry is never recorded
Same wrong amount on both sides	No	Wages and Cash both posted as 53,000
Wrong account of the same type	No	Wages debited to Rent
Two errors that cancel	No	A debit too high by 10,000 and another too low by 10,000

The wages example in the table is a special slip. A **transposition error** is a mistake in which two digits of a number are written in the wrong order, such as 53 for 35. It is common because the eye reads quickly and the hand writes quickly. Posted on one side only, it makes the totals differ by a number that always divides by 9. The reason is short. A two-digit number with digits a and b equals $10a + b$. With the digits reversed it equals $10b + a$. The difference is $9a - 9b = 9(a - b)$, which is a multiple of 9. For 53 and 35, the difference is $18 = 9 \times 2$.

The property has a limit. A difference that divides by 9 suggests a transposition, but other slips can also give such a difference, so the

A balanced list can still hide an error, as one missing entry shows.

The test finds errors that put the totals out of line.

A table sets six kinds of error beside their effect on the totals.

One common slip in copying digits has a name and a useful property.

Transposition error. A mistake in which two digits of a number are written in the wrong order, such as 53 for 35.

A difference divisible by 9 suggests a transposition but

MEGHNA AND SONS WORKSHEET, COLUMNS 5 TO 10 (TAKA)

ACCOUNT	ADJUSTED DR	ADJUSTED CR	INCOME DR	INCOME CR	BALANCE SHEET DR	BALANCE SHEET CR
Cash	150,000				150,000	
Accounts Receivable	110,000				110,000	
Supplies	15,000				15,000	
Prepaid Insurance	27,000				27,000	
Equipment	360,000				360,000	
Accumulated Depreciation		72,000				72,000
Accounts Payable		60,000				60,000
Wages Payable		14,000				14,000
Unearned Revenue		12,000				12,000
Capital		400,000				400,000
Drawings	80,000				80,000	
Service Revenue		788,000		788,000		
Wages	344,000		344,000			
Rent	120,000		120,000			
Utilities	26,000		26,000			
Supplies Expense	33,000		33,000			
Insurance Expense	9,000		9,000			
Depreciation Expense	72,000		72,000			
Totals	1,346,000	1,346,000	604,000	788,000	742,000	558,000
Net income			184,000			184,000
Final totals			788,000	788,000	742,000	742,000

Steps and answer: expenses Tk 604,000, net income Tk 184,000, both pairs balanced.

A second check with the accounting equation confirms the result.

Steps. Columns 5 and 6 each total Tk 1,346,000. Column 7 totals Tk $344,000 + \text{Tk } 120,000 + \text{Tk } 26,000 + \text{Tk } 33,000 + \text{Tk } 9,000 + \text{Tk } 72,000 = \text{Tk } 604,000$. Column 8 holds Service Revenue of Tk 788,000. So NI = $\text{Tk } 788,000 - \text{Tk } 604,000 = \text{Tk } 184,000$, which we enter under column 7 and under column 10. Column 9 totals Tk 742,000. Column 10 totals $\text{Tk } 72,000 + \text{Tk } 60,000 + \text{Tk } 14,000 + \text{Tk } 12,000 + \text{Tk } 400,000 = \text{Tk } 558,000$ before net income, and Tk 742,000 after it. *Answer:* Net income is Tk 184,000, the same as the case file.

Check. The worksheet checks pass, and we can test the result a second way. Assets less Accumulated Depreciation are $\text{Tk } 662,000 - \text{Tk } 72,000 = \text{Tk } 590,000$. Liabilities are $\text{Tk } 60,000 + \text{Tk } 14,000 + \text{Tk } 12,000 = \text{Tk } 86,000$. Owner's equity is $\text{Tk } 400,000 + \text{Tk } 184,000 = \text{Tk } 584,000$.

When the cause is found, the books are corrected by a new journal entry. The entry carries a narration and is posted like any other. The wrong figure is never rubbed out, because the books must stay readable to an auditor, who is a person that checks records for accuracy. A correcting entry usually has two parts. It removes the wrong amount from the account in which it was posted, and it clears the suspense account.

Before moving on. A difference is searched for in a fixed order: re-add, test for divisibility by 9, compare with one balance or half a balance, tick the postings, recalculate. A suspense account holds the difference for a short time. A correcting entry, never an erasure, fixes the error.

* * *

WORKED EXAMPLE: THE WAGES DIFFERENCE

Case file, illustrative. Given: the January 2026 ledger of Meghna Binding Service is correct except that the wages of 31 January were posted to the Wages account as Tk 53,000 instead of Tk 35,000. The credit to Cash was posted correctly. The correct trial balance totals Tk 660,000 on each side.

Find. The trial balance totals with the error, the difference, its probable cause, and the entries that clear it.

Formula. In words: the difference is the larger column total minus the smaller. In symbols: $d = DT - CT$, where DT is the debit total and CT is the credit total. A transposition of two digits gives $d = 9(a - b)$, where a and b are the two digits, in the units of their place.

Steps. Step 1: add the columns. The wrong debit balance for Wages is Tk 53,000, so $DT = 245,000 + 120,000 + 200,000 + 60,000 + 53,000 = \text{Tk } 678,000$. $CT = 120,000 + 500,000 + 40,000 = \text{Tk } 660,000$. Step 2: subtract. $d = 678,000 - 660,000 = \text{Tk } 18,000$, with the debits larger.

Step 3: test for 9. $18,000 \div 9 = 2,000$, so the difference divides by 9. Step 4: look for the digits. Two numbers that differ by 18 in the thousands are 53 and 35, which are a transposition of each other. The payroll record, which is the source document, shows Tk 35,000. The cause is therefore a transposition error in the debit posting to Wages.

Step 5: hold the difference. The credits are short by Tk 18,000, so a credit balance of that amount is entered in a suspense account. The table that follows shows the list.

A found error is corrected by a new journal entry and never by erasing.

A short review of the search, the suspense account and the correcting entry.

The case file gives an error to find and to fix.

The task is to find the difference, its cause and the correction.

The two formulas are a subtraction and a divisibility test.

Steps 1 and 2 find the totals and the difference.

Steps 3 and 4 test the difference and confirm the cause.

Step 5 shows the trial balance with the difference held in suspense.

WHERE EACH KIND OF ACCOUNT IS ENTERED		
KIND OF ACCOUNT	NORMAL SIDE	STATEMENT COLUMNS
Asset, for example Cash	Debit	Balance sheet, debit
Accumulated Depreciation	Credit	Balance sheet, credit
Liability, for example Wages Payable	Credit	Balance sheet, credit
Capital	Credit	Balance sheet, credit
Drawings	Debit	Balance sheet, debit
Revenue	Credit	Income statement, credit
Expense	Debit	Income statement, debit

Two accounts cause most confusion, and some offices use fewer columns.

The recap restates the five pairs and the four Steps.

Two rows need care. Accumulated Depreciation is not a liability; it is a credit balance that reduces the Equipment on the balance sheet, so it goes to the balance sheet credit column. Drawings is a debit balance, but it is not an expense, because it records money taken out by the owner. It reduces owner's equity, so it goes to the balance sheet debit column. Finally, some offices use fewer columns than ten.¹ We use ten because the adjusted trial balance then has its own pair of totals, and each pair is easier to check.

Before moving on. The worksheet has five pairs of columns: trial balance, adjustments, adjusted trial balance, income statement and balance sheet. Step 1 copies the trial balance. Step 2 enters the adjustments. Step 3 combines them into the adjusted trial balance. Step 4 sends each balance to one statement column. *You should now be able to say* what each pair of columns holds and which accounts go to the income statement columns.

* * *

WORKED WORKSHEET

We now complete the worksheet. This section has three tasks: to extend each balance, to total the columns and find the result of the year, and to check the worksheet. We begin with the stall again, because its figures are small, and then we turn to the running case. Take each task slowly, and notice that none of them needs a new idea. They need only care.

Given. The stall's adjusted trial balance from section 8.2. The debits are Cash 5,000, Supplies 400, Equipment 6,000, Wages 2,300, Supplies Expense 600 and Depreciation Expense 500. The credits are Capital 8,000, Repair Revenue 6,000, Wages Payable 300 and Accumulated Depreciation 500. All figures are in taka. *Find.* The income statement columns, the balance sheet columns and the result of the year.

¹ To extend a balance needs no calculation, only care. Wages
¹ Some offices use eight columns, leaving out the adjusted trial balance pair and extending each balance straight from the first two pairs. The method is shorter, but an error is harder to locate.

We finish the stall first, with small numbers, then complete the running case.

Given and Find: the stall's adjusted balances, and the statement columns to complete.

Extend Word in use. In daily

may be held in a suspense account, and a correcting entry then removes the error.

THINK IT THROUGH

6.T1 Name two errors that leave a trial balance in balance. For each, give an example from the January ledger of Meghna Binding Service and say what would still be wrong.

WHERE PEOPLE GO WRONG

Many students believe that a balanced trial balance proves that the books are correct. It proves only that the debit balances and the credit balances are equal. Missing entries and equal errors on both sides remain hidden.

A second error is to assume that the difference equals the error. A transposition error of 53 for 35 makes an error of 18, which equals the difference only because the credit was right. A balance in the wrong column makes the difference twice the balance.

A third error is to leave the difference in the suspense account and carry on. The account is temporary. Its balance must be explained and cleared before reports are prepared.

A fourth confusion is to treat the trial balance as the balance sheet. The trial balance is a working list for testing the books. The balance sheet is a formal report taught in Chapter 9, 'Financial Statements and Closing Entries'.

IN HISTORY

Simon Stevin was born in Bruges, in Flanders, in 1548 and died in 1620. He was a mathematician and engineer who served as tutor and adviser to Prince Maurice of Nassau, the military leader of the Dutch Republic. Stevin wrote on many subjects, among them bookkeeping. His book on the subject, usually called *Princely Bookkeeping*, was printed in Leiden in the first decade of the seventeenth century, with a French edition in 1608.

Stevin urged that the household and finances of a prince be recorded by the double-entry method that merchants already used, and he put the method into practice for Prince Maurice. The idea behind a trial balance is the same idea: the books of a business or a state should be tested against themselves at regular times, so that slips are found by method and not by chance.

REVIEW QUESTIONS

6.1 Prepare a trial balance as at 31 March 2026 from these balances: Cash Tk 50,000 debit; Equipment Tk 30,000 debit; Owner's Equity Tk 80,000 credit. State whether the totals are equal.

6.2 Prepare a trial balance from these balances and state whether it is balanced: Cash 187,450 debit; Inventory 96,300 debit; Equipment 150,000 debit; Rent 42,000 debit; Wages 78,500 debit; Publisher Payable 64,350 credit; Owner's Equity 400,000 credit; Service

A balanced trial balance does not prove the books are right.

*The difference is not always
the size of the error.*

The suspense account is not a place to leave a problem.

A trial balance is a test and not a financial statement.

*A Flemish mathematician
argued that a ruler's accounts
also need method.*

*Stevin wanted a slate to keep
its books by the method of
merchants.*

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two hatched pairs on the right are the statement columns. In the tables of this chapter, Dr stands for debit and Cr stands for credit.

[illegible]

FIG. 8.1. *Ten-column worksheet layout with column headings*

The first pair is the trial balance columns. In step 1, we copy every account title and balance from the trial balance, debits into column 1 and credits into column 2. Then we total both columns. The totals must agree before we go further, because an error here would travel through the whole worksheet. For the stall, Tk 14,000 equals Tk 14,000, so we continue. Below the last account we leave blank rows for accounts that the adjustments will create, such as Supplies Expense. Their cells in columns 1 and 2 stay empty, because the ledger holds no balance for them yet. If the totals disagree, stop and find the error first. A transposition error is a common cause.

The second pair is the adjustments columns, and here the worksheet shows its purpose. Under the accrual basis of accounting, each accounting period must carry its own revenue and expenses, and the matching principle places an expense in the period that earned the related revenue. The adjustments make this happen. In step 2, we enter each adjustment as a debit in column 3 and an equal credit in column 4. We label both parts with the same small letter, so that a reader can pair them. Adjustment (a): debit Supplies Expense Tk 600 and credit Supplies Tk 600. Adjustment (b): debit Wages Tk 300 and credit Wages Payable Tk 300. Adjustment (c): debit Depreciation Expense Tk 500 and credit Accumulated Depreciation Tk 500.

Now we total columns 3 and 4. Each equals Tk 1,400, and they must agree, because every adjustment has the same amount on both sides. Remember the limit from section 8.1: the worksheet only shows the adjustments. The entries must still be recorded in the journal and posted.

The third pair is the **adjusted trial balance**. An adjusted trial balance is a trial balance prepared after the adjusting entries, showing the correct balance of each account. For example, Supplies stand at Tk 1,000 in column 1; after the Tk 600 credit in column 4,

Recall: trial balance, Chapter 6, section 6.1 'Preparing a Trial Balance'.

Step 1 copies the trial balance into columns 1 and 2 and checks the totals.

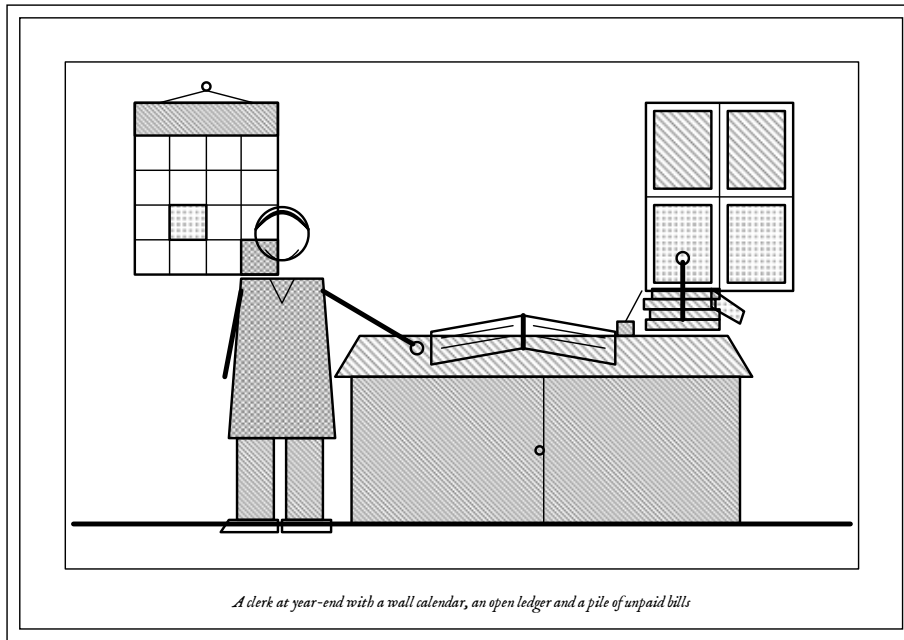
Recall: accrual basis of accounting and the matching principle.

Step 2 enters each adjustment as an equal debit and credit in columns 3 and 4.

The adjustments columns must total equally, and they are not yet in the ledger.

GO SLOWLY HERE.

The adjusted trial balance shows the correct balance of each account after the changes.



CHAPTER VII

Adjusting the Accounts

If money moves on one day and the benefit arrives on another, which day counts?



Records made as money moves do not always show what a period really earned and used. This chapter explains why accounts need adjustment at period end, teaches the two families of adjustment, deferrals and accruals, and shows how each is entered, using the year-end records of a small binding workshop.

YOU WILL NEED

Chapter 7 'Adjusting the Accounts', section 7.4 'Recording Adjusting Entries'; Chapter 6 'The Trial Balance', section 6.1 'Preparing a Trial Balance'; Appendix A1 'Maths You Will Use'.

A year-end clerk holds correct balances that are not yet ready to use.

Changing the records directly, and drafting the statements early, are both risky.

The question of the chapter: how can every change be tested on paper first?

It is the evening of 31 December, and a clerk in a small office is holding the trial balance. It lists eighteen accounts, and the debit total equals the credit total. Yet the balances are not ready to use. Some supplies have been consumed, some wages are owed, and the equipment has lost part of its value. Six changes must be made before anyone can learn what the business gained or lost during the year.

The clerk could write each change into the ledger at once. But suppose the third change contains an error. The ledger now holds a wrong figure, and correcting it leaves crossings-out on a record that other people must trust. The clerk could also draft the final statements straight away. But the statements need the changed figures, so a late discovery means writing every statement a second time.

So here is the question for this chapter. How can an accountant try every change, see every result and find every error, without writing in any permanent record? Think about a difficult sum at home. You take a spare sheet of paper, write the figures in columns and work there, and only then write the answer neatly. Accountants do the same. Their spare sheet has a name, and it has an exact layout. By the end of this chapter you will be able to build it, complete it and check it.

8.1 WHY USE A WORKSHEET

Before meeting the subject idea, consider a family that is planning a wedding. On a large sheet of paper, the family lists every cost. Beside the list is a column headed "first estimate". Next to it is a column headed "changes", where the family enters a rise in the hall price and a cousin who has cancelled. A third column shows the new estimate. Only when the whole sheet is complete does the family write the final budget on a clean page. The large sheet is rough, and mistakes on it cost little. The clean page is the one that other people will read.

The accountant's method follows the same pattern, and the rough sheet has a name. A **worksheet** is a working paper with columns that moves figures from the trial balance through the adjustments to the statements. For example, a worksheet may show Supplies as Tk 1,000 in one column and Tk 400 in a later column, so that a reader sees the figure before and after a change. Notice the word "working". The worksheet is where the work is done. It is not where the answer is published. The family's first estimate matches the trial balance, the changes match the adjustments, and the clean page matches the statements.

GO SLOWLY HERE.

A family budget on a rough sheet shows the idea before the subject.

A worksheet is a working paper that carries figures from the trial balance to the statements.

Sheet. Word in use. In daily life, a flat piece of paper or cloth. In accounting, the large ruled page, or its computer copy, on which a worksheet is drawn.

Worksheet. A working paper with columns that moves figures from the trial balance through adjustments

There are two ways to decide which period an item belongs to. The first follows the cash: record an item when money moves. The second follows the activity: record an item when the work is done or the benefit is received. The second way is the **accrual basis of accounting**: recording revenue when earned and expenses when incurred, whether or not cash has moved. An expense is incurred when the cost is used up, not when it is paid. A worker who labours on 28 December 2026 has earned wages in December, even if the pay arrives on 5 January 2027. On the accrual basis those wages belong to 2026. From here onward this book uses the accrual basis.

Think about fairness. If the workshop earns fees all year, the report for the year should carry the cost of the room, the labour and the insurance that made those fees possible. The **matching principle** is the rule that expenses are recorded in the same period as the revenue they helped to earn. Insurance for the months from October to December belongs beside the revenue of those months. Insurance for the following nine months belongs beside the revenue of 2027. The accrual basis says when to count an item; matching says where to place its cost.

PAUSE AND CHECK 7.P1 A workshop pays Tk 24,000 on 1 November for twelve months of rent. How much of this cost belongs to a year ending on 31 December, and which principle decides it?

The ordinary entries of the year, taught in Chapter 5, are made when something happens: cash is paid, a bill arrives, a customer is billed. Two problems follow. Some items are recorded too early or too late for the period: a payment may cover many months, and a fee may arrive before the work. Other items are never recorded, because no event pushed anyone to write them: wages that are earned but unpaid, and work that is done but not yet billed. So at the end of each period the accounts must be checked and corrected. The corrections fall into two families. In a deferral, cash moves first and the effect is put off to later. In an accrual, the effect comes first and the cash moves later. Chapter 8, 'The Worksheet', and Chapter 9, 'Financial Statements and Closing Entries', use the corrected figures.

The table below is a trial balance.² It was taken on 31 December 2026 before any correction. Read each row as an account name and its balance. The debit total and the credit total both equal Tk 1,240,000.

Revenue and expenses belong to the period of the activity, not the period of the cash.

GO SLOWLY HERE.

Accrual. Word in use. In daily life, a gradual build-up, such as interest growing in a savings account. In accounting, an amount earned or used up that cash has not yet settled.

Accrual basis of accounting. Recording revenue when earned and expenses when incurred, whether or not cash has moved.

Costs are placed beside the revenue they helped to earn.

Matching principle. The rule that expenses are recorded in the same period as the revenue they helped to earn.

At period end some items are wrongly timed and others are missing.

GO SLOWLY HERE.

The year-end figures of the workshop show what needs correction.

² Recall: trial balance, Chapter 6, 'The Trial Balance', section 6.1 'Preparing a Trial Balance'.

used up, move that part into an expense or into revenue, and leave the rest where it is.

On 1 October 2026 the workshop pays Tk 36,000 for twelve months of insurance. On that day nothing has been used up, and the whole payment is an asset: a right to be protected for a year. A **prepaid expense** is a cost paid in advance that is an asset until it is used up. Insurance, rent and subscriptions are common examples. Each day, a small part is used up and becomes an expense. At the end of the period, the entry moves the used part from the asset to an expense account.

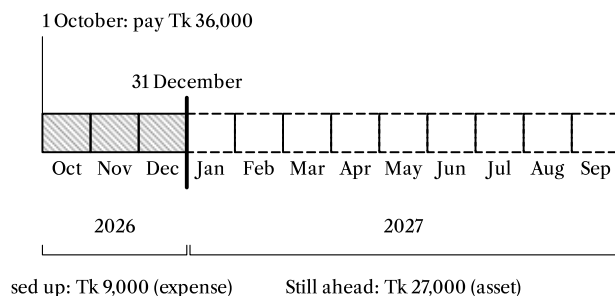


FIG. 7.1. Insurance paid on 1 October 2026 for twelve months: the part used by 31 December and the part still to come

To read the figure, follow the bar from left to right. Each box is one month. The hatched boxes, October to December 2026, are used up in the period. The dashed boxes, January to September 2027, are still ahead. The vertical line marks the end of the period. Each box stands for Tk 3,000.

Given. Insurance of Tk 36,000 paid on 1 October 2026 for twelve months; the period ends on 31 December 2026. *Find.* The insurance expense for 2026 and the prepaid expense that remains. *Formula.* In words: monthly cost equals the amount paid divided by the months covered; expense equals monthly cost times months used; remainder equals amount paid minus expense. In symbols: $m = P \div n$; $X = m \times k$; $R = P - X$, where P is the amount paid, n the months covered and k the months used. *Steps.* $m = 36,000 \div 12 = 3,000$. The months used are October, November and December, so $k = 3$. $X = 3,000 \times 3 = 9,000$. $R = 36,000 - 9,000 = 27,000$. Entry: debit Insurance Expense 9,000; credit Prepaid Insurance 9,000. *Answer.* The insurance expense for 2026 is Tk 9,000, and Tk 27,000 remains as a prepaid expense. *Check.* $9,000 + 27,000 = 36,000$. Also, $27,000 \div 3,000 = 9$, the nine months still ahead.

Some assets are used up without a clear daily rate. The workshop buys paper, thread and glue during the year, and records the cost in an asset account called Supplies. These are items bought to be used in the work, not goods held for sale. Nobody records each

A cost paid in advance is an asset until it is used up.

GO SLOWLY HERE.

Prepaid expense. A cost paid in advance that is an asset until it is used up.

The bar shows twelve months of cover, split at 31 December.

Worked example: the insurance adjustment, step by step.

Supplies are counted at the end, and the difference is the amount used.

Three slips are common when the table is built.

The worked case joins the earlier ideas into one routine.

The chapter in a few lines: periods, the accrual basis, matching and six corrections.

Adjusting entries do not correct mistakes; they update the records for the passing of time.

A prepaid cost is an asset when paid, not an expense.

Money received in advance is not yet revenue.

Depreciation does not set money aside.

Students often forget one side of an entry, so the totals differ by one amount. They carry a figure to the wrong side, such as treating Accumulated Depreciation as a debit. And they replace the old balance instead of changing it, for example writing 33,000 for Supplies instead of 48,000 less 33,000. Compare each total with the check before moving on.

Before moving on. The worked case applies all six adjustments to the unadjusted balances and shows that profit falls from Tk 280,000 to Tk 184,000. You should now be able to carry a trial balance through its adjustments and test the result.

* * *

SUMMARY

Results are measured for an accounting period. The accrual basis places revenue and expenses in the period of the activity, and the matching principle sets each cost beside the revenue it helped to earn. Because ordinary entries follow cash, the accounts need adjustment at period end. A prepaid expense is an asset that is used up, unearned revenue is a liability that is earned, an accrued expense is a cost owed, and accrued revenue is revenue owed to the business. Each correction is an adjusting entry dated on the last day of the period.

THINK IT THROUGH

7.T1 Suppose Meghna Binding Service made none of the six adjustments. State the profit that its records would show for 2026, and say who might be misled and how.

7.T2 Explain why an adjusting entry does not involve Cash, using the insurance example.

WHERE PEOPLE GO WRONG

Many readers believe that an adjusting entry corrects a mistake. It does not. The original entries were right on the day they were made. Adjusting entries bring the accounts up to date for time that has passed or work that has been done.

Another belief is that a cost is an expense on the day it is paid. When the workshop pays Tk 36,000 for a year of insurance, nothing is used up on that day, so the payment is an asset. Only the part used up becomes an expense.

A third belief is that money received is always revenue. The advance of Tk 24,000 from a school was a liability on the day it arrived, because the work was still owed. Only the work done, Tk 12,000, became revenue by 31 December. Revenue follows the work, not the cash.

A fourth belief is that depreciation puts cash aside for replacing equipment. It does not. It spreads a cost already paid across the years of use, and no cash moves when it is recorded.

happened, but the cash has not yet moved, and nothing has yet been written down. Two accruals concern the workshop, one for a cost and one for revenue.

The workshop pays its workers on the fifth day of each month. On 31 December 2026 the workers have worked for several days since their last pay, and Tk 14,000 is owed to them. Nothing has been written down, because no money has moved. An **accrued expense** is a cost already used up but not yet paid, shown as a liability. The workers' labour has been used in earning December's revenue, so the matching principle puts its cost in December. The liability is real: the workshop owes the workers Tk 14,000.

Given. The Wages account shows Tk 330,000 before adjustment; wages of Tk 14,000 have been earned by workers but not paid. *Find.* The wages expense for 2026 and the liability at 31 December. *Formula.* In words: wages expense equals wages recorded plus wages owed. *Steps.* $330,000 + 14,000 = 344,000$. *Entry:* debit Wages 14,000; credit Wages Payable 14,000. *Answer.* The wages expense for 2026 is Tk 344,000, and the workshop owes Tk 14,000 in Wages Payable. *Check.* The new liability, Tk 14,000, equals the increase in the expense, so the equation still balances.

Now consider the other side. A freelance designer finishes a job in the last week of December, but sends the invoice in January. The work is done, so the revenue belongs to December. In the workshop's case, binding work worth Tk 20,000 was finished for a school before 31 December, and the invoice has not yet been prepared. **Accrued revenue** is revenue already earned but not yet received or billed, shown as an asset. The asset is the right to be paid. An amount owed to the business by a customer is called an account receivable; some books say debtor. This book says account receivable.

Given. Binding work worth Tk 20,000 is complete on 31 December 2026 but not billed. Service Revenue stands at Tk 756,000 before this item and before the item for unearned revenue. *Find.* The entry for the unbilled work and its effect on Accounts Receivable. *Formula.* In words: revenue earned equals the value of work done; the customer's debt is added to Accounts Receivable. *Steps.* *Entry:* debit Accounts Receivable 20,000; credit Service Revenue 20,000. Accounts Receivable becomes $90,000 + 20,000 = 110,000$. *Answer.* Revenue rises by Tk 20,000, and Accounts Receivable becomes Tk 110,000. *Check.* Assets rise by 20,000 and owner's equity rises by 20,000 through revenue, so the equation balances.

PAUSE AND CHECK 7.P2 At year end a workshop owes its workers Tk 6,500 and has finished work worth Tk 11,000 that it has not yet billed. State the two amounts, and say which is an asset and which is a liability.

Before moving on. An accrued expense is a cost used up but unpaid, and it creates a liability. Accrued revenue is revenue earned but unbilled, and it creates an asset. In both, the cash moves later. You should now be able to say how an accrual differs from a deferral.

A cost already used up but not yet paid is recorded as a debt.

GO SLOWLY HERE.

Accrued expense. A cost already used up but not yet paid, shown as a liability.

Worked example: wages owed at the end of the year.

Revenue already earned but not yet billed is recorded as an asset.

Accrued revenue. Revenue already earned but not yet received or billed, shown as an asset.

Worked example: work done but not yet billed.

Both accruals record an effect before any cash moves.

	Cash moves first (deferral)	Cash moves later (accrual)
Cost side	Prepaid expense an asset Insurance: Tk 9,000 used	Accrued expense a liability Wages owed: Tk 14,000
Revenue side	Unearned revenue a liability Advance fee: Tk 12,000 earned	Accrued revenue an asset Unbilled work: Tk 20,000

Supplies used and depreciation follow the deferral pattern.

FIG. 7.2. The four kinds of adjustment, arranged by when cash moves and by cost or revenue

Each correction is written as a journal entry, as in Chapter 5. An **adjusting entry** is a journal entry made at period end to bring an account to its correct balance for the period. It is dated on the last day of the period, here 31 December 2026, and carries a short narration. Three habits apply to every adjusting entry. It is made after the unadjusted trial balance. It changes at least one revenue or expense account and at least one asset or liability account. And it does not involve Cash, because the adjustment is about timing, and the cash has already moved or will move later.³

The table lists the six adjustments for Meghna Binding Service. Read each row from left to right: the reference letter, the account debited, the account credited, the amount in taka, and the narration that explains the entry.

ADJUSTING ENTRIES OF MEGHNA BINDING SERVICE, 31 DECEMBER 2026 (TK)

REF	DEBIT	CREDIT	AMOUNT	NARRATION
(a)	Supplies Expense	Supplies	33,000	Supplies used in 2026
(b)	Insurance Expense	Prepaid Insurance	9,000	Insurance expired, 1 October to 31 December
(c)	Depreciation Expense	Accumulated Depreciation	72,000	One year of depreciation on equipment
(d)	Unearned Revenue	Service Revenue	12,000	Advance fees earned in 2026
(e)	Wages	Wages Payable	14,000	Wages earned but unpaid
(f)	Accounts Receivable	Service Revenue	20,000	Work done but not billed

Common slip: forgetting that

Several slips occur in this work. Students record the whole pay-
³ Recall: debit and credit, Chapter 4, 'Accounts, Debits and Credits', section 4.3 'Debit and Credit'.

* * *

SUMMARY

A special journal records one repeated kind of transaction in its own book. The sales journal holds credit sales, and the purchases journal holds credit purchases. Each line is posted to an individual account in a subsidiary ledger, and each total is posted once to the main ledger. A control account sums a subsidiary ledger, and the two totals must agree. For Meghna and Sons in November 2026, credit sales were Tk 269,000, credit purchases were Tk 300,000, and the accounts receivable control account closed at Tk 549,000.

THINK IT THROUGH

11.T1 A very small business makes five credit sales in a year. Explain why it may not need a sales journal, and say at what point the answer might change.

11.T2 A clerk posts every line of the sales journal to the customer accounts and then posts every line again to the accounts receivable control account. What goes wrong, and how would the agreement test reveal it?

11.T3 A clerk totals a sales journal as Tk 52,400, but the true total is Tk 54,200. The customer accounts have been posted correctly. Which test reveals the error, and what difference appears?

WHERE PEOPLE GO WRONG

Some students believe that a special journal replaces the ledger. It does not. A special journal is a book of first record, like the general journal, and its entries still reach the ledger by posting. Others believe that all transactions of a business go into special journals. They do not: cash transactions, adjusting entries and correcting entries have other books. A third belief is that agreement of the control account with the customer accounts proves that all figures are right. It proves only that the two totals are equal. Last, students sometimes read the word control as a power over customers. In accounting it means a check. A fifth belief is that special journals need new rules for debit and credit. They do not: the rules of Chapter 4 'Accounts, Debits and Credits' apply unchanged, and only the method of recording is different.

IN HISTORY

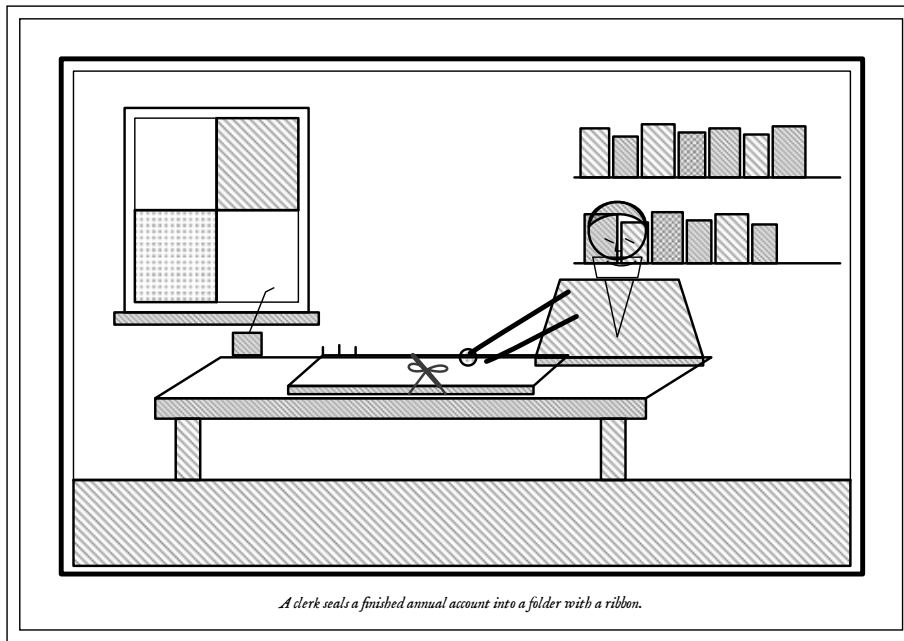
In 1890 the United States counted its people with machines designed by Herman Hollerith, an American engineer and former employee of the Census Bureau. The count of 1880 had taken about seven years to tabulate. The Census Bureau then held a con-

² The Census Bureau, the Smithsonian Institution and The Henry Ford museum each describe the contest and the machine, and their accounts agree on these points.

The summary restates the whole chapter in a few lines.

Five common misunderstandings are stated here, each with its correction.

A machine built for the census of 1890 started the mechanising of record-keeping.



A clerk seals a finished annual account into a folder with a ribbon.

CHAPTER IX

Financial Statements and Closing Entries

What must a business show at the end of the year, and what must it empty before the next year begins?



At the end of a period, the accounts must report their results and then prepare for a fresh start. This chapter teaches the three reports that present the results, the closing entries that reset the accounts, and the reversing entries that make the first payment of the new period easy to record.

The table below shows the income statement of the stall. Read it from the top downwards: revenue first, then expenses, and the result on the last line.

INCOME STATEMENT OF THE TEA STALL FOR ONE MONTH

ITEM	TK
Revenue: tea sold	5,000
Expenses: leaves, milk, sugar and fuel	3,000
Net income	2,000

Every statement has a heading of three lines: the name of the business, the name of the statement, and the period or date. The income statement covers a period, so its third line reads, for example, *For the year ended 31 December 2026*. Without the period, a reader cannot judge whether Tk 2,000 is a good result for one week or a poor result for one year. Because the books follow the accrual basis of accounting, the statement reports revenue when it is earned and matches expenses to that revenue, as the matching principle requires.

The second answer comes from the **statement of owner's equity**. A statement of owner's equity is a report showing how the owner's capital changed during a period. Owner's equity is the owner's claim on the business. Some books call it capital, net worth or proprietorship; this book uses owner's equity for the idea, and Owner's Capital as the name of the account. The stall began with capital of Tk 2,500, and the statement shows how that figure became a new one.

The calculation has three parts. Ending capital equals starting capital, plus net income, minus drawings. Drawings are amounts that the owner withdraws from the business for personal use. For the stall, the steps are $\text{Tk } 2,500 + \text{Tk } 2,000 - \text{Tk } 500 = \text{Tk } 4,000$. A net loss would be subtracted instead of added, and any extra money that the owner invests during the period would be added.

The third answer comes from the **balance sheet**. A balance sheet is a report of assets, liabilities and owner's equity at one date. Assets are things of value that the business owns. Liabilities are amounts that the business owes. The report is called a balance sheet because its two sides must be equal: assets equal liabilities plus owner's equity. Some countries and standards use the title statement of financial position for the same report; this book uses balance sheet.¹

The table below shows the stall's balance sheet. The left part lists what the stall owns, and the lower part lists what it owes and what belongs to the owner. The two totals must agree.

BALANCE SHEET OF THE TEA STALL AT THE END OF THE MONTH

ITEM	TK
Cash	3,700
Equipment	1,300

¹ The International Financial Reporting Standards use the title statement of financial position. The contents are the same as in a balance sheet. Which title is required depends on the law and standards of the country and on the kind of business.

The table lists revenue first, expenses next and the result last.

Every statement has a three-line heading: who, which statement, and when.

The second statement answers: how did the owner's stake change?

Statement of owner's equity. A report showing how the owner's capital changed during a period.

Ending capital equals starting capital plus net income minus drawings.

The third statement answers: what does the business own and owe on one date?

Balance sheet. A report of assets, liabilities and owner's equity at one date.

The balance sheet for the stall shows that both sides equal Tk 5,000.

* * *

11.3 CONTROL ACCOUNTS

The individual accounts of customers need a home of their own, apart from the main ledger.

Subsidiary ledger. A ledger holding the individual accounts of customers or suppliers, supporting a main ledger account.

The main ledger keeps a single summary account for each subsidiary ledger.

Control. Word in use. In daily life, power to direct something. In accounting, a check that a total agrees with the sum of its parts.

Control account. A main ledger account that sums up the balances of all the accounts in a subsidiary ledger.

Pause and check that you can tell the two ledgers apart.

Agreeing totals means comparing the control account with the sum of the individual balances.

The case file gives the opening balance, which both records must match.

The calculation follows the standard order, and the check is the agreement itself.

Agreement does not prove that every figure is correct, only

A business with hundreds of customers cannot keep every customer account among its main accounts, because the main ledger would become huge and hard to read. It keeps them in a separate book. A **subsidiary ledger** is a ledger holding the individual accounts of customers or suppliers, supporting a main ledger account. One subsidiary ledger holds the accounts of customers. Another holds the accounts of suppliers. Meghna and Sons posts each line of the sales journal to the customer's account in the first of these.

A **control account** is a main ledger account that sums up the balances of all the accounts in a subsidiary ledger. The accounts receivable control account sums up every customer account. The accounts payable control account sums up every supplier account. Its name explains its job. The word control here means a check, not power over customers. If the control account and the subsidiary ledger are correct, they must tell the same story. A difference shows that an error exists somewhere.

PAUSE AND CHECK 11.P1 Which holds the balance of one named customer: the accounts receivable control account or the subsidiary ledger? Which holds the total owed by all customers?

Agreeing the totals is a short calculation, and we perform it with the case. The control account receives the total of the sales journal as a debit. The subsidiary ledger receives the same sales as individual debits, one for each customer. Neither side may be changed without the other. A student might ask why a business keeps both records. The answer is that each one checks the other.

Case file, illustrative. At 1 November 2026, the accounts receivable control account of Meghna and Sons had an opening balance of Tk 280,000. During November, credit sales posted from the sales journal total Tk 269,000.

Given. The opening balance is Tk 280,000. We assume that the customer accounts brought forward also total Tk 280,000, as a correct ledger requires. The November credit sales are Tk 85,000, Tk 120,000 and Tk 64,000. *Find.* The closing balance of the control account, and of the sum of customer accounts. *Formula.* Closing balance equals opening balance plus credit sales posted. In symbols, $C = O + S$, where C is the closing balance, O is the opening balance and S is the sales posted. *Steps.* $\text{Tk } 85,000 + \text{Tk } 120,000 + \text{Tk } 64,000 = \text{Tk } 269,000$. Then $\text{Tk } 280,000 + \text{Tk } 269,000 = \text{Tk } 549,000$. *Answer.* Both the control account and the sum of customer accounts show Tk 549,000. *Check.* The two totals are equal, so the posting is consistent.

Exceptions need care. Agreement shows that the control account and the customer accounts hold the same total. It does not

Accountants answer by dividing accounts into two groups. A **temporary account** is an account for revenue, expense or drawings, closed to zero at the end of each period. Revenue and Expenses are examples. A **permanent account** is an account for assets, liabilities or capital, whose balance carries forward to the next period. Cash and Accounts Payable are examples. A temporary account measures one period only. A permanent account shows the position of the business and has no end date.

The table below sorts the common account types. Read each row from left to right: the kind of account, its group, and what happens to its balance at the end of the period.

WHICH ACCOUNTS ARE TEMPORARY AND WHICH ARE PERMANENT

KIND OF ACCOUNT	GROUP	AT THE END OF THE PERIOD
Revenue	Temporary	Closed to zero
Expense	Temporary	Closed to zero
Drawings	Temporary	Closed to zero
Asset	Permanent	Balance carries forward
Liability	Permanent	Balance carries forward
Owner's Capital	Permanent	Updated, then carried forward

The tool that resets a temporary account is the closing entry. A **closing entry** is a journal entry at period end that transfers a temporary account's balance to capital. Suppose the stall's Revenue account has a credit balance of Tk 5,000. A closing entry debits Revenue for Tk 5,000. That debit cancels the credit, and the account stands at zero. The same amount must be credited somewhere, and that place is capital, or an account that leads to capital.

Most books use a helper account for this purpose. The **income summary** is a temporary account used to collect revenue and expenses before the net result goes to capital. Think of it as a small table where the month's revenue and expenses are placed side by side, so that the difference is easy to see. After the net result is moved to capital, the income summary itself stands at zero.

Closing takes four steps, and each step is one journal entry. The steps are performed in this order, after all adjusting entries have been recorded.

Step 1. Close each revenue account: debit it and credit Income Summary. Step 2. Close each expense account: credit it and debit Income Summary. Step 3. Close Income Summary to Owner's Capital: a credit balance means net income, which is credited to capital. Step 4. Close Drawings to Owner's Capital: credit Drawings

Accounts that measure one period are temporary; accounts that show position are permanent.

Temporary account. An account for revenue, expense or drawings, closed to zero at the end of each period.

Permanent account. An account for assets, liabilities or capital, whose balance carries forward to the next period.

The table sorts the account types into temporary and permanent.

The word closing in accounting means moving a balance to capital so that an account starts again at zero.

Closing. Word in use. In daily life, shutting a shop for the night. In accounting, moving the balance of a temporary account to capital so that the account starts the new period at zero.

Closing entry. A journal entry at period end that transfers a temporary account's balance to capital.

A helper account collects revenue and expenses before the net result moves to capital.

Income summary. A temporary account used to collect revenue and expenses before the net result goes to capital.

Closing takes four steps, each one a journal entry, and the order is fixed.

The four steps close revenue, expenses, the income summary and drawings.

11.2 SALES AND PURCHASES JOURNALS

A small clean example shows the method before we meet the real records.

The sales journal records credit sales of goods, one line for each sale invoice.

Sales journal. A special journal that records only sales of goods on credit.

The case file shows three credit sales in November 2026, each with an invoice number.

To read the sales journal, follow each row across and then the amount column down.

Each line goes to the customer's own account; the total goes to the main ledger once.

We begin with the tea stall, because its numbers are small. The calculation follows the order used throughout this book. *Given.* Three credit sales, of Tk 100, Tk 200 and Tk 300. *Find.* The total to post to the sales account. *Formula.* The total equals the first amount plus the second plus the third. In symbols, $T = a_1 + a_2 + a_3$, where T is the total and a_1 , a_2 and a_3 are the three amounts. *Steps.* Tk 100 + Tk 200 = Tk 300. Then Tk 300 + Tk 300 = Tk 600. *Answer.* The total is Tk 600, posted once to sales. *Check.* Posting the three amounts one by one would also give Tk 600.

The **sales journal** is a special journal that records only sales of goods on credit. Its source document is the copy of the sales invoice that the business keeps. Each invoice becomes one line, and the usual columns are the date, the invoice number, the name of the customer and the amount.¹ The sales journal lists the credit sales of the business, nothing more.

Case file, illustrative. In November 2026, Meghna and Sons made three credit sales. On 4 November it sold goods to Dhanmondi Girls' School for Tk 85,000 (invoice 701). On 12 November it sold goods to Rose Valley School for Tk 120,000 (invoice 702). On 20 November it sold goods to Lalmatia College for Tk 64,000 (invoice 703). The total of credit sales for the month is Tk 269,000.

The table below is the sales journal for the case. Read each row from left to right: the date, the invoice number, the customer and the amount in taka. Then read the amount column downwards and total it. The heading does not name the two accounts for each line, because every line has the same debit and credit.

SALES JOURNAL OF MEGHNA AND SONS, NOVEMBER 2026

DATE	INVOICE	CUSTOMER	AMOUNT (TK)
4 November	701	Dhanmondi Girls' School	85,000
12 November	702	Rose Valley School	120,000
20 November	703	Lalmatia College	64,000
Total			269,000

The figure shows what happens next. Each line is posted to the account of the customer named on it, so that the business always knows how much each customer owes. At the end of the month, the total of Tk 269,000 is posted once to two main accounts: a debit to the account receivable and a credit to sales. Solid lines in the figure show the posting of each line. Dashed lines show the posting of the total.

¹ Some businesses add columns, for example for tax charged on sales. The rules differ from country to country, so this chapter leaves them out and shows only the main idea.

expenses directly to capital and skip the income summary.² The final capital is the same.

Permanent accounts are never closed; their balances continue. If the result of the period is a net loss, the income summary has a debit balance after step 2. Step 3 then credits Income Summary and debits Owner's Capital, which reduces capital.

IN SHORT

Temporary accounts measure one period and return to zero through closing entries. Permanent accounts show the position and carry their balances forward.

Before moving on. A temporary account is closed to zero at the end of each period; a permanent account carries its balance forward. The four closing entries move revenue, expenses and drawings into Owner's Capital, using the income summary as a helper for the net result. After closing, only permanent accounts hold balances.

* * *

9.3 REVERSING ENTRIES

Case file, illustrative. At 31 December 2026, Meghna Binding Service owes its workers Tk 14,000 for wages earned in 2026 but not yet paid. The adjusting entry of that date debited Wages Expense and credited Wages Payable for Tk 14,000. On 5 January 2027 the workshop pays wages of Tk 28,000. Half of this payment settles the old debt, and half pays for work done in 2027. How should the payment be recorded?

The difficulty is the split. If the bookkeeper records the full Tk 28,000 as an expense, the Tk 14,000 of 2026 is counted twice: once by the adjusting entry, and again by the payment. To avoid this, the bookkeeper must remember to debit Wages Payable and Wages Expense for Tk 14,000 each. With many such payments, memory is not a safe method.

The remedy is the **reversing entry**. A reversing entry is an entry on the first day of a new period that cancels an earlier accrual so that routine payment can be recorded in the usual way. Here the reversing entry is the exact opposite of the adjusting entry: it debits Wages Payable and credits Wages Expense for Tk 14,000. After it, the payment on 5 January needs no thought. The bookkeeper debits Wages Expense and credits Cash for the full amount.

The table below shows the two entries in 2027. Read the rows by date: the reversal first, then the payment.

² This shorter method gives the same ending capital. It is used less often in teaching, because the income summary shows the net result in one place before it moves to capital.

Permanent accounts are never closed, and a net loss reverses the direction of Step 3.

Before moving on: closing resets four kinds of balance and leaves the permanent accounts alone.

A payment that covers two periods creates extra work for the person who records it.

Without a reversal, the bookkeeper must split the payment by hand, and a slip counts the same wages twice.

A reversing entry cancels the earlier accrual so that payment can be recorded in the usual way.

Reversing entry. An entry on the first day of a new period that cancels an earlier accrual so that routine payment can be recorded in the usual way.

The table shows the reversal and the payment, with their dates.

YOU WILL NEED

You will need the idea of the journal from Chapter 5 'The Journal and the Ledger', section 5.2 'The Journal', and the idea of posting from Chapter 5 'The Journal and the Ledger', section 5.3 'The Ledger and Posting'. Appendix A1 'Maths You Will Use' covers rounding and units, which help when you check totals.

A busy shop faces a puzzle: hundreds of similar sales, each needing the same careful record.

Repeating the same entry many times costs time and invites mistakes.

The answer is to separate the kinds of transaction and to give each kind its own book.

This chapter builds that idea in three steps and then adds a check against error.

The single journal records every kind of transaction, so it repeats the same words again and again.

Recall: the journal, Chapter 5, 'The Journal and the Ledger', section 5.2 'The Journal'.

Journal Word is used in

Picture a stall in a market that sells tea and biscuits. Most customers pay at once. Three regular customers, however, take their goods now and promise to pay at the end of the month. Each of these sales must be recorded. The owner has learned that every transaction begins in the journal, so each credit sale receives its own entry: a date, the account to debit, the account to credit, the amount and a short explanation.

Now picture the stall growing into a large shop. It makes two hundred credit sales in one month. Two hundred journal entries must be written, and in each one the same two account names appear: the account receivable and the sales account. Then every entry must be posted to the ledger, which means two hundred postings to the sales account alone. A tired clerk might skip one entry, write a figure wrongly, or post the debit and forget the credit. Many students find this point puzzling, because the method works well for ten entries. The difficulty appears only at two hundred.

Think about how the shop could save effort. Every credit sale has the same shape. The debit is always the account receivable, and the credit is always sales. Only three things change from one sale to the next: the date, the customer and the amount. Suppose the shop prints those headings across a page and writes one line for each sale. The repeated words disappear. At the end of the month the clerk totals the amount column once and posts a single total to each of the two accounts. Two hundred postings shrink to two.

Section 11.1 names the idea and explains why a single journal reaches its limit. Section 11.2 shows the sales journal and the purchases journal and how their totals reach the ledger. Section 11.3 introduces a main account that sums many customer accounts, and a method to prove that the two agree. Our running example is a bookshop in Dhanmondi, Dhaka, where credit sales and credit purchases are frequent.

* * *

11.1 WHY SPECIAL JOURNALS

In Chapter 5 you learned that the journal is the first book of record. Each transaction, supported by a source document such as an invoice, is written there in date order before it reaches the ledger. A journal of this kind accepts any transaction: a purchase of furniture, a payment of rent, a loan from a bank. For that reason we shall call it the general journal in this chapter. Its strength is

Answer: the net income of the workshop for the year ended 31 December 2026 is Tk 184,000. *Check:* revenue is larger than expenses, so the result must be a net income; and Tk 184,000 is about 23 per cent of Tk 788,000, a reasonable share.

Case file, illustrative. Calculation 2: capital at the end of the year. *Given:* capital at the start Tk 400,000; net income Tk 184,000; drawings Tk 80,000. *Find:* capital at 31 December 2026. *Formula:* ending capital equals starting capital plus net income minus drawings; in symbols, $C_1 = C_0 + N - D$. *Steps:* Tk 400,000 + Tk 184,000 = Tk 584,000; then Tk 584,000 - Tk 80,000 = Tk 504,000. *Answer:* ending capital is Tk 504,000. *Check:* the owner added net income and withdrew less than that, so capital must have grown: Tk 504,000 is larger than Tk 400,000.

Case file, illustrative. Calculation 3: the balance sheet. *Given:* the assets, liabilities and ending capital above. *Find:* total assets, total liabilities, and whether the two sides agree. *Formula:* assets equal liabilities plus owner's equity; in symbols, $A = L + E$. *Steps:* assets are Tk 150,000 + Tk 110,000 + Tk 15,000 + Tk 27,000 + Tk 288,000 = Tk 590,000. Liabilities are Tk 60,000 + Tk 14,000 + Tk 12,000 = Tk 86,000. Then $L + E = \text{Tk } 86,000 + \text{Tk } 504,000 = \text{Tk } 590,000$. *Answer:* the balance sheet has total assets of Tk 590,000, and both sides agree. *Check:* the equity of Tk 504,000 also equals assets less liabilities: Tk 590,000 - Tk 86,000.

The table below sets the balance sheet out in full. Read it from the top: assets, then liabilities, then owner's equity, and last the two equal totals.

BALANCE SHEET OF MEGHNA BINDING SERVICE AT 31 DECEMBER 2026

ITEM	TK
Cash	150,000
Accounts receivable	110,000
Supplies	15,000
Prepaid insurance	27,000
Equipment (net)	288,000
Total assets	590,000
Accounts payable	60,000
Wages payable	14,000
Unearned revenue	12,000
Total liabilities	86,000
Owner's Capital	504,000
Total liabilities and owner's equity	590,000

The books are now closed. The table below shows the four closing entries of 31 December 2026. Revenue and expenses are shown as totals, because the case file gives totals only.

The second calculation finds ending capital for the Statement of owner's equity.

The third calculation totals assets and liabilities and tests the balance sheet.

The table presents the workshop's balance sheet at 31 December 2026.

The four closing entries, dated 31 December 2026, use the same figures.

IN HISTORY

Frank Winfield Woolworth was an American merchant, born in 1852. In 1879 he opened his first store in Utica, a city in New York State in the United States. It sold goods at a fixed price of five cents, and it failed within months. Later the same year he opened a second store in Lancaster, Pennsylvania, which succeeded and added goods at ten cents. A shop of this kind lives on the gap between the cost of its goods and the price at which they sell. A fixed price means that every customer pays the same amount for the same item. Woolworth's chain grew from that second store, and the gap between net sales and cost of goods sold, the gross profit, was the figure on which it depended.

REVIEW QUESTIONS

10.1 A shop has sales of Tk 5,000, sales returns of Tk 300 and sales allowances of Tk 100. Find net sales.

10.2 Prepare a classified income statement from this data (Tk): sales 80,000; sales returns 2,000; opening inventory 12,000; purchases 50,000; purchase returns 1,000; closing inventory 14,000; selling expenses 6,000; administrative expenses 7,000.

10.3 Prepare a classified balance sheet from this data (Tk): cash 3,000; accounts receivable 4,000; inventory 14,000; prepaid expenses 1,000; equipment (net) 20,000; accounts payable 5,000; loan due within one year 2,000; long-term loan 10,000; owner's equity 25,000.

10.4 Name the kind of error in each case: (a) a sale of Tk 900 never recorded; (b) a delivery van of Tk 40,000 debited to Repairs Expense; (c) electricity of Tk 700 debited to Telephone Expense; (d) Wages debited Tk 300 too much and Rent debited Tk 300 too little.

10.5 Write the correcting entries for errors (b) and (c) of question 10.4.

10.6 Classify each as a current asset, a non-current asset, a current liability or a non-current liability: Inventory; shop fittings; accounts payable; a loan due in four years; a loan due in six months.

WHERE PEOPLE GO WRONG

Some students close every account, including Cash. This is wrong. Only revenue, expense and drawings accounts are closed. Assets, liabilities and capital show the position of the business and must carry their balances forward.

Some students charge drawings to the income summary with the expenses. Drawings are the owner's withdrawal, not a cost of earning revenue. They are closed directly to Owner's Capital, and they never reduce net income.

Some students think the books are wrong without a reversal. They are not. A reversal only makes later recording easier. Deferrals recorded in asset or liability accounts usually need no reversal at all.

IN HISTORY

Before 1844, a company in the United Kingdom could usually gain a separate legal existence only by royal charter or by a private Act of Parliament. In 1844 Parliament passed the Joint Stock Companies Act, which received royal assent on 5 September 1844 and came fully into force on 1 November 1844. A company could now register with an official called the Registrar of Joint Stock Companies. The Act also placed duties on registered companies. They had to keep books of account, and their directors had to prepare a "full and fair" balance sheet. Auditors then examined it, and a copy went to the Registrar for the public to inspect. Parliament later replaced the Act with the Joint Stock Companies Act 1856. Today the law of each country sets how a company reports to its owners.

REVIEW QUESTIONS

- 9.1 State the question that each of the three statements answers, and name the third line of each heading.
- 9.2 A small shop has this adjusted trial balance at 31 December 2026 (in Tk): Cash 6,000; Accounts Receivable 2,000; Equipment 9,000; Drawings 1,000; Expenses 7,000 (debits); Accounts Payable 3,000; Owner's Capital 10,000; Revenue 12,000 (credits). Prepare the three statements.
- 9.3 Write the four closing entries for the shop in question 9.2.
- 9.4 Sort these accounts into temporary and permanent: Revenue, Prepaid Insurance, Wages Expense, Drawings, Wages Payable, Owner's Capital.
- 9.5 Wages of Tk 6,000 were earned but unpaid at 31 December 2026, and Tk 8,000 was paid on 4 January 2027. Write the reversing entry and the payment, and state the wages expense of 2027.
- 9.6 A business has revenue of Tk 9,000 and expenses of Tk 11,000 for the year. Write the third closing entry and say what happens to capital.

Mistake one: closing permanent accounts, which would erase the position of the business.

Mistake two: treating drawings as an expense.

Mistake four: believing that reversing entries are always required.

The Joint Stock Companies Act 1844 made a balance sheet a legal duty of registered companies.

CORRECTING ENTRIES FOR THE FOUR KINDS OF ERROR

KIND OF ERROR	DEBIT	CREDIT	TK
Omission: unrecorded credit sale	Accounts Receivable	Sales	800
Commission: electricity in Telephone Expense	Electricity Expense	Telephone Expense	700
Principle: van in Repairs Expense	Vehicles	Repairs Expense	40,000
Compensating: Wages too high, Rent too low	Rent Expense	Wages Expense	300

A bookkeeper finds omission, principle and compensating errors by checking, not by the trial balance.

Before moving on: four kinds of error, and each is fixed by a new entry.

How are such errors found? A disagreeing trial balance warns of a one-sided mistake, and the bookkeeper searches for it. Errors of omission, principle and compensating errors do not disturb the totals, so only careful checks reveal them: comparing records with invoices, reviewing the accounts, and asking whether each entry used the right class. A good bookkeeper never relies on the trial balance alone. Because the books are permanent records, corrections are made by a new entry, dated and explained, so that the history stays visible.

Before moving on. An error of omission leaves out a whole transaction. An error of commission uses the wrong account of the right class, or a wrong amount. An error of principle uses the wrong class. A compensating error is two mistakes that cancel. Each is corrected by a new journal entry.

* * *

A WORKED EXAMPLE: MEGHNA AND SONS FOR 2025

Case file, illustrative. Meghna and Sons, 2025. Net sales Tk 14,400,000. Opening inventory Tk 3,300,000; purchases Tk 10,200,000; purchase returns Tk 200,000; closing inventory Tk 3,500,000. Selling expenses Tk 1,990,000; administrative expenses Tk 1,860,000. Capital at the start Tk 2,860,000; drawings Tk 500,000.

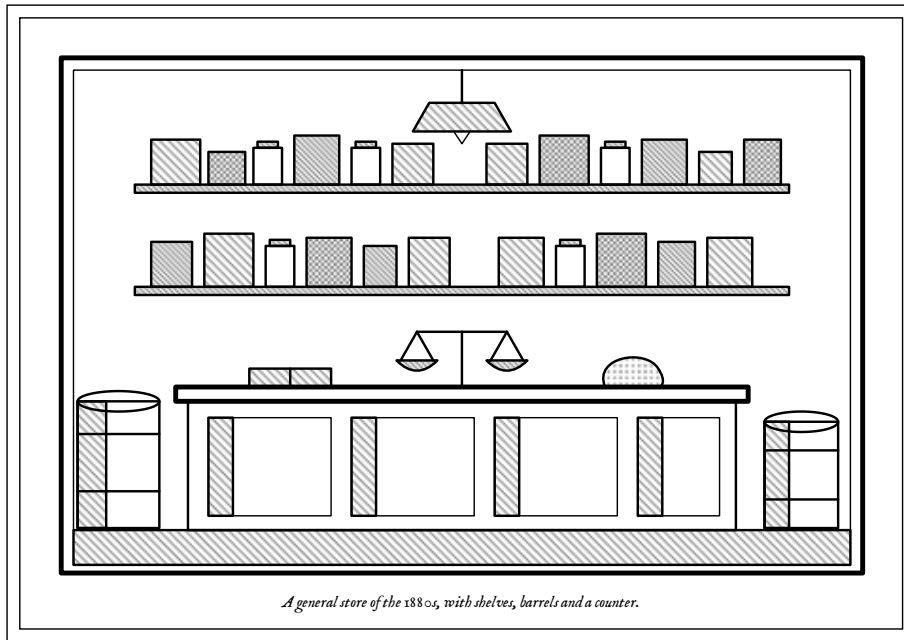
Case file, illustrative. Calculation 1. *Given:* the inventory and purchase figures above. *Find:* cost of goods sold and gross profit. *Formula:* $C = O + P - R - E$, then gross profit = net sales - C. *Steps:* $Tk\ 3,300,000 + Tk\ 10,200,000 = Tk\ 13,500,000$; less $Tk\ 200,000 = Tk\ 13,300,000$; less $Tk\ 3,500,000 = Tk\ 9,800,000$. Then $Tk\ 14,400,000 - Tk\ 9,800,000 = Tk\ 4,600,000$. *Answer:* cost of goods sold is Tk 9,800,000 and gross profit is Tk 4,600,000. *Check:* gross profit is about 32 per cent of net sales, a reasonable share for a bookshop.

Case file, illustrative. Calculation 2. *Given:* selling expenses Tk 1,990,000; administrative expenses Tk 1,860,000; gross profit Tk 4,600,000. *Find:* operating expenses and profit. *Formula:* profit = gross profit - operating expenses. *Steps:* $Tk\ 1,990,000 + Tk\ 1,860,000 = Tk\ 3,850,000$; then $Tk\ 4,600,000 - Tk\ 3,850,000 = Tk\ 750,000$.

The data for the bookshop's 2025 statements is given, first.

The first calculation finds the cost of goods sold and gross profit.

The second calculation finds operating expenses and profit.



CHAPTER X

Accounts of a Merchandising Company

If a shop buys goods all year, why is its cost of goods not the amount it spent?



A business that buys goods and sells them again needs accounts that a service business does not. This chapter teaches how sales, purchases and returns are recorded, how the income statement and balance sheet are arranged in classes, and how mistakes in the books are classified and corrected.

expenses. The classes tell the reader where the business gains and where it spends.

* * *

10.3 CLASSIFIED BALANCE SHEET

Suppose you run a shop and wish to know what money you can count on in the next twelve months, and what you must pay in the same time. A single list of assets and liabilities will not tell you, because a shop counter and a coin in the till are both assets, yet only one of them can pay a bill next month. The classified balance sheet solves this by separating items that are near to cash, or near to payment, from those that are not.

A **current asset** is an asset expected to be turned into cash or used up within one year. Cash is the plainest example. Accounts receivable, the amounts that customers owe, will be collected soon. Inventory will be sold. Prepaid expenses, such as insurance paid in advance, will be used up. Current assets are listed in the order in which they come closest to cash.

A **current liability** is a debt that must be paid within one year. Accounts payable, the amounts the business owes to suppliers, are the usual example. A long loan can also contribute a current part: the instalments due in the next twelve months are shown as current, and the rest is shown separately.

Items that last or fall due after more than one year are non-current. A non-current asset, such as shop fittings, serves the business for several years. A non-current liability, such as a long-term loan, is paid after more than a year. Owner's equity is shown in a group of its own. Fittings are shown net, meaning after the fall in value that Chapter 16, 'Depreciation', teaches.

Case file, illustrative. The table below shows the balance sheet of Meghna and Sons at 31 December 2025. Read it in three parts: assets first, then liabilities, then owner's equity. The two totals at the foot must agree.

CLASSIFIED BALANCE SHEET OF MEGHNA AND SONS AT 31 DECEMBER 2025 (TK)

ITEM	TK
Cash	420,000
Accounts receivable	280,000
Inventory	3,500,000
Prepaid expenses	60,000
Total current assets	4,260,000
Fittings (net)	600,000
Total assets	4,860,000
Accounts payable	1,150,000
Loan due within one year	120,000
Total current liabilities	1,270,000
Long-term loan	480,000
Owner's capital (2 860 000 + 750 000 - 500 000)	3 110 000

The word current in accounting means within one year.

Current. Word in use. In daily life, happening now. In accounting, expected to be used, collected or paid within one year.

Current assets are turned into cash or used up within a year.

Current asset. An asset expected to be turned into cash or used up within one year.

Current liabilities are debts that must be paid within a year.

Current liability. A debt that must be paid within one year.

Non-current items last longer than a year, and owner's equity forms its own group.

Meghna and Sons' balance sheet at 31 December 2025 follows the classes.

Given: sales Tk 10,000; sales returns Tk 500; sales allowances Tk 200. *Find:* net sales. *Formula:* net sales equal sales minus sales returns minus sales allowances; in symbols, $NS = S - R - A$. *Steps:* Tk 10,000 - Tk 500 = Tk 9,500; then Tk 9,500 - Tk 200 = Tk 9,300. *Answer:* net sales are Tk 9,300. *Check:* net sales must be smaller than sales, and Tk 9,300 is.

On the buying side the same pattern appears. The Purchases account records goods bought for resale. It does not record shop fittings or supplies, because those are not sold. A purchase return is goods that the business sends back to a supplier, recorded in its own account. For the stall, goods bought during the month cost Tk 6,000, and notebooks worth Tk 300 were sent back to the supplier. Purchases less purchase returns give net purchases of Tk 5,700.

Now we can solve the opening puzzle. The **cost of goods sold** is the cost of the goods that were sold during the period. The business does not record the cost of each sale as it happens. It counts the goods that remain at the end of the period, and from that count it calculates the cost of the goods sold.¹ Goods that remain are called inventory, and they are an asset, not an expense, until they are sold.

The formula has four parts. Opening inventory is the inventory held at the start of the period. Add purchases and subtract purchase returns. The result is the goods available for sale. Then subtract closing inventory, the inventory still held at the end. In symbols, $C = O + P - R - E$, where C is cost of goods sold, O is opening inventory, P is purchases, R is purchase returns and E is closing inventory.

Given: opening inventory Tk 2,000; purchases Tk 6,000; purchase returns Tk 300; closing inventory Tk 1,700. *Find:* cost of goods sold. *Formula:* $C = O + P - R - E$. *Steps:* Tk 2,000 + Tk 6,000 = Tk 8,000; Tk 8,000 - Tk 300 = Tk 7,700; Tk 7,700 - Tk 1,700 = Tk 6,000. *Answer:* the cost of goods sold is Tk 6,000. *Check:* goods available for sale were Tk 7,700, and Tk 1,700 of them remain, so Tk 6,000 must have been sold.

PAUSE AND CHECK 10.P1 A shop starts the month with inventory of Tk 500, buys goods for Tk 2,000, and counts Tk 700 of inventory at the end. What is the cost of goods sold?

Case file, illustrative. In 2025, Meghna and Sons began with inventory of Tk 3,300,000 and bought books for Tk 10,200,000. It sent books worth Tk 200,000 back to suppliers and counted inventory of Tk 3,500,000 at the end. So Tk 3,300,000 + Tk 10,200,000 - Tk 200,000 - Tk 3,500,000 = Tk 9,800,000. The bookshop bought Tk 10,200,000 of books, yet the cost of the books it sold was Tk 9,800,000.

Each person with an interest sees something different here. The owner wants to know how much revenue the business keeps. A manager watches the returns, because a high figure suggests faulty goods or poor service. A customer sees only the refund but the

¹ Some businesses update their inventory records with every sale and so do not need to calculate the cost of goods sold at the end. This chapter uses the count at the end of the period.

The first calculation finds net sales for the stall.

Purchases are goods bought for resale, and purchase returns are goods sent back to the supplier.

Cost of goods sold takes the goods available for sale and removes the goods still held.

Cost of goods sold. The cost of the goods that were sold during the period.

The formula starts with opening inventory, adds net purchases and removes closing inventory.

The second calculation finds the cost of goods sold for the stall.

The bookshop's 2025 figures resolve the opening puzzle.

Owners, managers and customers each care about returns for a different reason.

Before moving on: net sales and cost of goods sold both remove what the business did not really sell.

A classified income statement groups items so that the reader can see the shop's two kinds of result.

Gross profit is the first result: net sales less the cost of the goods sold.

Gross profit. Net sales less cost of goods sold, before operating expenses are deducted.

Before moving on. Net sales are sales less returns and allowances. The cost of goods sold equals opening inventory plus purchases less purchase returns less closing inventory. Unsold goods remain an asset until they are sold.

* * *

10.2 CLASSIFIED INCOME STATEMENT

Return to the stationery seller. Her net sales are Tk 9,300 and her cost of goods sold is Tk 6,000. She also paid for advertising, delivery, rent and wages. If all these costs were placed in one long list, she could not tell whether her trading with customers or her running costs were the problem. A classified income statement answers by grouping items into classes. It first shows the result of buying and selling goods, and then the cost of running the business.

The first result is the **gross profit**. Gross profit is net sales less cost of goods sold, before operating expenses are deducted. For the stationery seller, Tk 9,300 - Tk 6,000 = Tk 3,300. The figure below shows gross profit as the gap between two bars. The left bar is net sales, and the right bar is the cost of goods sold. The height of each bar shows the amount, and the gap at the top of the right bar is the gross profit.

Height of a bar shows the amount.

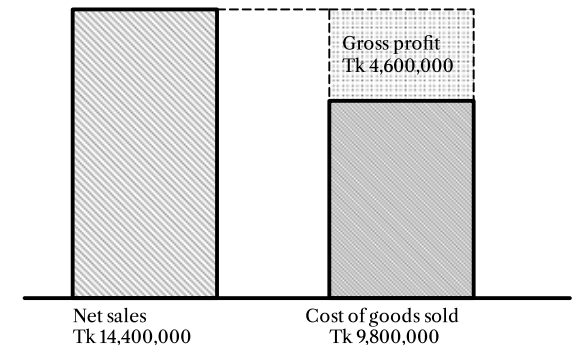


FIG. 10.1. Gross profit as the gap between net sales and cost of goods sold

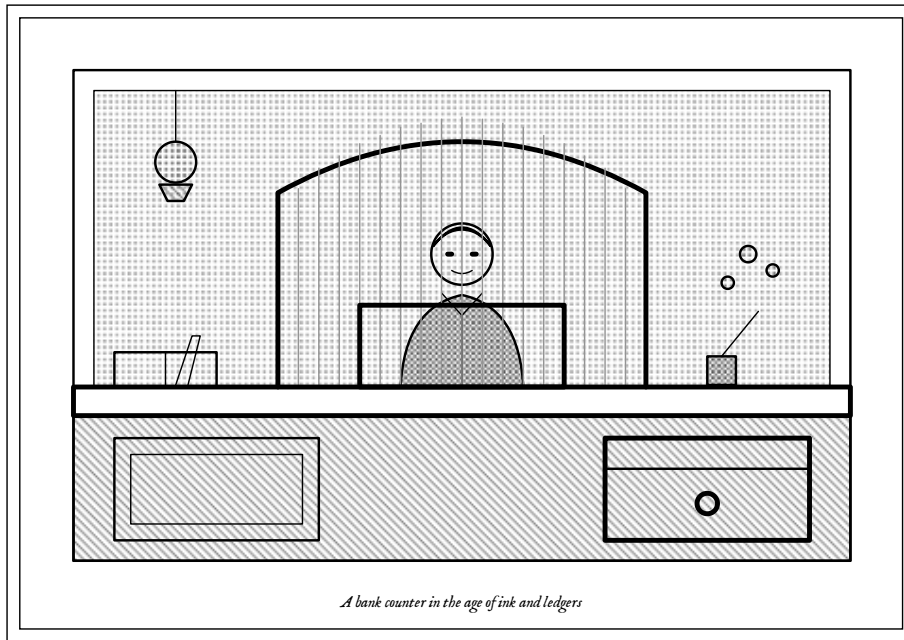
The second class contains the operating expenses. An **operating expense** is a cost of running the business, such as wages or rent, apart from the cost of goods sold. Most statements divide operating expenses into two groups. Selling expenses are costs of finding customers and delivering goods, such as advertising. Administrative expenses are costs of managing the business as a whole, such as office rent and salaries. The groups show where the money goes.

Given: gross profit Tk 3,300; selling expenses Tk 900; administrative expenses Tk 1,200. *Find:* profit. *Formula:* profit equals gross

² In this chapter the case file calls the final result profit. It is the same figure that Chapter 9 calls net income.

REVIEW QUESTIONS

- 14.1 A shop has an opening inventory of 10 units at Tk 10 and buys 10 units at Tk 12. It sells 10 units for Tk 20 each, net sales Tk 200. Find ending inventory, cost of goods sold and gross profit under FIFO, LIFO and weighted average cost.
- 14.2 A shop has an opening inventory of 40 units at Tk 45. It buys 60 units at Tk 48 and then 100 units at Tk 52. It sells 130 units at Tk 80 each, net sales Tk 10,400. Find ending inventory, cost of goods sold and gross profit under all three methods, and check each result.
- 14.3 Prices fall. A shop has an opening inventory of 25 boxes at Tk 80 and buys 25 boxes at Tk 70. It sells 30 boxes at Tk 100 each, net sales Tk 3,000. Find ending inventory, cost of goods sold and gross profit under all three methods, and state which method gives the highest profit.
- 14.4 Use your answers to question 14.2. Express each gross profit as a percentage of net sales, to one decimal place. Then state which method gives the highest ending inventory and explain why.



CHAPTER XII

The Cash Book and Bank Reconciliation

Two honest records of one account, and two different totals. Which one is right?



This chapter shows how a business records its cash and bank money in a cash book, and why its bank balance rarely matches the figure the bank reports. You learn the common reasons for a difference and how to prepare a bank reconciliation statement that explains every taka of it. You also learn which differences need a journal entry afterwards.

A cash book page looks like a ledger account with two sides. The left side is the debit side and holds money received. The right side is the credit side and holds money paid. Each side is split into columns. A cash book with a Cash column and a Bank column on each side is called a two-column cash book. The Cash column follows the notes and coins in the drawer. The Bank column follows the money in the bank account. The table below shows four small entries and where each one is placed. Read each row from left to right: the event, the side, the column, and the reason.

FOUR ENTRIES IN A TWO-COLUMN CASH BOOK

EVENT	SIDE	COLUMN	REASON
---	---	---	---
Customer pays Tk 500 in notes	Debit	Cash	Money received into the drawer
Customer's cheque for Tk 2,000 deposited	Debit	Bank	Money received into the bank account
Rent of Tk 1,000 paid by cheque	Credit	Bank	Money paid out of the bank account
Tk 300 of notes paid for postage	Credit	Cash	Money paid out of the drawer

Sometimes the owner moves money between the drawer and the bank. Suppose the owner takes Tk 3,000 in notes from the drawer and deposits it. The Bank column receives money, so it is debited. The Cash column loses money, so it is credited. Both halves are in the same book. We call this a contra entry. The word *contra* is Latin for "against": the two halves face each other across the page. A contra entry has an important feature. It does not touch any other ledger account, because both sides are already in the cash book.

At the end of each period you total each column. On the debit side the total shows all money received. On the credit side the total shows all money paid. The difference is the balance. The Cash column balance should equal the notes and coins in the drawer. The Bank column balance is what the books say is in the bank. The bank reports its own figure, and the rest of this chapter compares the two. For the owner, this balance answers a daily question: how much can we spend today?

Before moving on. The cash book is a journal and a ledger in one. Its two sides hold receipts and payments, and its columns separate cash from bank. A contra entry moves money between the two columns and touches no other account. You should now be able to say where each of four events is written in a cash book.

12.2 WHY BANK AND BOOK DIFFER

The bank account has two keepers. The business keeps the Bank column of its cash book. The bank keeps its own record of the same account and sends the business a statement from time to time.

The page is divided into receipts on the left and payments on the right, with a column for each kind of money.

One kind of entry moves money between the two columns and needs no other ledger account.

Totalling a column gives its balance, and the Bank column balance is the figure we shall later compare with the bank.

Before moving on, we recall the three ideas of this section.

The bank keeps a second record of the same account, so two records must be compared.

Statement. Word in use. In daily life, something said or written, such as a statement

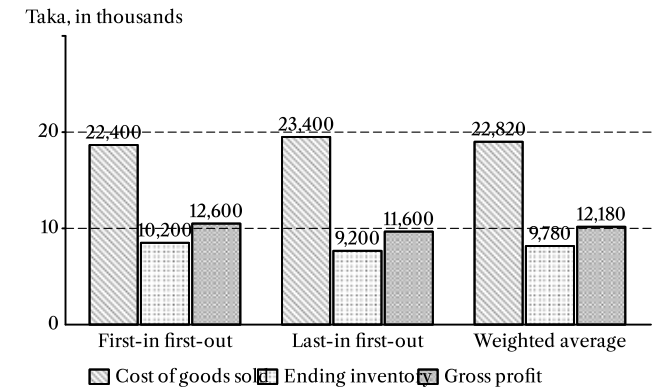


FIG. 14.2. Results of the three methods side by side

Recap: weighted average cost uses one unit cost, and its results lie between the other two methods.

Before moving on. Weighted average cost divides the cost available by the units available and gives every unit that cost. For the dictionary sets the unit cost is Tk 326. You should now be able to say why its profit lies between the profits of FIFO and LIFO.

* * *

WORKED CASE: PRICES THAT FALL

Situation. A different shop sells identical goods. It held 20 units at Tk 340, bought 30 units at Tk 320, and later bought 50 units at Tk 300. It sold 70 units at Tk 500, net sales Tk 35,000. These figures are for practice, and are not part of the case file. The prices fall from Tk 340 to Tk 300. **Question.** What are ending inventory, cost of goods sold and gross profit under each method?

Reasoning. The cost available is $20 \times 340 + 30 \times 320 + 50 \times 300 = 6,800 + 9,600 + 15,000 = \text{Tk } 31,400$, for 100 units. Under FIFO, the 30 units left are the newest, all at Tk 300, so $E = \text{Tk } 9,000$, $C = \text{Tk } 22,400$ and $GP = \text{Tk } 12,600$. Under LIFO, the 30 units left are the oldest: 20 at Tk 340 and 10 at Tk 320, so $E = 6,800 + 3,200 = \text{Tk } 10,000$, $C = \text{Tk } 21,400$ and $GP = \text{Tk } 13,600$. Under weighted average, $u = 31,400 \div 100 = \text{Tk } 314$, so $E = 30 \times 314 = \text{Tk } 9,420$, $C = 70 \times 314 = \text{Tk } 21,980$ and $GP = \text{Tk } 13,020$.

Conclusion. When prices fell, LIFO gave the highest profit, Tk 13,600, and FIFO the lowest, Tk 12,600. When prices rose, as in the case file, the order was the reverse. Weighted average lay between the two in both situations. **Check.** In each method, cost of goods sold plus ending inventory is Tk 31,400: $22,400 + 9,000$, then $21,400 + 10,000$, then $21,980 + 9,420$.

Common slips deserve a warning. The first is to take the sold units from the wrong end of the pile, which swaps the results of FIFO and LIFO. The second is to average the three unit costs, Tk 300, Tk 320 and Tk 340, instead of dividing the total cost by the total units. That error gives Tk 320, not Tk 326, because it ignores

This worked case repeats the method with prices that fall, to show that the ranking reverses.

The reasoning first finds the cost available, and then applies each rule in turn.

The conclusion is that the ranking reverses when prices fall, while the cost available never changes.

Three slips occur often in this work, and each one can be prevented.

from country to country, so a business must check the law that applies where it trades.

One more situation fits group two. A customer may pay the bank account directly by transfer, with no cheque and no visit. The bank records the receipt at once. The business learns of it only from the statement. The same can happen for a payment out, such as a standing order to pay a bill. We treat a direct receipt like a credit memo and a direct payment like a debit memo.

The table below gathers the items. Read each row as a question: who records the event first, and what must the business do? The two items at the top need no entry, because the bank will catch up. The four at the bottom need an entry in the cash book.

WHY THE BANK STATEMENT AND THE CASH BOOK DIFFER

ITEM	RECORDED FIRST BY	IS AN ENTRY NEEDED IN OUR BOOKS?
---	---	---
Deposit in transit	Business	No; the bank catches up
Outstanding cheque	Business	No; the bank catches up
Credit memo, or direct receipt	Bank	Yes; we add to the cash book
Debit memo, or direct payment	Bank	Yes; we subtract from the cash book
Dishonoured cheque	Bank	Yes; we subtract and restore the account receivable

Case file, illustrative. On 31 October 2026 the items were as follows. Cash book bank balance: Tk 612,000. Bank statement balance: Tk 655,000. Deposit in transit: Tk 48,000. Outstanding cheques: no. 4417 for Tk 36,000 and no. 4421 for Tk 22,000. Bank service charge: Tk 1,500. Dishonoured customer cheque: Tk 8,000. Interest credited by the bank: Tk 2,500. School payment made directly to the bank and not yet in the books: Tk 40,000.

Before moving on. The bank statement and the cash book differ because one record learns of an event before the other. A deposit in transit and an outstanding cheque are known to the business first, and need no entry. A credit memo, a debit memo and a dishonoured cheque are known to the bank first, and need an entry. You should now be able to say, for any item, who knows first and what follows.

12.3 PREPARING THE RECONCILIATION

You now know the six causes of difference. The next question is how to arrange them so that nothing is forgotten. The answer is a working paper called the **bank reconciliation statement**.

Payments made straight into or out of the bank account are also learned from the statement.

A table sorts the items by who records first and whether an entry is needed.

The case file lists seven items at 31 October 2026, and each fits one row of the table.

Before moving on, we restate the principle of this section.

The reconciliation turns two different totals into two matching totals, with every step shown.

Bank reconciliation statement. A statement that explains the difference between the bank balance in the cash book and the balance on the bank's statement.

should now be able to say why LIFO gives a lower profit than FIFO when prices rise.

* * *

14.4 WEIGHTED AVERAGE COST

Think of a tea seller who pours tea bought at different prices into one large jar. After the pouring, no one can say which tea came from which purchase. Every spoonful of the mixture has the same average cost. The seller prices and counts the tea by that single figure. The third method treats the goods in the same way.

The **weighted average cost** method is a method that gives every unit the average cost of all units available. The word weighted means that each layer counts in proportion to its number of units. A layer of 50 units has more effect on the average than a layer of 20 units. The method needs no choice of oldest or newest layer, because every unit receives the same cost.

We apply the method to the case. *Given.* Cost available Tk 32,600 for 100 units, and a sale of 70 units for Tk 35,000. *Find.* The unit cost, the ending inventory, the cost of goods sold and the gross profit. *Formula.* Unit cost equals cost available divided by units available. In symbols, $u = A \div N$, where u is the unit cost, A is the cost available and N is the number of units available. Then $E = n \times u$ and $C = x \times u$, where n is the units left, x is the units sold, and $GP = S - C$. *Steps.* $u = \text{Tk } 32,600 \div 100 = \text{Tk } 326$ a unit. $E = 30 \times \text{Tk } 326 = \text{Tk } 9,780$. $C = 70 \times \text{Tk } 326 = \text{Tk } 22,820$. $GP = \text{Tk } 35,000 - \text{Tk } 22,820 = \text{Tk } 12,180$. *Answer.* Under weighted average cost, ending inventory is Tk 9,780, cost of goods sold is Tk 22,820 and gross profit is Tk 12,180. *Check.* $\text{Tk } 22,820 + \text{Tk } 9,780 = \text{Tk } 32,600$, the cost available. The unit cost Tk 326 lies between Tk 300 and Tk 340, as an average must.

PAUSE AND CHECK 14.P2 Using the unit cost found above, what is the cost of goods sold by weighted average if only 25 sets are sold from the 100 available?

The first figure shows the pile of Meghna and Sons as a column of three layers. The height of each block is in proportion to its number of units. Read a column from the bottom, where the oldest layer lies, to the top, where the newest layer lies. Hatched blocks show costs treated as sold. Blank blocks show costs that remain in ending inventory. In the third column all units are mixed, so the pile is one block at the average cost.

GO SLOWLY HERE.

Mixing tea of different prices in one jar shows the idea: the jar has one average price.

Weighted average cost gives every unit the average cost of all units available.

Weighted average cost. A method that gives every unit the average cost of all units available.

The unit cost is the total cost available divided by the total units available.

Pause and check that you can find the average cost of the units available.

The figure shows how each method takes cost from the pile of layers.

Given. The cash book shows a bank balance of Tk 5,000. The statement shows Tk 5,700. A deposit of Tk 400 is in transit. One cheque for Tk 900 is outstanding. The bank added a credit memo for Tk 300 and a debit memo for Tk 100.

Find. The adjusted balance on each side, and whether they agree.

Formula. Adjusted bank balance = S + D – C. Adjusted book balance = B + M – N – H, with no dishonoured cheque here, so H = 0.

FIRST EXAMPLE: SMALL NUMBERS

BANK STATEMENT SIDE	TK	CASH BOOK SIDE	TK
---	---	---	---
Statement balance	5,700	Cash book balance	5,000
Add deposit in transit	400	Add credit memo	300
Less outstanding cheque	900	Less debit memo	100
Adjusted bank balance	5,200	Adjusted book balance	5,200

Steps. Left: $5,700 + 400 = 6,100$, and $6,100 - 900 = 5,200$. Right: $5,000 + 300 = 5,300$, and $5,300 - 100 = 5,200$.

Answer. Both adjusted balances are Tk 5,200, so the two records agree and the true balance is Tk 5,200.

Check. The original difference was $5,700 - 5,000 = 700$. The bank side fell by 500 ($400 - 900$) and the book side rose by 200 ($300 - 100$), and $500 + 200 = 700$. The puzzle is solved: neither record was wrong.

PAUSE AND CHECK 12.P1 Suppose the adjusted bank balance is Tk 645,000 and the adjusted book balance is Tk 644,000. What should you do before you make any entry?

After the reconciliation, one question remains: which items need a journal entry? The reconciliation statement is not a book of account. It is a working paper, so writing it changes no ledger account. The deposit in transit and the outstanding cheque need no entry, because the bank will record them without our help. The credit memos, debit memos and dishonoured cheques need entries, because our books are missing them. Each entry is posted to the cash book on one side and to the matching ledger account on the other. After posting, the cash book balance equals the adjusted balance.

Before moving on. A bank reconciliation statement has two halves, one from the bank statement and one from the cash book. Timing items adjust the bank side. Items from the bank adjust the book side. Only the book side leads to journal entries. You should now be able to say why a deposit in transit needs no entry and a debit memo does.

WORKED EXAMPLE: THE OCTOBER RECONCILIATION

We now reconcile the seven items of the case file, with larger numbers and more varied items.

The data of the small example.

The question to answer.

The two rules, in symbols.

Every line of arithmetic.

The result in a sentence, and a test of it.

A test that the result is sensible.

The reconciliation Statement is a working paper, so only the book-side items lead to entries.

Recall: ledger and posting, Chapter 5 'The Journal and the Ledger', section 5.3 'The Ledger and Posting'.

Before moving on, we restate the method.

The case file for 31 October 2026 is reconciled in full, with every line shown.

of Tk 6,000. It then bought 30 sets at Tk 320 a set, a cost of Tk 9,600, and later bought 50 sets at Tk 340 a set, a cost of Tk 17,000. During the period it sold 70 sets at Tk 500 a set, net sales of Tk 35,000.

The table below shows the layers in the order in which they arrived. Read each row from left to right: the layer, the number of units, the unit cost and the total cost of that layer. The last row adds the units and the costs.

LAYERS OF BOXED DICTIONARY SETS AVAILABLE FOR SALE

LAYER	UNITS	UNIT COST (TK)	TOTAL COST (TK)
Opening inventory	20	300	6,000
First purchase	30	320	9,600
Second purchase	50	340	17,000
Available for sale	100		32,600

Given. The three layers in the table. *Find.* The units and the cost available for sale, and the units left after the sale. *Formula.* Cost available equals the sum of the layer costs. In symbols, $A = L_1 + L_2 + L_3$, where A is the cost available and L_1 , L_2 and L_3 are the layer costs. *Steps.* $Tk\ 6,000 + Tk\ 9,600 = Tk\ 15,600$. Then $Tk\ 15,600 + Tk\ 17,000 = Tk\ 32,600$. Units: $20 + 30 + 50 = 100$, and $100 - 70 = 30$. *Answer.* Cost available is Tk 32,600, and 30 sets remain. *Check.* Each layer cost equals its units times its unit cost: 20×300 , 30×320 and 50×340 .

Before moving on. Goods bought at different prices form layers. A cost flow assumption decides which layers are treated as sold, and ending inventory is the cost that remains. Cost of goods sold plus ending inventory always equals the cost available. You should now be able to say why two businesses with identical goods can report different profits.

* * *

14.2 FIRST-IN FIRST-OUT

Think of a queue at a canteen. The person who arrived first is served first, and the person who arrived last waits longest. Many shops follow the same order with their goods. They sell the oldest goods first, because old goods may spoil or fall out of fashion. A cellar of casks works in this way too: the cooper marks the oldest cask, and that cask is opened first.

The **first-in first-out** method, often written FIFO, is a method that treats the oldest costs as sold first, leaving the newest costs in inventory. In our pile, the opening layer is the oldest, so its cost leaves first. Then the cost of the first purchase leaves, and so on, until the number of units sold is reached. What remains is made of the newest layers.

We apply the method to the case. *Given.* The layers in the table of section 14.1, and a sale of 70 units for Tk 35,000. *Find.* The ending inventory, the cost of goods sold and the gross profit.

The cost of goods available is Tk 32,600, and 30 sets remain after 70 are sold.

Recap: a cost flow assumption divides the cost available between cost of goods sold and ending inventory.

GO SLOWLY HERE.

A queue at a canteen shows the idea: the first person to arrive is the first to be served.

First-in first-out treats the oldest costs as sold first.

First-in first-out. A method that treats the oldest costs as sold first, leaving the newest costs in inventory.

We take 70 units from the oldest layers, and 30 units at Tk 340 remain.

SUMMARY

The cash book is a journal and a ledger in one, with a Cash column and a Bank column. The bank statement and the cash book differ because one learns of an event before the other. Deposits in transit and outstanding cheques are timing differences and need no entry. Credit memos, debit memos and dishonoured cheques come from the bank and need entries. A bank reconciliation statement brings both records to one adjusted balance.

THINK IT THROUGH

12.T1 In the case file of 31 October 2026, which items need a journal entry and which do not? Give one reason for each group.

12.T2 Cheque no. 4417 for Tk 36,000 is still outstanding three months later. What should the business do, and what does the reconciliation show each month until then?

WHERE PEOPLE GO WRONG

Some students think the statement is always right. It is not. The statement lacks the business's deposits in transit and outstanding cheques, so it is also incomplete. Neither record is correct until both are adjusted. The adjusted balance is correct, and neither raw total is.

Only the items that came from the bank need entries. A deposit in transit and an outstanding cheque are already in the cash book. Entering them again would record them twice.

The names follow the bank's books. A credit memo means the bank owes you more, so your balance rises. A debit memo means the bank owes you less, so your balance falls.

A dishonoured cheque is a loss of money that was recorded as received. It is subtracted from the cash book, and the customer's account receivable is restored.

Agreement is strong evidence, not proof. Two errors of equal size in opposite directions could cancel. The agreement shows that the known items explain the difference.

IN HISTORY

In the City of London, the old centre of banking in the English capital, banks once sent clerks from door to door to exchange the cheques they held against each other. From the 1770s the clerks began to meet each day in one place. Each clerk brought the cheques drawn on the other banks, and the clerks swapped them and settled only the differences, in cash. This daily meeting is the origin of cheque clearing. In the early nineteenth century a formal Bankers' Clearing House grew from it.

This history matters for the reconciliation. A cheque must travel through the clearing process before the bank pays it, and the journey takes days. In those days the business has recorded the payment and the bank has not. That day is the outstanding cheque.

The chapter's ideas in a few lines.

Belief: the bank statement is the correct record, so the cash book must be forced to match it.

Belief: every reconciling item needs a journal entry.

Belief: a credit memo reduces the balance, because credit sounds like a reduction.

Belief: a dishonoured cheque is a gain, because it is a cheque.

Belief: an agreeing reconciliation proves there are no mistakes.

Cheque clearing began when bank clerks in London began meeting to exchange cheques.

The history explains why outstanding cheques exist at all.

YOU WILL NEED

You will need the difference between a perpetual record and a periodic record from Chapter 13 'Inventory: Meaning and Measurement', section 13.3 'Recording Inventory', and the idea of cost of goods sold from Chapter 10 'Accounts of a Merchandising Company', section 10.1 'Sales, Purchases and Returns'. Appendix A1 'Maths You Will Use' covers averages and rounding, which section 14.4 uses.

A shopkeeper holds two identical sacks of rice bought at different prices, and a customer buys one.

The profit on the sale is Tk 50 or Tk 30, depending on which cost we say the customer took away.

The choice looks small for one sack, but it grows large when thousands of units are sold.

This chapter teaches the three rules one at a time, using one set of figures throughout.

Picture a shopkeeper in a market. In the first week she buys one sack of rice for Tk 100. In the second week the price rises, and she buys a second sack for Tk 120. The two sacks stand side by side. They look the same, they weigh the same, and no label shows the week of purchase. A customer arrives and buys one sack for Tk 150.

Now the shopkeeper wants to know her profit. If the customer took the cheaper sack, the profit is Tk 150 less Tk 100, which is Tk 50. If the customer took the dearer sack, the profit is Tk 150 less Tk 120, which is Tk 30. Both answers are reasonable, and the sacks cannot tell us which is true. Many students find this puzzling at first, because they expect a fact to settle the matter. Here no physical fact can settle it. The business must choose a rule.

For one sack the difference is Tk 20. For a bookshop that buys thousands of books at changing prices, the difference can reach hundreds of thousands of taka. The owner, the tax collector and the bank that lends money to the business all read the profit figure. A rule that changes the profit changes what each of them sees. For that reason accountants name the possible rules, apply them carefully, and use one rule consistently.

Section 14.1 explains the problem and builds the vocabulary. Section 14.2 teaches the first rule, section 14.3 the second, and section 14.4 the third, with a comparison. A worked case then shows what happens when prices fall instead of rise. Go slowly. Each section finishes one idea before the next begins, and the same figures return every time so that only the rule changes.

* * *

14.1 THE PROBLEM OF CHANGING PRICES

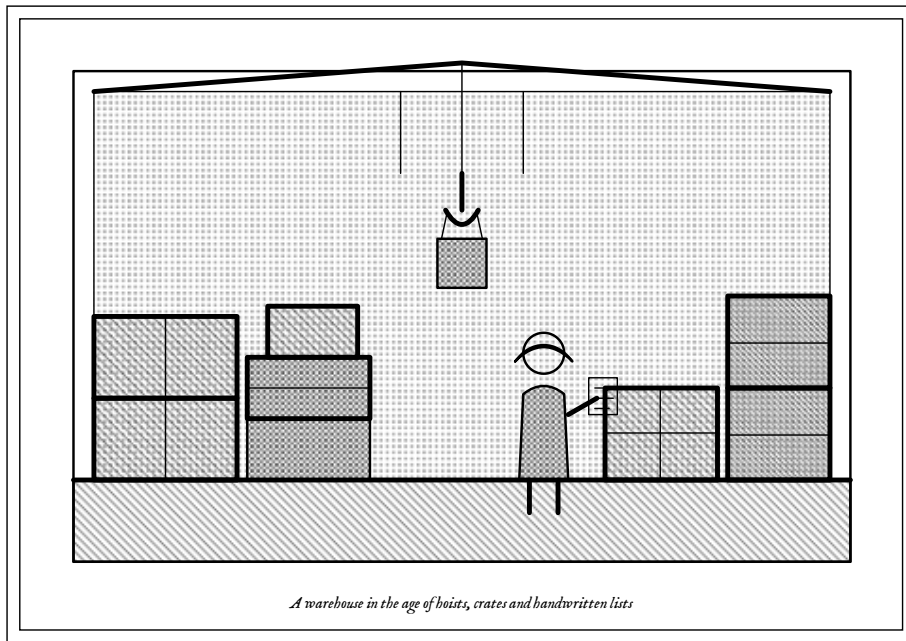
GO SLOWLY HERE.

Prices of goods change, so units that look alike can carry different costs.

Recall: perpetual and periodic records, Chapter 13, 'Inventory: Meaning and Measurement', section 13.3

A business rarely buys all its goods at one price. A supplier raises or lowers a price, a season ends, or a change in the exchange rate makes imported goods dearer. The business therefore holds goods that look the same but cost different amounts. Chapter 13 explained that the business records the goods it holds as inventory, and that it may count them at the end of the period or record every movement. In this chapter we count at the end of the

¹ Where each unit can be told apart, for example a car with its own serial number, a business may use the actual cost of that unit. This chapter treats only goods that cannot be told apart. The same rules also apply to raw materials and work in process, which face the same problem.



CHAPTER XIII

Inventory: Meaning and Measurement

A shelf that is empty and a store room that is full: which mistake costs more?



This chapter explains the goods a business holds for sale and the materials it holds to make them. You learn the three kinds of inventory in a factory, why a business holds inventory at all, and what it costs to hold too little or too much. You then learn the two systems for recording inventory, how a physical count works, and how to reduce the value of goods that will sell for less than they cost.

Many students find the word "stock" confusing, because shopkeepers use it for goods and investors use it for shares in a company. Both uses are correct in their own settings. To avoid mixing them, this book uses inventory for goods and stock only for company shares. One phrase keeps the older word for tradition, and we meet it in section 13.3.

Inventory has a life in two steps. While the goods wait on the shelf, their cost is an asset on the balance sheet. When a customer buys them, the cost leaves the balance sheet and becomes the expense called cost of goods sold, which Chapter 10, section 10.1 'Sales, Purchases and Returns' explains. The sale itself is revenue. So the same Tk 100 of cost appears in two places in two different times: first as an asset, then as an expense. Gross profit is the revenue from the sale less this cost. If the cost is wrong, the profit is wrong.

What does not count? The test is the purpose. A cupboard in the shop is owned and useful, but the owner does not intend to sell it. It is equipment, not inventory. A box of carrier bags for customers is a supply used in the business, not a good held for sale. Books that belong to a publisher and are only stored in the shop do not count either, because the shop does not own them. Inventory is owned goods held for sale, or owned materials held to make goods for sale.

A shop such as a bookshop buys goods already finished and sells them in the same condition. It therefore holds one kind of inventory: finished goods bought for resale. A business that makes things holds three kinds, because making takes time and the product passes through stages. Imagine a book-binding workshop, such as the separate workshop Meghna Binding Service. It buys paper, board, thread and glue. It folds, sews and covers books over several days. It then holds finished volumes until a customer collects them. At any moment the workshop has materials waiting, books half-made, and books complete.

The first kind is **raw materials**. Raw materials are materials a manufacturer holds to turn into products. For example, the reels of paper and the sheets of board in the workshop are raw materials. They have not yet been changed in any way.

The second kind is **work in process**. Work in process means goods in a factory that have been started but are not yet finished. For example, a pile of printed sheets that are sewn but have no cover is work in process. Its cost includes the materials and the labour used so far.

The third kind is **finished goods**. Finished goods are completed products held for sale. For example, a bound and trimmed volume on the collection shelf is a finished good. A bookshop's inventory consists wholly of finished goods, bought ready-made from a publisher.

The figure below shows the chain. Read it from left to right. Each box is one kind of inventory, and the arrows show the order

Two different words in common use cause confusion, so one of them is set aside.

Stock. Word in use. In daily life, a supply of goods kept for later. In this book, stock means shares in a company; goods are called inventory.

Inventory moves from the balance sheet to the income statement at the moment of sale.

Not everything in a shop is inventory, because the test is the purpose for which the item is held.

A merchandising shop holds one kind of inventory; a factory holds three.

The first stage is the materials waiting at the start of the chain.

Raw materials. Materials a manufacturer holds to turn into products.

The second stage is the unfinished products in the middle of the chain.

Work in process. Goods in a factory that have been started but are not yet finished.

The third stage is the completed products waiting for a customer.

Finished goods. Completed products held for sale.

The figure shows the chain from materials to customer, and how each stage differs in

The write-down is recorded as a loss and reduces the inventory account by the same amount.

The common slips are in the units and in the direction of the comparison.

The chapter's ideas in a few lines.

Belief: any item the business owns and holds is inventory.

Belief: the periodic system is wrong because it does not update the account.

Belief: net realisable value is the selling price.

Belief: when the market value of an item rises, the item should be written up.

Belief: a count that differs from the records means the count is wrong.

A famous modern case shows

180 = Tk 21,600. *Answer.* The write-down is Tk 14,400. *Check.* 36,000 - 21,600 = 14,400, which agrees.

The entry follows from the answer. Debit an expense account, here called Inventory Write-down Expense, with Tk 14,400, and credit Inventory with Tk 14,400. The income statement bears the loss in 2025, the year in which the value fell, and the balance sheet shows the lower figure.

Three slips are common. The first is to multiply by the wrong number of copies. The second is to use the selling price in place of net realisable value, forgetting to subtract the cost of selling. The third is to write up an item whose value has risen. A write-down is permitted; a write-up above cost is not.

SUMMARY

Inventory is owned goods held for sale, or owned materials held to make them. A shop holds finished goods; a factory holds raw materials, work in process and finished goods. Holding too little costs sales, and holding too much costs money and risk. A perpetual system updates Inventory at every purchase and sale; a periodic system finds it by a count at period end. Inventory is reported at the lower of cost and net realisable value.

THINK IT THROUGH

13.T1 The 800 unsold titles of Meghna and Sons cost Tk 380,000 and the holding cost is 10% a year. If they stay unsold for a further 12 months, what extra cost will the shop bear, and what other loss might it face?

13.T2 A bookseller says: "I never hold more than one week of goods, so I never lose money on inventory." Name two costs that this policy might create.

WHERE PEOPLE GO WRONG

The test is purpose. Equipment and office supplies are owned, but they are not held for sale, so they are not inventory.

Both systems are correct. They give the same cost of goods sold when the records are right. They differ in what they show during the year.

Net realisable value is the selling price less the cost of selling. Forgetting to subtract that cost raises the value wrongly.

Inventory is not reported above its cost. A fall below cost is recognised; a rise above cost is not.

The count is the final authority on quantity. A difference must be investigated, but if no cause is found the records are corrected to the count.

IN HISTORY

In April 2001 the American networking company Cisco Systems,

never happened leaves no record. They are real all the same, and that is why a shortage is easy to overlook.

Now suppose the owner decides to hold a great deal. The goods must be stored, insured and guarded. Paper can spoil in damp air. Fashions change, and a book may fall out of demand. Above all, the money spent on the goods could have been used elsewhere, for example to pay a debt or to buy goods that sell faster. The total of these burdens is called the holding cost, and it is usually stated as a percentage of the goods' cost for one year.

Case file, illustrative. On 31 December 2025, 800 titles had a total cost of Tk 380,000 and had not sold for 18 months. Meghna and Sons estimated its holding cost at 10% a year. The holding cost for one year is therefore $\text{Tk } 380,000 \times 10\% = \text{Tk } 38,000$. That is Tk 38,000 spent every year to keep goods that did not sell.

The table below compares the two mistakes. Introduce each row as a question: what is lost, and who sees it? Read across a row to compare the two cases.

THE COST OF TOO LITTLE AND THE COST OF TOO MUCH

QUESTION	TOO LITTLE INVENTORY	TOO MUCH INVENTORY
---	---	---
What is lost?	Sales and customers	Money tied up and goods that age
Is it recorded in the books?	No; the sale leaves no entry	Yes; storage and insurance appear as expenses
Who notices first?	The customer	The owner, when cash runs short
Typical remedy	Reorder sooner	Reduce purchases; sell at a lower price

The right amount cannot be set by a rule that fits every business. A grocer who sells fresh fish needs very little, because fish spoils in a day. A bookseller who sells textbooks needs more, because titles keep their value for a year or longer. There is also an exception to the idea that more is safer: if demand collapses suddenly, a large stock of goods becomes a large loss. We meet a famous case in the closing pack.

Before moving on. A business holds inventory because customers do not wait, suppliers need time, bulk buying saves money and demand follows the season. Holding too little costs sales. Holding too much costs storage, insurance, spoilage and the use of money. You should now be able to say one cost of each mistake and why the first is harder to see.

13.3 RECORDING INVENTORY

A business must know two things about its inventory: the quantity it holds and the value of that quantity. The value of each item is its cost, which Chapter 14 explains in detail. This section teaches how

Holding too much costs money in many small amounts that grow large together.

Cost. Word in use. In daily life, the price you pay for something. In accounting, the amount given up to acquire or hold an item; a holding cost is a different amount, spent to keep goods.

The running case gives a holding cost for a group of titles that did not sell.

A table sets the two mistakes side by side so that their different natures can be compared.

The right amount lies between the two mistakes and depends on the business.

Before moving on, we collect the ideas of this section.

To record inventory a business must answer two questions: how many, and

The perpetual system keeps a running record that changes after every purchase and every sale.

Perpetual inventory system. A system that updates the inventory account after every purchase and sale.

The periodic system leaves the account unchanged all year and finds the figure by counting at the end.

Periodic inventory system. A system that finds inventory only by a physical count at period end.

A small example shows that both systems reach the same cost of goods sold.

Under the perpetual system, every movement is recorded as it happens.

Under the periodic system, the answer comes from a formula applied after a count.

The two systems differ in what they reveal during the year, and a table makes the difference plain.

The first system is the **perpetual inventory system**. A perpetual inventory system is a system that updates the inventory account after every purchase and sale. For example, when the shop buys five books at Tk 100 each, the Inventory account rises by Tk 500 at once. When it sells eight books, the account falls by their cost, and the cost of goods sold rises by the same amount. At any moment the account shows what the shop should be holding. The word *perpetual* means "going on without end".

The second system is the **periodic inventory system**. A periodic inventory system is a system that finds inventory only by a physical count at period end. For example, a small stall records every purchase in a Purchases account and every sale in a Sales account, but does not touch Inventory until the last day of the year. The stall then counts what remains, values it at cost, and calculates the cost of goods sold by subtraction. The word *periodic* means "at regular periods".

Take small, clean numbers so that the method is the only thing to learn. *Given.* A shop begins the year with 10 books at Tk 100 each. It buys 5 more at Tk 100. It sells 8. All books cost Tk 100, so no question arises about which book is sold. *Find.* The cost of goods sold, and the inventory at the end, under each system.

Perpetual. Start with $10 \times 100 = \text{Tk } 1,000$. The purchase adds $5 \times 100 = \text{Tk } 500$, so the account shows Tk 1,500. The sale of 8 books reduces it by $8 \times 100 = \text{Tk } 800$, so the account shows Tk 700. The cost of goods sold, recorded at each sale, is Tk 800.

Periodic. The year's records show the opening inventory of Tk 1,000 and purchases of Tk 500. A count at year end finds 7 books, which are valued at $7 \times 100 = \text{Tk } 700$. The cost of goods sold is the opening inventory plus purchases less the closing inventory: $1,000 + 500 - 700 = \text{Tk } 800$. *Answer.* Both systems give cost of goods sold of Tk 800 and closing inventory of Tk 700. *Check.* $10 + 5 - 8 = 7$ books, and $7 \times 100 = 700$. The two systems agree, as they should.

PAUSE AND CHECK 13.P1 A shop with a periodic system loses 3 books to theft during the year. Will the loss show as a separate item in its records, or will it be hidden? Give a reason.

If both systems give the same answer, why choose between them? The difference is in what they show along the way. Read each row of the table below as a question about the system.

PERPETUAL AND PERIODIC SYSTEMS COMPARED

QUESTION	PERPETUAL	PERIODIC
---	---	---
When is Inventory updated?	After every purchase and sale	Once, at period end
Can the owner see the quantity at any time?	Yes, from the records	No, only after a count
Is a loss from theft visible?	Yes, as a difference from the count	No; it hides inside cost of goods sold
Work needed during the year	More	Less

a building all pass the test. A pack of paper fails the fourth question, because the business uses it up within weeks.

Think about what this means for the statements. When the business buys a plant asset, the money does not disappear as an expense. The business gives up cash and receives an asset of equal value. So the balance sheet shows the asset, and the income statement shows no expense on the day of purchase. The business then uses the asset over many periods. This is the same idea as the deferrals of Chapter 7, section 7.2 'Deferrals': a cost is paid now, and its benefit comes later. Chapter 16 'Depreciation' teaches how the cost is spread over the years of use.

The test depends on purpose, so the same object may fall into either group. A book-cutting machine on the floor of Meghna and Sons is a plant asset, because the shop uses it. The same machine in the showroom of a company that sells such machines is inventory, because that company holds it for sale. The question is always: for what purpose does this business hold the item?

Land deserves a short note. It is physical, owned and used, and it lasts. So it counts as a plant asset. But land does not wear out the way a machine does, and Chapter 16 'Depreciation' explains why this matters. For now, remember only that the class includes land, buildings, machines, vehicles and furniture.

Not every long-lived asset has a physical form. An **intangible asset** is a long-lived asset with no physical form, such as a patent or a trademark. For example, a patent is a legal right, given by the state, to be the only maker of a new invention for a number of years. A trademark is a name or sign that identifies a product. The owner cannot touch either one, yet each can earn money for years. The word *intangible* means "not able to be touched". Rules for intangible assets differ from country to country, so a business follows the standards of the place where it reports.

The table below compares the two kinds of long-lived asset with an ordinary item that is neither. Read each row as a question, and each column as an answer for one item.

PLANT ASSETS, INTANGIBLE ASSETS AND ORDINARY SUPPLIES

QUESTION	PLANT ASSET	INTANGIBLE ASSET	PACK OF PAPER
---	---	---	---
Physical form?	Yes	No	Yes
Used in the business, not held for sale?	Yes	Yes	Yes
Lasts more than one period?	Yes	Yes	No
Example	Book-cutting machine	Patent	Paper for printing

The class matters to different people. The owner wants to know how much money is tied up in equipment. A bank that lends money may accept the equipment as security, so it asks what the equipment cost. The manager decides when a machine must be replaced. The employee uses the machine every day and wants it

A plant asset moves from the business's cash to its balance sheet, and then it serves for years.

An exception: the same object can be a plant asset in one business and inventory in another.

Land is also a plant asset, but it behaves differently from the others.

Some valuable assets cannot be touched, and they form a separate group.

Intangible asset. A long-lived asset with no physical form, such as a patent or a trademark.

A table sets the two groups side by side so that you can compare them.

The viewpoints of owner, bank and manager show why the class matters.

YOU WILL NEED

Chapter 9, 'Financial Statements and Closing Entries', section 9.1 'The Three Statements'; Chapter 5, 'The Journal and the Ledger', section 5.2 'The Journal'; Appendix A1, 'Maths You Will Use'.

Three friends earn one profit from one stall, and each believes a different split is fair.

Equal shares and shares by money both seem unfair, so a better rule is needed.

This chapter answers four questions about people who own one business together.

A business with several owners raises three questions that a single owner never faces.

Partnership. A business owned by two or more people who agree to share its profit and loss.

A partnership is a business with two or more owners who share profit and loss.

Partnership deed. A

Three friends run a market stall together. Anwar paid for most of the first goods, but he visits only on Sundays. Babita paid very little, yet she works at the stall every day. Chandra paid a middle sum and keeps the accounts. At the end of the first month the stall has earned a profit of Tk 30,000. How should the three friends divide it?

Divide it into three equal parts, and Anwar feels cheated, because his money carried most of the risk. Divide it by the money each person paid in, and Babita feels cheated, because her daily work counts for nothing. Both friends have a fair point. Neither rule is wrong in itself; each is incomplete. Friends who sit together can argue until they agree. A business that lasts for years cannot rely on memory and goodwill, so its owners write the agreement down, and the accountant keeps books that follow it.

Accountants have a standard answer to the puzzle, and this chapter teaches it step by step. You will learn how the money each owner contributes is recorded, how profit is divided in layers so that both money and work are rewarded, what changes in the books when a new owner joins, and how the business is closed and the owners are paid when it ends. We begin, as the friends must, with the question of who owns what.

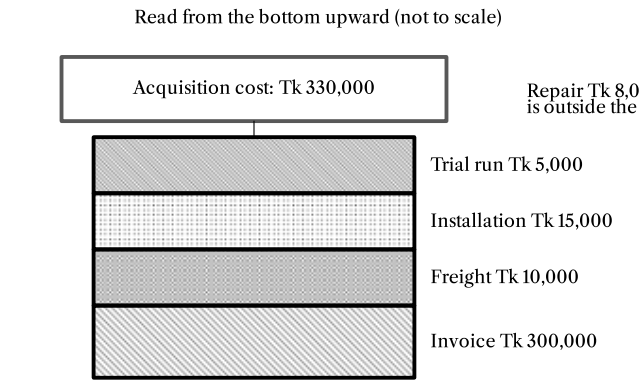
* * *

18.1 FORMING A PARTNERSHIP

A business with one owner is easy to describe. The owner puts money in, takes profit out, and the books show one amount, the owner's equity. That amount is also called capital, net worth or proprietorship, and this book uses the term owner's equity. When two or more people own a business together, three questions arise at once. How much has each person put in? How will the profit be divided? Who answers for the debts? The owners must agree on these answers before the first sale is recorded.

The law gives a name to such an arrangement. A **partnership** is a business owned by two or more people who agree to share its profit and loss. Each owner is called a partner. Three friends who open a bookshop together, and agree to share what it earns and loses, form a partnership. In many countries the law holds every partner personally responsible for the firm's debts, so the choice of partners is a serious one.¹

Partners protect themselves by writing their agreement down. A ¹ Partnership law differs by country. In the United Kingdom the Partnership Act 1890 supplies rules for partners who have not agreed their own; in Bangladesh the Partnership Act 1932 has the same purpose. Read the law of your own country before applying any rule from this chapter.

FIG. 15.1. *Cost build-up of one machine as stacked parts*

We now apply the method. *Given.* A shop buys a machine with an invoice price of Tk 200,000, less a trade discount of Tk 10,000. Freight is Tk 4,000. Installation is Tk 6,000. The staff training course costs Tk 3,000. *Find.* The acquisition cost.

Formula. Acquisition cost equals the invoice price less trade discount, plus freight, plus installation, plus testing and any tax that cannot be recovered. In symbols, let P be the invoice price, D the trade discount, F the freight, I the installation and T the other payments needed for readiness. Then the acquisition cost is $P - D + F + I + T$.

Steps. The price after discount is $200,000 - 10,000 = \text{Tk } 190,000$. Add freight: $190,000 + 4,000 = 194,000$. Add installation: $194,000 + 6,000 = 200,000$. There is no testing cost, so $T = 0$. The training course of Tk 3,000 is excluded, because it does not make the machine ready. *Answer.* The acquisition cost is Tk 200,000. *Check.* The total paid for the machine and its readiness is $190,000 + 4,000 + 6,000 = 200,000$, and the Tk 3,000 is treated as an expense.

PAUSE AND CHECK 15.P1 A delivery van is bought for Tk 900,000. The business also pays Tk 20,000 for the first year's road licence and Tk 12,000 for painting its name on the side before use. Which payments, if any, belong in the acquisition cost? Give a reason for each.

Now we apply the method to the machine of Meghna and Sons. The price is Tk 300,000, with no discount stated. Freight is Tk 10,000, installation is Tk 15,000 and the trial run is Tk 5,000. The acquisition cost is $300,000 + 10,000 + 15,000 + 5,000 = \text{Tk } 330,000$. The repair of Tk 8,000 is excluded, because it happened more than four months after the machine was working.

The entry follows. Suppose the shop paid by bank. The business debits Machinery with Tk 330,000 and credits Bank with Tk 330,000. The machine appears on the balance sheet at Tk 330,000, and no expense is recorded on the day of purchase. Why record one figure rather than five? The five payments together are the

A first example, with small and clean numbers, shows the method before the case.

The formula is stated in words first and then in symbols.

Each line of arithmetic is shown, and the training course is left out with a reason.

Now the case file is worked in the same order, with the same method.

The entry records the whole acquisition cost as one asset, not as five separate items.

CAPITAL EXPENDITURE AND REVENUE EXPENDITURE COMPARED

QUESTION	CAPITAL EXPENDITURE	REVENUE EXPENDITURE
---	---	---
What does it do?	Creates or improves a lasting asset	Keeps the asset working
Where is it recorded first?	Balance sheet, as an asset	Income statement, as an expense
How long is the benefit?	More than one period	This period
Example from the case	Machine, Tk 330,000	Repair, Tk 8,000

There is one exception that avoids waste of effort. A business may buy a stapler that lasts five years. Strictly it is a plant asset. But recording it as an asset and tracking its cost for five years is not worth the effort. Most businesses therefore set a minimum amount, below which any such item is charged as an expense at once. The size of the minimum is a business decision and differs from one business to another.

Suppose the Tk 8,000 repair were wrongly added to the machine. The income statement would show Tk 8,000 less expense, so the profit would be Tk 8,000 higher. The balance sheet would show the machine at Tk 338,000, which is Tk 8,000 too much. Both statements would be wrong in the same direction. Now suppose the opposite error: a Tk 15,000 installation were charged as an expense. The profit would be Tk 15,000 too low and the machine Tk 15,000 too low in the balance sheet.

Before moving on. Capital expenditure creates or improves an asset lasting more than one period, and it is recorded as an asset. Revenue expenditure keeps an asset in working order, and it is an expense of the period. A repair restores; an improvement adds. A mistake in the choice moves profit and the asset by the same amount. You should now be able to classify any payment and say what happens if it is wrongly classified.

WORKED EXAMPLE: THE BOOK-CUTTING MACHINE

Case file, illustrative. Meghna and Sons paid the following for the book-cutting machine in 2026: invoice Tk 300,000; freight Tk 10,000; installation Tk 15,000; trial run Tk 5,000; repair after a breakdown on 14 August Tk 8,000.

Find. (a) the acquisition cost and the amount of revenue expenditure; (b) the balance sheet figure and the profit effect if the repair were wrongly added to the machine.

Part (a), Steps. Payments up to readiness: $300,000 + 10,000 = 310,000$. Then $310,000 + 15,000 = 325,000$. Then $325,000 + 5,000 = 330,000$. The repair of Tk 8,000 is after readiness and is a revenue expenditure. *Answer.* The acquisition cost is Tk 330,000 and the revenue expenditure is Tk 8,000. *Check.* The total of all pay-

The exception: a small payment may be treated as an expense even if the item lasts.

The effect of an error shows why the rule matters.

Before moving on, we collect the ideas of this section.

The whole bill for the machine is split into capital and revenue spending, with every line shown.

The question has two parts: the acquisition cost, and the effect on profit if the repair is wrongly capitalised.

Part (a) adds only the payments needed to make the machine ready.

THINK IT THROUGH

17.T1 A club reports a deficit in a year in which its cash balance rose. Give two reasons, using the ideas of this chapter, why both can be true.

17.T2 A trader's owner's equity fell from Tk 50,000 to Tk 46,000 during a year in which she took Tk 9,000 for private use and brought in no new money. Was her business profitable? Explain.

WHERE PEOPLE GO WRONG

Cash shows only money that moved. It leaves out fees owed, fees paid early and the cost of lasting assets, so it can mislead in either direction.

A surplus belongs to the members as a group. It is added to the accumulated fund and used for the purposes of the club.

Payments for lasting assets, such as a table, become assets in the balance sheet. Only the costs that belong to the year appear in the income and expenditure account.

The rise must be corrected. Withdrawals reduced the equity and new money raised it, so both must be put back or taken out before profit appears.

It is built by counting and by memory when no ledger exists. It looks like a balance sheet, but it is less exact, because nothing proves it.

IN HISTORY

On 5 October 1942, during the Second World War, a group of citizens met in the Old Library of the University Church in Oxford, England, and formed the Oxford Committee for Famine Relief. Its aim was to help civilians starving in Greece under occupation. The committee later became known as Oxfam. Its first big appeal, held in Oxford in 1943, raised £12,700 for the Greek Red Cross, and £2,300 of that came from a temporary gift shop.

A charity such as this one has no owner who takes profit. Its accounts show money received from donors and money spent on its cause, which is the pattern of the receipts and payments account and the income and expenditure account of this chapter. Its test of success is not profit but how much of the money reaches the people it was raised for.

REVIEW QUESTIONS

17.1 A club began 2026 with cash of Tk 6,000. It received subscriptions of Tk 40,000 and donations of Tk 3,000, and paid rent of Tk 15,000, wages of Tk 12,000 and Tk 8,000 for a new cupboard. Prepare its receipts and payments account and find the closing cash.

17.2 In question 17.1 members owed Tk 2,000 of fees for 2025 at 1 January 2026, and these were paid in 2026. Tk 1,500 was received

The effect was large. Reports at the time said that, without the transfers, WorldCom would have shown a loss for 2001 and for the first quarter of 2002. Its internal audit team, led by Cynthia Cooper, traced the entries. The company filed for bankruptcy in July 2002. The lesson is the one you have learned: spending that is used up in the period must be charged to the period, and only spending that creates a lasting asset can be added to the balance sheet.

REVIEW QUESTIONS

15.1 A shop buys a printing machine. The invoice is Tk 450,000 and a trade discount of Tk 20,000 is allowed. Freight is Tk 8,000, installation Tk 12,000, a trial run Tk 4,000, and operator training Tk 6,000. Find the acquisition cost and the amount treated as an expense.

15.2 Classify each payment as capital or revenue expenditure: (a) new blades of the same kind, Tk 3,000; (b) a motor that raises output by half, Tk 40,000; (c) a concrete base laid for a new machine, Tk 9,000; (d) the yearly service of a van, Tk 7,500; (e) painting a building before first use, Tk 25,000.

15.3 A business treats a repair of Tk 12,000 as capital expenditure by mistake. State the effect on profit for the year and on the plant asset on the balance sheet.

15.4 A van costs Tk 600,000. The business pays Tk 30,000 to bring it from the port and Tk 18,000 to paint its name on it, both before use. In the first year it pays Tk 22,000 for repairs and Tk 9,000 for a licence. Find the acquisition cost, and the total revenue expenditure for the year.

The case shows why the choice of capital or revenue spending can decide profit or loss.

A first example with small numbers shows the method.

The steps build the two statements of affairs and then apply the formula.

The final accounts are then built with the profit as the balancing figure.

The method has limits, and an honest accountant states them.

A deficit is not always a failure, and a surplus is not always a success.

Before moving on, we collect the ideas of this section.

owner's equity, plus withdrawals, less new money brought in by the owner. In symbols, with E_2 for closing equity, E_1 for opening equity, W for withdrawals and N for new money: $\text{profit} = E_2 - E_1 + W - N$.

Given. At 1 January the trader owned equipment Tk 20,000, goods Tk 12,000, money owed by customers Tk 6,000 and cash Tk 2,000, and she owed suppliers Tk 8,000. At 31 December she owned equipment Tk 18,000, goods Tk 15,000, money owed by customers Tk 7,000 and cash Tk 5,000, and she owed suppliers Tk 6,000. During the year she took Tk 10,000 for private use and brought in Tk 5,000 of her own savings. *Find.* The profit for the year.

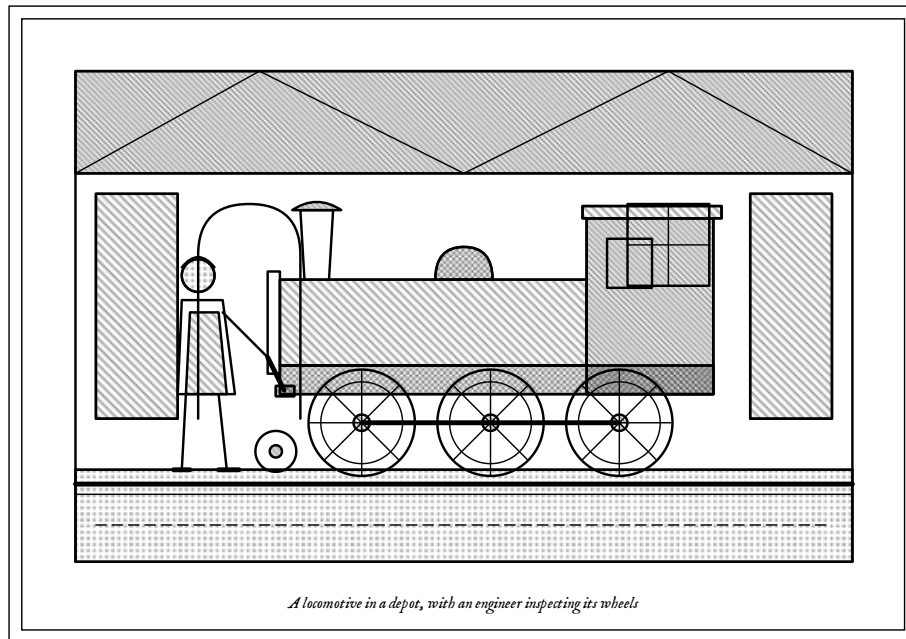
Steps. Opening assets: $20,000 + 12,000 + 6,000 + 2,000 = \text{Tk } 40,000$. Opening owner's equity: $40,000 - 8,000 = \text{Tk } 32,000$. Closing assets: $18,000 + 15,000 + 7,000 + 5,000 = \text{Tk } 45,000$. Closing owner's equity: $45,000 - 6,000 = \text{Tk } 39,000$. Profit = $39,000 - 32,000 + 10,000 - 5,000 = \text{Tk } 12,000$. *Answer:* The profit for the year is Tk 12,000. *Check.* Start from Tk 32,000, add the profit of Tk 12,000 and the new money of Tk 5,000, and subtract withdrawals of Tk 10,000: the result is Tk 39,000, which equals the closing owner's equity.

The trader can now prepare her final accounts. The balance sheet at 31 December lists the closing assets and liabilities from the statement of affairs, and its owner's equity is the opening owner's equity plus profit and new money, less withdrawals. The income statement shows the profit of Tk 12,000 as one figure. If she also has records of cash received and paid, she can show sales and expenses in more detail, by the methods of this chapter and Chapter 7. When she does not, the profit alone is the best result available.

The method has limits. It gives the amount of profit but not how it arose: you cannot see sales, expenses or gross profit. It depends on honest counting, since a missed asset or an unrecorded debt changes the answer. It also depends on correct figures for withdrawals, which small traders often do not record. For these reasons the result is an estimate that rests on the lists, and a trader should keep fuller records as the business grows. Reports that rely on such figures should state how they were prepared.

A word of judgement is useful here. A deficit in one year may be planned, for example when a club spends savings from earlier years on repairs. A surplus may be a warning, for example when a club saves money by delaying repairs that must be made later. The members of a club, the donors to a charity and the officials who inspect them read the accounts for these reasons. The figure matters less than the story behind it, and the accounts help the reader to find that story.

Before moving on. A statement of affairs lists the assets and liabilities at one date, and the difference is the owner's equity. Profit equals the change in owner's equity, plus withdrawals, less new money brought in. The method gives profit but not its parts. You should now be able to find a trader's profit from two statements of affairs and two further facts



CHAPTER XVI

Depreciation

If a machine serves for five years, why should one year pay for all of it?



This chapter teaches why the cost of a plant asset is spread over the years in which it helps to earn revenue. You will learn three methods of spreading cost, record the charge in the accounts, and find the gain or loss when an asset is sold.

The remedy is to apply the accrual basis, and the result has its own name.

Income and expenditure account. The non-trading organisation's equivalent of an income statement, showing surplus or deficit on an accrual basis.

The figure shows how the cash record is turned into the accrual record, step by step.

accumulated fund. A subscription is a regular payment by a member. The receipts and payments account summarises the cash only, so it cannot say whether the year went well. You should now be able to say what the account shows and the three things it leaves out.

17.2 INCOME AND EXPENDITURE ACCOUNT

The remedy for the three limits is a different statement, the **income and expenditure account**. An income and expenditure account is the non-trading organisation's equivalent of an income statement, showing surplus or deficit on an accrual basis. For example, it records the fees that belong to 2025, whether they were paid in 2025 or not. The accrual basis, which you met in Chapter 7, section 7.1 'Why Adjust', counts income and expense in the period to which they belong. The matching principle then puts the two together.

The figure below shows the change. Read it from left to right. The left box holds the receipts and payments account, which counts cash. The arrows mark the three changes you make: adjust for fees owed or paid early, adjust for costs owed or paid early, and take out the purchase of lasting assets. The right box holds the income and expenditure account, which counts what belongs to the year.

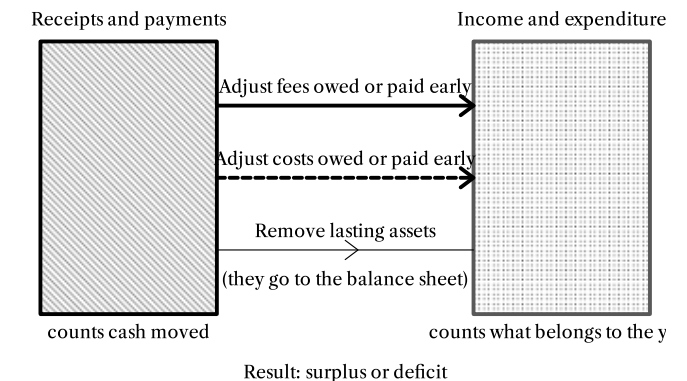


FIG. 17.1. Receipts and payments account turned into income and expenditure

The first change deals with accrued and prepaid items. An accrued item belongs to this period but has not been paid in cash. A prepaid item has been paid in cash but belongs to a later period. Take subscriptions as the main example. Suppose some members owe fees for the year; those fees are income of the year, though no cash has come in. Suppose some members paid next year's fee; that cash is not income of this year. Suppose some members paid last year's arrears this year; that cash belongs to the earlier year. A single rule covers all three. In words: income for the year equals cash received, less arrears from earlier years received this year, less

The first change adjusts items that belong to a different period from the cash.

Before we continue, notice that two ordinary words carry special meanings here. The word value in accounting often means a figure in the records, and the word life means the period of expected service. The two notes beside this paragraph give both meanings. We shall use each word in its accounting sense from now on.

The **useful life** of an asset is the period, or the amount of use, over which the asset is expected to serve the business. The bicycle has a useful life of four years. A delivery van might have a useful life of 200,000 kilometres. Notice that useful life is an expectation, not a fact, and it can be shorter than the time the asset physically lasts. A business may sell a machine after five years even though the machine could run for ten.

The **residual value** of an asset is the amount the business expects to receive when it disposes of the asset at the end of its useful life. The bicycle has a residual value of Tk 2,000. If the business expects to receive nothing, the residual value is zero. Only the part of the cost that will be used up needs to be spread, so the residual value is not charged to the years of use.

We now use the smallest numbers possible. *Given.* A bicycle costs Tk 10,000, has a residual value of Tk 2,000, and has a useful life of four years. *Find.* The amount to be spread, and an equal share for each year. *Formula.* The amount to be spread equals cost minus residual value. An equal yearly share equals the amount to be spread divided by the number of years. In symbols, $S = C - R$ and $D = S \div n$, where C is the cost, R is the residual value, S is the amount to be spread, n is the useful life in years and D is the yearly share. *Steps.* $S = \text{Tk } 10,000 - \text{Tk } 2,000 = \text{Tk } 8,000$. $D = \text{Tk } 8,000 \div 4 = \text{Tk } 2,000$. *Answer.* Tk 8,000 is spread, at Tk 2,000 a year. *Check.* Four years at Tk 2,000 make Tk 8,000, and Tk 10,000 less Tk 8,000 leaves the residual value of Tk 2,000.

Several factors lead a business to choose a life and a residual value. Wear from use reduces the usefulness of a machine. The passing of time rots wood and rusts iron, even when the machine stands idle. Changes in the market can make an asset out of date: a faster machine appears, or the books the shop sells change. A manager considers all three. The owner then chooses an estimate. Different people may reach different estimates, and tax rules in some countries use their own lives and rates, which may differ from those used in the accounts.¹

Exceptions need care. Land is a plant asset, but it is not depreciated. Intangible assets, such as a licence, are also spread over their useful lives, but this chapter treats plant assets only.

Two everyday words, value and life, have special meanings when used with assets.

Value. Word in use. In daily life, what something is worth to a person. In accounting, a figure in the records, such as cost less depreciation, which need not equal the price in the market.

Life. Word in use. In daily life, the time a person lives. In accounting, the period over which an asset is expected to serve the business, which need not equal its physical age.

Useful life is the period, or the amount of use, over which an asset is expected to serve.

Useful life. The period, or the amount of use, over which an asset is expected to serve the business.

Residual value is the amount the business expects to receive when the asset is disposed of.

Residual value. The amount a business expects to receive when it disposes of an asset at the end of its useful life.

A first calculation with small numbers shows the amount to spread and the share for each year.

Three factors affect the yearly charge: wear from use, passing of time, and change in the market.

Land is the usual exception,

A small puzzle: the cash grew, but the members are not sure the club is better off.

The puzzle asks what the cash figure leaves out.

The chapter has two settings, and both use the principles you know.

Imagine a village reading room, run by its members. They pay a yearly fee, and with the money they rent a room, buy newspapers and heat the room in winter. At the end of the year the treasurer opens the cash box and announces good news. A year ago it held Tk 4,000. Today it holds Tk 8,000. The cash has doubled.

A member asks three questions. Some members have not yet paid this year's fee; does that not count? Some members paid next year's fee early; is that money really this year's? The club bought a new table for Tk 10,000; is that a cost of this year only, or something that will serve for years? The treasurer's cash figure answers none of these questions. It records money moving in and out. It does not say whether the club has done well. This chapter shows how to answer with the tools you already know, in a new setting where nobody takes a profit.

There is a second puzzle, in a different setting. A small trader has kept no full books for a year. She has no journal, no ledger and no income statement. She knows only what she owns and owes at the start and at the end. Can she find her profit? She can, and the method is a pleasant surprise: it needs only two lists and some subtraction.

YOU WILL NEED

This chapter uses the three statements from Chapter 9 'Financial Statements and Closing Entries', section 9.1 'The Three Statements'. It also uses the accrual basis from Chapter 7 'Adjusting the Accounts', section 7.1 'Why Adjust'. The arithmetic is addition and subtraction; Appendix A1 'Maths You Will Use' holds a short reference.

* * *

17.1 NON-TRADING ORGANISATIONS

Until now this book has described businesses that are run to earn profit for an owner. Many organisations exist for another reason. A village reading room, a sports club, a school parents' association and a charity are examples. Nobody puts money in to take profit out. The members join to enjoy something together, or the donors give to help others.

The accounting name for this group is **non-trading organisation**. A non-trading organisation is an organisation, such as a club or charity, that exists to serve its members or a cause rather than to earn profit. For example, the village reading room serves its members by providing newspapers and a quiet place to read. The name does not mean that the organisation never sells anything. A club may sell tea or tickets. It means that the main aim is not profit.

Because no owner takes profit, some words change. The result of a year is not called profit but *surplus* when income is larger than expenditure, and *deficit* when it is smaller. A surplus is not a gain for an owner. It is money kept for the use of the members in the

Most of this book has described businesses that exist to earn a profit, and some organisations do not.

The accounting name for this group describes its purpose.

Non-trading organisation. An organisation, such as a club or charity, that exists to serve its members or a cause rather than to earn profit.

The accounts follow the same principles, but some names change to suit the purpose.

Surplus. Word in use. In daily life, an amount left over

Before moving on. The straight-line method charges Tk 60,000 a year for the machine, and Tk 45,000 for 2026. The units-of-production method charges Tk 0.60 a book, and Tk 54,000 for 90,000 books. You should now be able to say which method suits an asset that wears by use.

* * *

16.3 DECLINING-BALANCE

Some assets lose most of their usefulness early. A new machine cuts faster and breaks down less than an old one, and a new vehicle is worth more in its first year than in its fifth. For such assets a larger charge in the early years is a fairer picture. The third method provides it.

The **declining-balance method** is a depreciation method that charges a fixed percentage of the asset's remaining cost each year. We call the remaining cost the *book value*, a term that section 16.4 defines fully: it is the cost less all depreciation charged so far. Because the base falls each year, the charge falls each year. The balance that declines gives the method its name. The method uses the cost in the first year, and the residual value is not subtracted when the percentage is applied.

Given. Cost Tk 330,000, residual value Tk 30,000, a rate of 40% a year. *Find.* The charge and the remaining cost at the end of year 1 and year 2. *Formula.* The charge equals the rate times the remaining cost at the start of the year. The remaining cost at the end equals the start less the charge. In symbols, $D = r \times B$ and $B' = B - D$, where r is the rate, B is the remaining cost at the start and B' is the remaining cost at the end. *Steps.* Year 1: $D = 40\% \times \text{Tk } 330,000 = \text{Tk } 132,000$, and $B' = \text{Tk } 330,000 - \text{Tk } 132,000 = \text{Tk } 198,000$. Year 2: $D = 40\% \times \text{Tk } 198,000 = \text{Tk } 79,200$, and $B' = \text{Tk } 198,000 - \text{Tk } 79,200 = \text{Tk } 118,800$. *Answer.* The charges are Tk 132,000 and Tk 79,200, and the remaining cost after two years is Tk 118,800. *Check.* $\text{Tk } 132,000 + \text{Tk } 79,200 = \text{Tk } 211,200$, and $\text{Tk } 330,000 - \text{Tk } 211,200 = \text{Tk } 118,800$.

PAUSE AND CHECK 16.P2 An asset costs Tk 80,000 and is depreciated at 25% by the declining-balance method. Find the charge and the remaining cost at the end of year 1 and year 2.

The method has a limit that students often miss. The remaining cost must never fall below the residual value. The machine's remaining cost after year 3 is Tk 71,280, and after year 4 it is Tk 42,768. A charge of 40% in year 5 would be Tk 17,107, which would leave Tk 25,661, below the residual value of Tk 30,000. The year 5 charge is therefore cut to $\text{Tk } 42,768 - \text{Tk } 30,000 = \text{Tk } 12,768$.

The table below compares the yearly charges by the straight-line method and by the declining-balance method. Read each row across to compare the two methods for one year. The last row shows that both methods charge the same total, Tk 300,000, over five years. Only the timing differs.

Recap: Straight-line gives equal shares; units-of-production follows use.

GO SLOWLY HERE.

A larger charge in early years suits an asset that loses usefulness quickly.

The declining-balance method charges a fixed percentage of the remaining cost each year.

Declining-balance method. A depreciation method that charges a fixed percentage of the asset's remaining book value each year.

The machine's charge at 40% is Tk 132,000 in year 1 and Tk 79,200 in year 2.

Pause and check that you can apply a percentage to the remaining cost.

The method must stop at the residual value, so the last charge is cut to fit.

The table compares the yearly charges, and both methods spread the same Tk 300,000.

1859.² The railways were then about twenty years old, and their owners had to decide how to treat the wearing out of locomotives, carriages and track. Lardner discussed how parts wear with use and must be replaced, and he described the cost of keeping a railway in repair.

Railway companies owned costly equipment that served for many years, which is the same situation as that of the book-cutting machine. Lardner's discussion is an early case for the idea of this chapter: when equipment wears out, a fair record charges a part of its cost to each year of service. The methods in this chapter, straight-line, declining-balance and units-of-production, are later ways of making that idea exact.

REVIEW QUESTIONS

16.1 A machine costs Tk 50,000, has a residual value of Tk 10,000 and a useful life of four years. Find the straight-line charge for a year and the book value after two years.

16.2 Equipment costs Tk 275,000 on 1 October 2026. Its residual value is Tk 25,000, its useful life is eight years, and its capacity is 400,000 units. The business closes its books on 31 December. Find the straight-line charge for 2026, and the units-of-production charge for 2026 if 36,000 units were produced.

16.3 An asset costs Tk 150,000 and is depreciated at 20% by the declining-balance method. Find the charge and the remaining cost for each of the first three years. After three years it is sold for Tk 80,000. Find the gain or loss.

16.4 A press costs Tk 240,000 on 1 January 2026, with a residual value of Tk 40,000 and a useful life of five years. Using straight-line depreciation, write the entry for 2026. At 31 December 2028 the press is sold for Tk 100,000. Find the gain or loss and write the entry for the sale.

² Accounts of the book and its author are given in articles on the history of accounting and railways. This chapter relies on those summaries and not on the text of the book itself.

Accumulated depreciation is the total depreciation charged to date on an asset, shown as a deduction from its cost. After one year it equals one charge, after two years two charges, and so on. The business keeps the original cost visible, so that a reader can see both what the asset cost and how much has been charged. For the machine, the 2026 charge by straight line is Tk 45,000, and the entry is shown below.

ENTRY ON 31 DECEMBER 2026 FOR THE STRAIGHT-LINE CHARGE

ACCOUNT	DEBIT (TK)	CREDIT (TK)
Depreciation expense, machine	45,000	
Accumulated depreciation, machine		45,000

The **book value** of an asset is its cost less its accumulated depreciation. At 31 December 2026 the book value of the machine is Tk 330,000 - Tk 45,000 = Tk 285,000. The balance sheet shows the three figures together: cost, accumulated depreciation and book value. Book value is a figure in the records. It is not the price at which the machine could be sold, which may be higher or lower.

THE MACHINE IN THE BALANCE SHEET AT 31 DECEMBER 2026 (TK)

ITEM	AMOUNT
Machine, at cost	330,000
Less accumulated depreciation	45,000
Book value	285,000

The second figure shows the book value of the machine at the end of each year. The horizontal axis shows the years, and the vertical axis shows taka. Each horizontal step shows the book value that holds through a year, and each drop shows one yearly charge. After year 3 the book value is Tk 150,000, the amount used in the next calculation.

Book value at the end of each year, taka

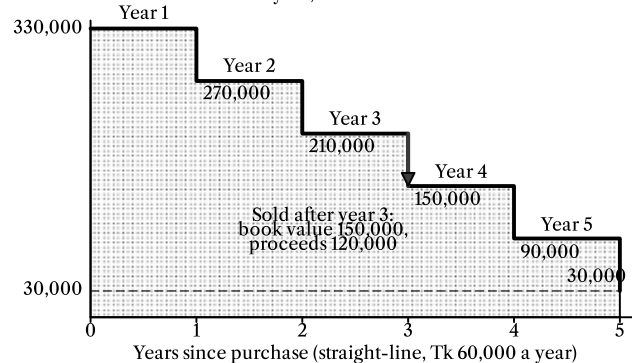


FIG. 16.2. Book value of the machine stepping down year by year

Case file, illustrative. After three years of straight-line depreciation, Meghna and Sons sells the book-cutting machine for Tk 120,000.

Accumulated depreciation is the running total of the charges made on one asset.

Accumulated depreciation. The total depreciation charged to date on an asset, shown as a deduction from its cost.

Book value is cost less accumulated depreciation, and it shows the cost not yet charged.

Book value. An asset's cost less its accumulated depreciation.

The figure shows book value stepping down once a year under straight-line.

A sale after three years brings a gain or a loss, found by comparing proceeds with book

The calculation first finds the book value at the date of sale and then compares it with the proceeds.

The entry removes the asset and its accumulated depreciation and records the cash and the loss.

Depreciation must be brought up to the date of sale first, and a sale above book value gives a gain.

Recap: accumulated depreciation and book value are linked, and a sale compares proceeds with book value.

This worked case shows that the method chosen changes the gain or loss on sale.

The reasoning finds the book value under each method and compares it with the proceeds.

Given. Cost Tk 330,000, charge Tk 60,000 a year for three years, proceeds Tk 120,000. *Find.* The gain or loss on sale. *Formula.* Gain or loss equals proceeds minus book value. A positive result is a gain and a negative result is a loss. In symbols, $G = P - B$, where P is the proceeds and B is the book value. *Steps.* Accumulated depreciation = $3 \times \text{Tk } 60,000 = \text{Tk } 180,000$. $B = \text{Tk } 330,000 - \text{Tk } 180,000 = \text{Tk } 150,000$. $G = \text{Tk } 120,000 - \text{Tk } 150,000 = -\text{Tk } 30,000$. *Answer.* The sale gives a loss of Tk 30,000. *Check.* The machine was carried at Tk 150,000 but brought only Tk 120,000, a shortfall of Tk 30,000.

The entry for the sale removes the machine from the records. It credits the machine for its full cost. It debits accumulated depreciation, so that the charges made so far are cleared. It debits cash for the proceeds, and it debits a loss account for the loss. The debits and credits must be equal, and the table shows that they are.

ENTRY FOR THE SALE OF THE MACHINE AFTER THREE YEARS

ACCOUNT	DEBIT (TK)	CREDIT (TK)
Cash	120,000	
Accumulated depreciation, machine	180,000	
Loss on sale of machine	30,000	
Machine, at cost		330,000

Two exceptions complete the picture. If the sale falls during a year, the charge for the part of the year must be recorded first, so that the book value is correct on the date of sale. If the machine had sold for Tk 170,000, the result would be $\text{Tk } 170,000 - \text{Tk } 150,000 = \text{Tk } 20,000$, a gain, and the entry would credit a gain account instead of debiting a loss account.

Before moving on. Accumulated depreciation is the total charged so far, and book value is cost less that total. At sale, gain or loss equals proceeds minus book value. You should now be able to say why the machine's sale gave a loss of Tk 30,000.

* * *

WORKED CASE: A VAN SOLD AFTER TWO YEARS

Situation. A different shop bought a delivery van on 1 January 2026 for Tk 480,000. It estimated a residual value of Tk 60,000 and a useful life of six years. At 31 December 2027, after two years, it sold the van for Tk 250,000. These figures are for practice and are not part of the case file. *Question.* What is the gain or loss on sale if the shop used straight-line depreciation, and what is it if the shop used declining-balance at 30%?

Reasoning. Straight line: the yearly charge is $(480,000 - 60,000) \div 6 = \text{Tk } 70,000$. After two years the accumulated depreciation is Tk 140,000, so the book value is $\text{Tk } 480,000 - \text{Tk } 140,000 = \text{Tk } 340,000$. Declining balance: year 1 is $30\% \times 480,000 = \text{Tk } 144,000$, leaving Tk 336,000. Year 2 is $30\% \times 336,000 = \text{Tk } 100,800$, leaving Tk 235,200. The accumulated depreciation is Tk 244,800. The proceeds are Tk 250,000 in both cases.

18.3 ADMISSION OF A PARTNER

What must be agreed, and what must be recorded, when a fourth person asks to join a firm of three? The two questions have different answers, and many students mix them up. The first belongs to the partners and their deed. The second belongs to the accountant and the journal.

When a new partner is admitted, the people who own the firm change. In many places the law treats the firm before and after the change as two different partnerships. For this reason the partners agree new terms: the date, the new partner's capital and the profit-sharing ratio that applies from then on. The deed is rewritten or added to. The accountant records the new capital only when it actually arrives.

Case file, illustrative. On 1 January 2027, Nusrat Meghna is admitted as a partner. She pays in cash of Tk 700,000, which is credited as her capital. The case file gives no new profit-sharing ratio, so none is used in this chapter.

Recording the new capital uses the rule from section 18.1. Cash comes in, so we debit Cash. The new partner's claim on the firm grows, so we credit her capital. Each journal entry has a date, the accounts, the amounts and a short narration. The table shows the entry for the case.

JOURNAL ENTRY FOR THE ADMISSION OF NUSRAT MEGHNA

DATE	ACCOUNT	DEBIT (TK)	CREDIT (TK)
1 January 2027	Cash	700,000	
	Nusrat Meghna, Capital		700,000
	Cash paid in by the new partner on admission		

Check the entry in two ways. The debit of Tk 700,000 equals the credit of Tk 700,000, as every entry requires. The capital accounts of Nurul, Rafiq and Tariq are untouched, because their claims did not change on that day. If the new partner brings a delivery van instead of cash, the debit goes to the van, and the credit to her capital stays the same.

Before moving on. Admission of a partner changes the owners, so the partners agree new terms. The accountant records only the new capital: debit Cash, or the asset brought, and credit the new partner's capital. Existing capital accounts stay as they were.

* * *

18.4 LIQUIDATION

Every business ends one day. How does an owner recover money from a shop full of shelves, counters and unsold books? The answer is to sell everything, pay what is owed and divide what remains. Accountants give a name to this process and to its first

A new partner changes who owns the firm, so the agreement and the records must both change.

Admission of a partner means a new person becomes an owner, usually after new terms are agreed.

The case file records Nusrat Meghna joining on 1 January 2027 with cash.

Recall: journal, Chapter 5, section 5.2 'The Journal'.

Recording a new partner's capital follows the rule for any cash paid in: debit Cash, credit capital.

Debits equal credits, and the existing partners' accounts do not change.

New terms are agreed by the partners; new capital is recorded by the accountant.

Ending a business means turning assets into cash, paying debts and sharing what is left.

Capital and, where there is a premium, Share Premium. The Share Application account is then closed. (*Chapter XIX*)

Asset. Something of value that a business owns or controls, such as cash, goods held for sale or equipment. For example, the shelving bought by a workshop is an asset, and so is the cash in its drawer. (*Chapter III*)

Authorised capital. The largest amount of share capital a company is allowed to issue under its founding documents. It is a ceiling and not money received. A company with authorised capital of Tk 50,000 may sell shares worth up to Tk 50,000 and no more. (*Chapter XIX*)

Balance sheet. A report of assets, liabilities and owner's equity at one date. Its two sides must agree, because assets equal liabilities plus owner's equity. Its heading says *At* and then the date. Some standards call it the statement of financial position. (*Chapter IX*)

Bank reconciliation statement. A statement that explains the difference between the bank balance in the cash book and the balance on the bank's statement. It has two halves, each ending in an adjusted balance, and the two must be equal. It is a working paper, not a ledger account. (*Chapter XII*)

Book value. An asset's cost less its accumulated depreciation. It is a figure in the records and need not equal the price at which the asset could be sold. For example, Tk 330,000 less Tk 180,000 gives a book value of Tk 150,000. (*Chapter XVI*)

Bookkeeping. The recording part of accounting: writing down each transaction in the right book. A bookkeeper writes each sale and payment as it happens, while an accountant also sorts the records, checks them and prepares reports for readers who need them. (*Chapter II*)

Business entity. The rule that a business is treated as separate from its owner when records are kept. The owner's household spending stays out of the business records, and two businesses of one owner keep separate books, even when the law treats the owner and business as one person. (*Chapter II*)

Capital expenditure. Spending that creates or improves an asset lasting more than one period. It is recorded as an asset on the balance sheet. The purchase of a machine is capital expenditure, and so is a motor fitted to make it work faster. (*Chapter XV*)

Cash book. A book that serves as both journal and ledger for cash and bank receipts and payments. It has a debit side for money received and a credit side for money paid, usually with a Cash column and a Bank column. A customer's Tk 500 in notes is written once, in the Cash column. (*Chapter XII*)

Chart of accounts. A list of every account a business uses, with a code for each. It stops the business from opening duplicate accounts or forgetting one. Meghna Binding Service, for example, uses eight accounts, so its chart is short. (*Chapter IV*)

Closing entry. A journal entry made at the end of a period that moves the balance of a temporary account to capital, usually through the income summary. After it, the temporary account

EXAMPLE 2: PAYING THE PARTNERS AT LIQUIDATION

PART	WORKING
Given	Capital: Nurul Tk 2,500,000, Rafiq Tk 1,000,000, Tariq Tk 600,000. Ratio 5:3:2. Assets, book amount Tk 5,300,000, sold for Tk 4,400,000. Liabilities Tk 1,200,000.
Find	The loss on realisation, each partner's share of it, and the cash paid to each partner.
Formula	In words: loss = book amount – amount received; payment = capital – share of loss. In symbols: $L = B - A$ and $X = C - D$, where B is the book amount, A is the amount received, C is the capital and D is the partner's share of the loss.
Step 1	Loss $L = 5,300,000 - 4,400,000 = \text{Tk } 900,000$.
Step 2	Shares of loss: Nurul $(5 \div 10) \times 900,000 = \text{Tk } 450,000$. Rafiq $(3 \div 10) \times 900,000 = \text{Tk } 270,000$. Tariq $(2 \div 10) \times 900,000 = \text{Tk } 180,000$.
Step 3	Payments X: Nurul $2,500,000 - 450,000 = \text{Tk } 2,050,000$. Rafiq $1,000,000 - 270,000 = \text{Tk } 730,000$. Tariq $600,000 - 180,000 = \text{Tk } 420,000$.
Step 4	Cash after debts = $4,400,000 - 1,200,000 = \text{Tk } 3,200,000$.
Answer	The loss is Tk 900,000. Nurul receives Tk 2,050,000, Rafiq Tk 730,000 and Tariq Tk 420,000.
Check	$2,050,000 + 730,000 + 420,000 = \text{Tk } 3,200,000$, which equals the cash after debts.

Two slips are usual. Some students pay the partners before the outside debts, which is wrong, because outsiders have the first claim. Others share the loss in proportion to capital instead of the profit-sharing ratio. Capital and ratio are different things, and the deed names the ratio.

Occasionally a loss is larger than a partner's capital. That partner's capital account then has a debit balance, which means the partner owes the firm the shortage. The partner must pay it in before the final payments are made. The case file has no such partner, because each capital is larger than that partner's share of the loss.

Before moving on. Liquidation sells the assets, pays the debts and shares what remains. Realisation is the sale and its gain or loss, which the partners share in the ratio. Outsiders are paid before partners, and each partner receives capital less his or her share of the loss.

* * *

Paying partners before debts, or sharing the loss by capital, are the two usual slips.

A partner whose capital falls below zero must pay in the shortage.

Sell, share the loss, pay debts, then pay the partners: this order is the technique of liquidation.

WORKED CASE: DIVIDING SIX MONTHS OF PROFIT

Section 18.2 used a full year. The case file is messier, because the profit covers only six months. Read the figure below from top to bottom. The top bar is the whole profit. The dashed layer shows interest on capital, and the dotted layer shows the remainder divided by the ratio. The bottom layer gives each partner's total.

The first six months of the case give a realistic division over a period shorter than a year.

APPROPRIATION ACCOUNT FOR THE SIX MONTHS ENDED 31 DECEMBER 2026
(TK)

ITEM	NURUL	RAFIQ	TARIQ	TOTAL
Profit for the period				420,000
Interest on capital	75,000	30,000	18,000	123,000
Remainder, shared 5:3:2	148,500	89,100	59,400	297,000
Total credited to partners	223,500	119,100	77,400	420,000

A common slip is to charge a full year of interest. The interest would then be 150,000, 60,000 and 36,000, a total of Tk 246,000, and the remainder would fall to Tk 174,000. Every total would be wrong. Always ask first: how long is the period?

Tariq holds the smallest capital. Interest gives him Tk 18,000, and the ratio gives him a fifth of the remainder. Nurul holds the largest capital, and for him the interest layer protects the reward for his money. The ratio then gives him half of what remains, as the deed provides.

* * *

SUMMARY

A partnership is a business owned by two or more people who share profit and loss under a deed. Each partner has a capital account. Profit is divided in two layers, interest on capital and then the remainder by the profit-sharing ratio, and both are recorded in the appropriation account. A new partner's capital is recorded by debiting Cash and crediting that partner's capital. In liquidation, the assets are realised, the loss is shared by the ratio, debts are paid first, and each partner receives capital less his or her share of the loss.

THINK IT THROUGH

18.T1 Tariq adds Tk 400,000 to his capital, and the ratio stays 5:3:2. What changes in his share of the remainder, and what changes in his interest at 6% a year?

18.T2 After a firm sells all its assets, its cash is smaller than its debts. Who is paid first, and what happens to the partners' capital?

WHERE PEOPLE GO WRONG

Belief: the partner with the most capital must have the biggest share of profit. **Correction:** the ratio is fixed by the deed and need not match the capital. Interest on capital rewards money separately, in the first layer.

Belief: interest on capital is a cost of the firm, like rent. **Correction:** it is a way of dividing profit among the partners, so it appears in the appropriation account and not in the income statement.

Charging a full year of interest gives wrong totals, even though the arithmetic looks neat.

Each partner sees the two layers differently, and both layers serve a purpose.

A partnership shares profit and loss by agreement and records each partner's capital separately.

A larger capital does not give a larger ratio; the deed sets the ratio.

Interest on capital divides profit; it is not an expense paid to an outsider.

the costs together: Tk 2,490,000 + Tk 915,000 = Tk 3,405,000. Then Tk 3,900,000 - Tk 3,405,000 = Tk 495,000, which equals the profit found above.

Now a second example with new figures. A small bakery has net sales of Tk 800,000 for a quarter. It bought flour and other goods for Tk 310,000 and paid freight in of Tk 12,000. It paid rent Tk 90,000, wages Tk 160,000 and delivery to customers Tk 18,000. Product costs are Tk 310,000 + Tk 12,000 = Tk 322,000. Delivery to customers is freight out, so it is a period cost; period costs are Tk 90,000 + Tk 160,000 + Tk 18,000 = Tk 268,000. Gross profit is Tk 800,000 - Tk 322,000 = Tk 478,000, and profit is Tk 478,000 - Tk 268,000 = Tk 210,000. Check: Tk 800,000 - Tk 322,000 - Tk 268,000 = Tk 210,000.

Watch for three slips. The first is to place freight in with the period costs, which would understate gross profit. The second is to mix periods, for example by comparing a quarter's profit with a full year's profit. The third is to add the profit instead of subtracting the costs, which gives a figure larger than sales.

Situation. The owner of Meghna and Sons wants to know whether the quarter's profit of Tk 495,000 can stand for a full year. **Question.** Is it safe to multiply the quarter's profit by four? **Reasoning.** The quarter is one interim period of a calendar year. Trade may be higher or lower in the other quarters, and some adjustments in an interim report are estimates. **Conclusion.** The figure is a useful signal but not a forecast of the year. **Check.** The case file does not give figures for the other quarters, so no firm yearly figure can be drawn from this one.

* * *

SUMMARY

A fiscal year is a chosen twelve-month reporting period, and an interim period is a shorter one. A product cost is tied to goods and waits in inventory until the sale, while a period cost is charged to the time in which it is incurred. The statements are linked: profit flows from the income statement to owner's equity to the balance sheet. A payable is owed by the business and a receivable is owed to it. The trial balance tests the accounts, and the balance sheet reports them.

THINK IT THROUGH

20.T1 Suppose Meghna and Sons changed its fiscal year to run from 1 July to 30 June, the same as the Bangladesh government. Name two things that would change in its reports, and one thing that would stay the same.

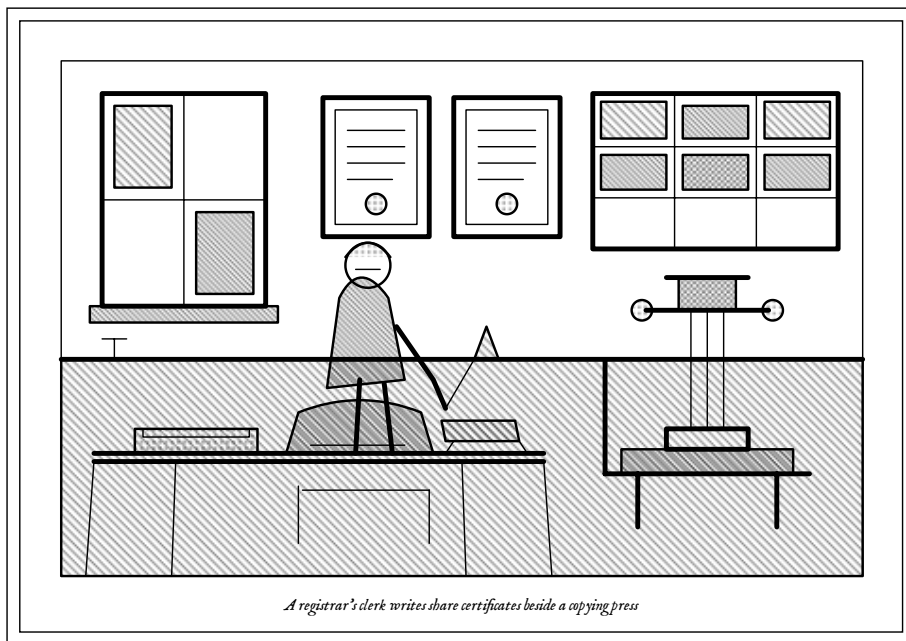
20.T2 A seller of school uniforms trades most heavily in the weeks before term starts. Which twelve-month period would you suggest for this business, and why?

A second, messier example uses new figures and one cost that is easy to misplace.

Common slips in this calculation arise from mixing classes, periods or signs.

A worked case applies the ideas to a decision instead of a calculation.

A short summary of the whole chapter.



CHAPTER XIX

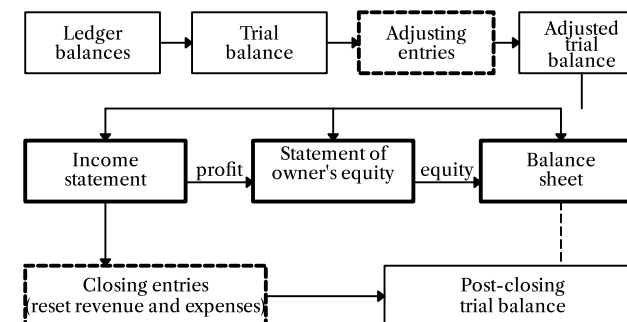
Company Accounts: Shares

How can a thousand strangers own one business together?



This chapter explains how a company raises money by selling shares. It shows what share capital is, how ordinary and preference shares differ, and how an issue of shares is recorded from application to allotment, including the share premium. Every entry is worked by hand, from small numbers to a realistic case.

income statement, the statement of owner's equity and the balance sheet are prepared. Fifth, closing entries reset the revenue and expense accounts to zero for the next period.



Dashed boxes are the steps that change account balances.

FIG. 20.1. How the trial balance, the adjustments, the statements and the closing entries connect

The figure shows the cycle as boxes joined by arrows, with dashed boxes for the steps that change balances.

How the three statements connect is the heart of the cycle.

An account payable is an amount we owe; an account receivable is an amount owed to us.

The trial balance and the balance sheet look alike but serve different purposes.

The complete cycle can be reviewed by following one

To read the figure, follow the arrows. Each box is a step or a report, and each arrow means that the figures pass on to the next box. The dashed boxes mark the two steps that change account balances: the adjusting entries and the closing entries. The solid boxes only collect and report the balances that already exist.

The three statements pass figures along a chain. The income statement finds the profit for the period. That profit goes into the statement of owner's equity, where it is added to the opening owner's equity, after any owner's withdrawals are subtracted. The closing owner's equity then goes into the balance sheet. If any link is wrong, the balance sheet will not balance.⁴

A common confusion lies between two accounts. An account payable means an amount the business owes to a supplier. An account receivable means an amount a customer owes to the business. Some books call these creditors and debtors, but this book uses payable and receivable. Both appear on the balance sheet: the payable among the liabilities, and the receivable among the assets. A payable is money going out later; a receivable is money coming in later.

The trial balance and the balance sheet also look alike. Both list accounts with amounts. The trial balance lists every ledger account, including revenue and expense accounts, to test that total debits equal total credits. It is a working test, and nobody outside the business needs to see it. The balance sheet lists only assets, liabilities and owner's equity at one date, and it is a report for readers. A trial balance can balance and still contain an error that affects the profit, such as a cost put in the wrong account.

Follow one quarter through the whole cycle. In July, August and September, each purchase of books, payment of rent and sale of

⁴ Chapter 9, 'Financial Statements and Closing Entries', section 9.3 'Reversing Entries', describes the balance sheet and the entries that follow the statements.

Every share is created with an amount printed on it. This amount is called the face value, and it is the same for every share of the same kind. A company might create shares with a face value of Tk 10. The face value is a fixed label. It is not the price at which people trade the shares afterwards, and it is not what the company's books record when one shareholder sells to another. Those later sales happen between shareholders, so the company's own accounts do not change.

A company cannot create unlimited shares. The people who found it choose a ceiling, and write that ceiling into its founding documents.¹ Think of a theatre. The building has a fixed number of seats, whatever the size of tonight's audience. The ceiling of a company works as the number of seats does.

Authorised capital is the largest amount of share capital a company is allowed to issue under its founding documents. A company with authorised capital of Tk 50,000 may sell shares worth up to Tk 50,000 and no more. If the founders later want to sell shares worth more, they must first change the founding documents by the legal method of their country.

Most companies do not sell all their authorised shares at once. **Issued capital** is the part of authorised capital that has actually been sold to shareholders. If the company above has sold shares worth Tk 30,000, then its issued capital is Tk 30,000. The remaining Tk 20,000 is authorised but not issued, and the company may sell it later. Issued capital can never be larger than authorised capital.

Example 1 applies these ideas to a small company. The first line of each step names the rule, and every line of arithmetic follows it.

EXAMPLE 1: AUTHORISED CAPITAL AND ISSUED CAPITAL

PART	WORKING
Given	Authorised capital Tk 50,000, in shares of face value Tk 5. The company has issued 6,000 shares.
Find	The number of shares authorised, the issued capital, and the capital not yet issued.
Formula	In words: shares authorised = authorised capital ÷ face value; issued capital = shares issued × face value; capital not yet issued = authorised capital – issued capital. In symbols: $N = A \div F$, $I = n \times F$ and $U = A - I$, where A is authorised capital, F is face value and n is the number of shares issued.
Step 1	$N = 50,000 \div 5 = 10,000$ shares.
Step 2	$I = 6,000 \times 5 = \text{Tk } 30,000$.

¹ The rule differs by country. Under the Companies Act 1994 of Bangladesh, the founding documents state a ceiling of share capital. In the United Kingdom, the Companies Act 2006 ended the need for most companies to state one. Read the law of your own country before applying any rule from this chapter.

Each share carries a face value, the amount printed on it, which is fixed when the share is created.

A company may not create shares without limit; its founding documents set a ceiling.

Authorised capital. The largest amount of share capital a company is allowed to issue under its founding documents.

Authorised capital is the ceiling on the shares a company may sell.

Issue. Word in use. In daily life, a topic of concern, or a copy of a magazine. In accounting, to issue shares means to sell and deliver them to the people who apply for them.

Issued capital. The part of authorised capital that has actually been sold to shareholders.

Issued capital is the part of the ceiling that has actually been sold.

Example 1 uses small round numbers, so that the method is the only thing to learn.

Recap of fiscal year and interim period.

The reader should now be able to state the difference between the two kinds of period.

A shop owner pays for books and for rent, yet only one of these payments waits on the shelf.

Product cost. A cost of buying or making goods held for sale, such as the price paid for books and freight in.

A product cost is the cost of buying goods that are held for sale, including freight in.

Cost. Word in use. In daily life, cost is the price you pay for something. In accounting, a cost is an amount given up to obtain goods or services, and it is later either held as an asset or charged as an expense.

The word cost has an everyday meaning and a narrower accounting meaning.

Period cost. A cost charged to the period in which it is incurred, such as rent or wages.

A period cost belongs to the time in which it is incurred, not to any particular goods.

The case file lists the product costs and period costs of one quarter.

The two classes of cost differ in what they are tied to and in when they reach the income statement.

taken place. Treat interim figures as a useful early signal, and not as the last word.

Before moving on. A fiscal year is any twelve months a business chooses, and the calendar year is one such choice. An interim period is shorter than a year, and a quarter is the usual example. Always ask of a report: from which date to which date?

You should now be able to say: "A fiscal year is a chosen twelve months, and an interim period is a shorter slice inside it."

20.2 PRODUCT COSTS AND PERIOD COSTS

Return to the second puzzle. The owner of Meghna and Sons has paid for books, for freight in, for rent and for wages. All four are costs, and all four reduce cash. But the first two are tied to goods, and the last two are tied to time. That difference decides when each cost reaches the income statement.

A **product cost** is a cost of buying or making goods held for sale, such as the price paid for books and freight in. Freight in means the carrier's charge for bringing the goods to the shop. A product cost first becomes part of inventory. It stays there until the goods are sold, and only then does it move to cost of goods sold in the income statement.²

The word *cost* needs care. In daily life it means the price paid. In accounting it means an amount given up to get goods or services, and the later treatment depends on what was bought. A cost for goods held for sale is held as inventory until the sale. A cost for something used up with time is charged at once.

A **period cost** is a cost charged to the period in which it is incurred, such as rent or wages. The shop must pay rent for July, August and September whether it sells ten books or ten thousand. The cost belongs to the time, and not to a particular book. So the income statement for the quarter carries the rent in full, and nothing waits on the shelf.

Case file, illustrative. For the quarter ended 30 September 2026, Meghna and Sons recorded these product costs: books bought Tk 2,450,000 and freight in Tk 40,000. It recorded these period costs: rent Tk 180,000 and wages Tk 735,000.

The table below compares the two classes. Read down each column to see how a cost behaves. The first row shows what the cost is tied to, the second shows where it first appears, and the third shows when it reaches the income statement.

² Which cost is attached to which unit sold depends on the cost flow assumption. Chapter 14 teaches this in section 14.1, 'The Problem of Changing Prices'. This chapter assumes first-in first-out, but any method would classify the costs in the same way.

We begin with the common kind. An **ordinary share** is a share that gives a holder a vote and a claim on profit after other claims are paid. The vote lets the holder help to choose the people who direct the company. The claim on profit has no fixed amount, so it is large when the company does well and may be nothing when it does badly. Babita would choose ordinary shares.

The second kind suits Anwar. A **preference share** is a share that gets a fixed dividend before ordinary shareholders, usually without a vote. A dividend is a share of profit that the company pays out to its shareholders. If Anwar holds preference shares with a fixed rate of 8% a year, he receives 8% of their face value first, before any ordinary shareholder is paid, but the company has not promised that it will earn that profit.

We now have the rights of each kind. The table below sets them side by side. To read a row, find the right in the left-hand column and move across to see how each kind treats it.

THE RIGHTS OF THE TWO KINDS OF SHARE

RIGHT	ORDINARY SHARE	PREFERENCE SHARE
Vote at meetings	Yes	Usually no
Dividend	Not fixed; paid from profit that remains	Fixed rate; paid first
Order of payment	After the preference dividend	Before the ordinary dividend
Gain when profit is large	Large, because the claim is on what remains	Limited to the fixed rate
Loss when profit is small	May receive nothing	Paid first, if profit is enough

The table shows a trade between safety and reward. The preference holder gains safety and gives up the vote and the chance of a large dividend. The ordinary holder accepts risk and gains a vote and a larger claim when profit is large. Neither kind is better for everyone, because the right choice depends on what the investor needs.

Example 2 uses clean figures, so that the method is clear. The company has 100,000 preference shares and 200,000 ordinary shares, all with a face value of Tk 10. These numbers are chosen to teach the method and do not belong to the case file.

Ordinary share. A share that gives a holder a vote and a claim on profit after other claims are paid.

An ordinary share gives a vote and a claim on whatever profit remains.

Preference share. A share that gets a fixed dividend before ordinary shareholders, usually without a vote.

A preference share gives a fixed dividend, paid before the ordinary shareholders, usually without a vote.

The two kinds differ in their rights, and a table makes the differences easy to compare.

In the table, the fixed rate protects the preference holder, and the remainder rewards the ordinary holder.

Example 2 divides one year's dividend between the two kinds of share.

A friend's question about a shop can be answered with three different stretches of time.

A second puzzle: two payments that look alike are treated in different ways.

A third puzzle: the many records and reports of earlier chapters must join into one system.

The chapter rests on earlier work on the balance sheet, the classified income statement and cost flow.

GO SLOWLY HERE.

A business never stops, so accountants cut its life into equal slices of time and measure each slice.

A fiscal year is any twelve-month reporting period that a business chooses; it may start in any month.

Fiscal year. Any twelve-month period a business chooses for its yearly reports, which may start in any month.

Year. Word in use. In daily life, a year is the twelve months from 1 January to 31

Imagine that you own a small bookshop. On the evening of 30 September, a friend who has lent you money asks how the shop is doing. You open your records. You could describe the whole of last year. You could describe the three months that ended today. Or you could describe the time since 1 July, when the government's own twelve-month period began. All three answers are true, and each gives different figures. Which one should you give, and who decides?

Now consider a second puzzle. During those three months the shop paid Tk 2,450,000 for new books and Tk 180,000 for rent. Both payments are money that left the business, yet an accountant handles them differently. The books are not charged against profit until a customer buys them; until then they wait on the shelf as inventory. The rent belongs to the three months in which the shop used the building, whether or not a single book is sold. Why the difference?

The third puzzle is larger. By now you have met the journal, the ledger, the trial balance, adjusting entries, the worksheet, the financial statements and closing entries. Each has its own chapter. This chapter asks how they fit together. If one number changes in the ledger, which reports must change with it, and in what order? When you can answer that, you hold the whole accounting cycle in your mind, and not only its parts.

You will need: Chapter 9, 'Financial Statements and Closing Entries', section 9.3 'Reversing Entries'; Chapter 10, 'Accounts of a Merchandising Company', section 10.2 'Classified Income Statement'; Chapter 14, 'Costing Inventory: FIFO, LIFO and Weighted Average', section 14.1 'The Problem of Changing Prices'; and Appendix A1, 'Maths You Will Use'.

* * *

20.1 FISCAL YEAR AND INTERIM PERIODS

Think first of a family budget. A family that earns a monthly salary does not wait until old age to check its spending. It looks at one month, sees what came in and what went out, and plans the next month. A school does the same with its school year, which begins when classes begin and not on 1 January. A business needs the same habit. It may trade for fifty years, but its owner cannot wait fifty years to learn whether it earns a profit. So the owner chooses a slice of time and measures it.

The usual name for this slice is the **fiscal year**. A fiscal year means any twelve-month period that a business chooses for its yearly reports, and it may start in any month. A shop might choose 1 April to 31 March, or 1 July to 30 June. The word *fiscal* comes from the Latin *fiscus*, a basket for money, and it now means "to do with money and accounts".

Notice that the word *year* now has two meanings. In daily life it usually means 1 January to 31 December. In accounting it means

account holds the money until the company decides who receives shares. It is a temporary account, closed at the next step.

The second step belongs to the company. **Allotment** is the company's formal decision to give shares to applicants. When the company allots 500 shares to the neighbour, he or she becomes a shareholder. From this moment the company may credit its share capital, because the shares now exist in the hands of their owners. The Share Application account is closed, and its balance moves into the permanent accounts.

Some companies sell shares for more than their face value. **Share premium** is the amount paid for a share above its face value. A share with a face value of Tk 10 sold for Tk 25 carries a premium of Tk 15. The premium is not part of the share capital, which holds only the face value. It is credited to a separate account named Share Premium. In many countries the law limits what a company may do with that account.

Why would anyone pay more than the face value? Because the business may already be worth more than the amounts printed on its shares. A shop with a good name, loyal customers and years of trading is worth more than a new shop with the same number of shares. In a partnership, a new partner may pay for such a gain, as Chapter 18 showed. In a company, the share premium records the same idea.

PAUSE AND CHECK 19.P1 A share of face value Tk 10 is issued at Tk 18. What is the premium on each share, and what is the total premium on 1,000 shares?

The entries follow the order of events, and each has a debit total equal to its credit total. When the application money is received, we debit Bank and credit Share Application. When the shares are allotted, we debit Share Application, and we debit Bank for any further money received at that time. We credit Share Capital with the face value of the shares and credit Share Premium with the premium. This chapter assumes that all money is paid when it is due. A real company must also record money that arrives late.

Example 3 uses small numbers. The company issues 1,000 shares of face value Tk 10 at Tk 12. Applicants pay Tk 12 a share with the application, and all 1,000 shares are allotted.

Allotment. The company's formal decision to give shares to applicants.

Allotment is the company's decision to give shares to the applicants.

Share premium. The amount paid for a share above its face value.

Share premium is the amount paid above face value, and it is kept in its own account.

Recall: partnership, Chapter 18, section 18.1 'Forming a Partnership'.

A company may charge a premium because the business is already worth more than the face value of its shares.

Two journal entries record an issue: one when applications arrive and one on allotment.

Example 3 records a small issue with a premium, so that the method is the only thing to learn.

EXAMPLE 3: RECORDING A SMALL ISSUE WITH A PREMIUM

PART	WORKING
Given	1,000 shares, face value Tk 10, issued at Tk 12. All the money is paid with the application, and all 1,000 shares are allotted.
Find	The share capital, the share premium, and the two journal entries.
Formula	In words: share capital = shares × face value; share premium = shares × (issue price – face value). In symbols: $K = n \times F$ and $M = n \times (P - F)$, where n is the number of shares, F is the face value and P is the issue price.
Step 1	Application money = $1,000 \times 12 = \text{Tk } 12,000$. Debit Bank 12,000; credit Share Application 12,000.
Step 2	$K = 1,000 \times 10 = \text{Tk } 10,000$.

EXAMPLE 4: THE PUBLIC ISSUE OF MEGHNA AND SONS LIMITED

PART	WORKING
Given	200,000 shares, face value Tk 10, issue price Tk 25. Application money Tk 10 a share. Allotment money Tk 15 a share. All money is paid when due.
Find	The money at each step, the share capital, the share premium, and the journal entries.
Formula	In words: money at a step = shares $\times$ amount a share; share capital = shares $\times$ face value; share premium = shares $\times$ (issue price – face value). In symbols: $K = n \times F$ and $M = n \times (P - F)$.
Step 1	Application money = $200,000 \times 10 = \text{Tk } 2,000,000$.
Step 2	Allotment money = $200,000 \times 15 = \text{Tk } 3,000,000$.
Step 3	Total money = $2,000,000 + 3,000,000 = \text{Tk } 5,000,000$, which also equals $200,000 \times 25$.
Step 4	Share capital $K = 200,000 \times 10 = \text{Tk } 2,000,000$.
Step 5	Share premium $M = 200,000 \times (25 - 10) = \text{Tk } 3,000,000$.
Answer	Share capital credited is Tk 2,000,000 and share premium credited is Tk 3,000,000.
Check	$2,000,000 + 3,000,000 = \text{Tk } 5,000,000$, which equals the total money received.

The journal entries follow. The first records the money received with the applications. The second records the allotment and the money received at that stage.

JOURNAL ENTRIES FOR THE PUBLIC ISSUE

STEP	ACCOUNT	DEBIT (TK)	CREDIT (TK)
Application	Bank	2,000,000	
	Share Application		2,000,000
Allotment	Share Application	2,000,000	
	Bank	3,000,000	
	Share Capital		2,000,000
	Share Premium		3,000,000

Now consider the effect on the company. Before this issue, its issued capital was Tk 4,100,000. After the public issue it is Tk 4,100,000 + Tk 2,000,000 = Tk 6,100,000, which is still below the authorised capital of Tk 10,000,000. The share premium of Tk 3,000,000 is held in its own account, outside the share capital. The company has raised Tk 5,000,000 in cash in all. The balance sheet shows both accounts among the owners' equity, in the way that Chapter 9 described.

* * *

SUMMARY

A company is owned by shareholders through shares. Share capital is the money it receives for them. Authorised capital is the ceiling

The two journal entries put the money into Bank first and into capital and premium second.

After the issue, the company has issued capital of Tk 6,100,000 and a share premium account of Tk 3,000,000.

A company raises share capital by selling shares, and the entries follow application

usually without a vote. In an issue, a share application is recorded in a temporary account, allotment moves it into Share Capital at face value, and share premium is credited to its own account.

THINK IT THROUGH

19.T1 The company offers 200,000 shares, but applications arrive for 300,000. Suggest two ways the company could treat the extra applications, and say what must happen to the money of any applicant who receives no shares.

19.T2 A company with shares of face value Tk 10 sells new shares for Tk 40. Explain, in two or three sentences, why the buyers may be willing to pay Tk 30 more than the face value.

WHERE PEOPLE GO WRONG

Belief: authorised capital is money the company already has. **Correction:** authorised capital is a ceiling. Only the part that has been issued, and paid for, has become share capital.

Belief: the whole issue price goes to Share Capital. **Correction:** only the face value is credited to Share Capital. The amount above face value is credited to Share Premium.

Belief: a preference dividend is a promise that is always paid. **Correction:** it is a fixed rate and the first claim on profit. The company can pay it only if there is profit, and the rules of each company decide what happens when there is none.

Belief: when shares are sold between shareholders, the company's share capital changes. **Correction:** the company's books change only when the company itself issues shares and receives money for them. Later sales are between shareholders.

IN HISTORY

In the nineteenth century, a person who invested in a business could, in many cases, lose more than the money paid in. The United Kingdom Parliament changed this. The Joint Stock Companies Act 1856, which received royal assent on 14 July 1856, allowed seven or more people to register a company by signing a memorandum of association. The company could be registered with limited liability, which means that a shareholder could lose no more than the amount he or she had agreed to pay for shares.

A company with limited liability had to carry the word "Limited" at the end of its name, so that everybody who dealt with it knew the limit. The neighbours in our opening puzzle would have understood the value of this rule. A person could put in Tk 5,000, lose at most Tk 5,000, and share in the profit. This is the reason that share ownership spread, as the Act made small investments safer for people who did not run the business themselves.

REVIEW QUESTIONS

Fiscal year. Any twelve-month period a business chooses for its yearly reports. It may start in any month, and the calendar year is one choice among many. Meghna and Sons uses 1 January to 31 December, while the Bangladesh government uses 1 July to 30 June. (*Chapter XX*)

Going concern. The assumption that a business will continue to operate for the foreseeable future. Because of it, a shop's shelves are treated as items used for years. The assumption fails when the business is about to close, and records must then show sale values. (*Chapter II*)

Gross profit. Net sales less cost of goods sold, before operating expenses are deducted. It shows what the business earns from buying and selling goods. Example: Tk 14,400,000 - Tk 9,800,000 = Tk 4,600,000. (*Chapter X*)

Historical cost. The rule of recording an asset at the price paid for it, not at its present value. The price paid can be proved by a receipt, while a present value is an opinion. A shelf bought for a price stays at that price in the records. (*Chapter II*)

Income and expenditure account. The non-trading organisation's equivalent of an income statement, showing surplus or deficit on an accrual basis. It records income and costs in the period to which they belong, and leaves out the purchase of lasting assets. (*Chapter XVII*)

Income statement. A report of revenue and expenses for one period, such as a year, ending in net income or net loss. Its heading says *For the year ended*. Example: revenue Tk 788,000 less expenses Tk 604,000 gives net income of Tk 184,000. (*Chapter IX*)

Income summary. A temporary account used to collect revenue and expenses at period end. Its balance equals net income or net loss, which is then moved to capital. After that move, the account stands at zero. It exists only during closing. (*Chapter IX*)

Intangible asset. A long-lived asset with no physical form, such as a patent or a trademark. The business cannot touch it, yet it can earn money for years. A patent is a legal right to be the only maker of an invention for a number of years. (*Chapter XV*)

Interest on capital. A payment credited to partners for their capital before the remaining profit is divided. It rewards money put in. For a period shorter than a year, it is capital times the yearly rate times the fraction of the year. It divides profit; it is not an outside cost. (*Chapter XVIII*)

Interim period. A reporting period shorter than a year. A quarter of three months is the usual example, and months and half-years are others. Interim reports use the same methods as yearly reports but depend more on estimates. (*Chapter XX*)

Inventory. Goods held for sale, or materials held to make goods for sale, in the ordinary course of business. The business must own them. It is a current asset until sale, when its cost becomes cost of goods sold. Books on a bookshop's shelves are inventory. (*Chapter XIII*)

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Net sales. Sales less sales returns and allowances. It is the revenue a merchandising business really keeps. Example: sales of Tk 10,000, less returns of Tk 500 and allowances of Tk 200, give net sales of Tk 9,300. (*Chapter X*)

Non-trading organisation. An organisation, such as a club or charity, that exists to serve its members or a cause rather than to earn profit. Its result is called a surplus or a deficit, and its equity is the accumulated fund. A village reading room is an example. (*Chapter XVII*)

Normal balance. The side of an account, debit or credit, on which its increases are recorded and where its balance usually ends. Cash has a debit normal balance, and Accounts Payable has a credit normal balance. A balance on the other side is a signal to look closer. (*Chapter IV*)

Operating expense. A cost of running the business, such as wages or rent, apart from the cost of goods sold. Operating expenses are grouped as selling expenses and administrative expenses. Example: advertising and office rent. (*Chapter X*)

Ordinary share. A share that gives a holder a vote and a claim on profit after other claims are paid. The dividend is not fixed, so it is large when profit is large and may be nothing when profit is small. It carries more risk and more possible reward. (*Chapter XIX*)

Outstanding cheque. A cheque written and recorded in the books but not yet paid by the bank, because the payee has not yet presented it or it has not yet cleared. For example, a cheque for Tk 900 written on 30 June and paid on 4 July. It is subtracted on the bank side. (*Chapter XII*)

Owner's equity. The owner's claim on the business: what is left of the assets after all liabilities are paid. For example, assets of Tk 565,000 less liabilities of Tk 120,000 leave owner's equity of Tk 445,000. (*Chapter III*)

Partnership. A business owned by two or more people who agree to share its profit and loss. The owners are called partners. Three friends who open a bookshop and agree to share what it earns and loses form a partnership. The law of each country adds its own rules. (*Chapter XVIII*)

Partnership deed. A written agreement setting out each partner's capital, duties and share of profit. It prevents disputes by recording what was agreed. Where the deed is silent on a point, the law of the place supplies a rule, which partners often wish to replace. (*Chapter XVIII*)

Period cost. A cost charged to the period in which it is incurred, because it belongs to time and not to particular goods. It reaches the income statement at once. Rent and wages are examples. (*Chapter XX*)

Periodic inventory system. A system that finds inventory only by a physical count at period end. During the year only purchases are recorded, and cost of goods sold is found by subtraction: opening inventory plus purchases less closing inventory. (*Chapter XIII*)

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Reversing entry. An entry made on the first day of a new period that cancels an earlier accrual. It lets the later payment be recorded in the usual way. Example: debit Wages Payable and credit Wages Expense for Tk 14,000 on 1 January 2027. (*Chapter IX*)

Sales journal. A special journal that records only sales of goods on credit. Each line comes from a copy of a sales invoice and shows the date, invoice number, customer and amount. Its total is posted once to the main ledger. Cash sales are not recorded in it. (*Chapter XI*)

Service business. A business that earns money by doing work for customers rather than by selling goods. The customer pays for the work performed, such as a repaired book, a haircut or a lesson, and takes home no goods that the business bought or made. (*Chapter I*)

Share application. A request to buy shares, sent with money, to a company that is offering them. The sender is not yet a shareholder. The company holds the money in a temporary account called Share Application until it decides who receives shares. (*Chapter XIX*)

Share capital. The money a company receives from shareholders in exchange for shares. It records the face value of the shares issued. If shareholders pay Tk 1,000 for 100 shares of face value Tk 10, share capital is Tk 1,000. It appears among the owners' equity. (*Chapter XIX*)

Share premium. The amount paid for a share above its face value. A share of face value Tk 10 sold for Tk 25 carries a premium of Tk 15. The premium is credited to its own account and is not part of share capital. (*Chapter XIX*)

Shareholder. A person who owns one or more shares of a company. A person who holds one tenth of the shares owns one tenth of the company. More shares give a larger part of the ownership. (*Chapter I*)

Source document. Paper or electronic proof of a transaction, such as an invoice, receipt or cheque stub. Records begin from the document, not from memory, because a bank, a tax authority or an auditor can inspect it. A publisher's invoice for books is an example. (*Chapter V*)

Special journal. A journal used for only one kind of repeated transaction, such as credit sales. It names the two accounts once in its heading and records only the changing parts, such as date, name and amount, on each line. Using one saves writing and reduces postings. (*Chapter XI*)

Statement of affairs. A list of a business's assets and liabilities at one date, used to find capital when records are incomplete. The assets less the liabilities give the owner's equity. It is built by counting, so it is less exact than a balance sheet from a ledger. (*Chapter XVII*)

Statement of owner's equity. A report showing how the owner's capital changed during one period. It starts with capital at the beginning, adds net income, subtracts drawings, and ends with capital at the end. Example: Tk 100,000 + Tk 184,000 – Tk 80,000 = Tk 204,000.

statements. It is private, it is not a financial statement, and it changes no ledger account. Example: Supplies show Tk 1,000 before an adjustment and Tk 400 after it. (*Chapter VIII*)

ANSWERS TO QUESTIONS

CHAPTER I · BUSINESS ACTIVITY AND COMPANIES

1.p1 It is a merchandising business, because it buys finished goods (rice) and sells them again without changing them.

1.p2 One quarter, because 25 is one quarter of 100.

1.t1 Its customers pay for work, so it holds no goods bought for resale. It still records fees, wages, rent and any money owed, and the materials it uses are a cost of the work.

1.t2 The salon combines a service business (hair work) with a merchandising business (shampoo bought and resold). If shampoo is a small part of its income, the service kind brings in most of its money.

1.t3 Not in the sense of this chapter, because the club does not supply goods or services to earn a profit. It still needs records of the fees it collects and the money it spends.

1.t4 It combines manufacturing (bread baked from materials) and merchandising (tea bought and resold). Ownership has the shape of a partnership, with three owners. Any one of the three friends, a lender or the tax authority would need its records.

1.1 The tailor is a service business, the grocer is a merchandising business, and the furniture workshop is a manufacturing business.

1.2 Examples: the owner (whether the bakery earns a profit), a lender (whether a loan can be repaid), the tax authority (how much tax is due), and an employee (whether wages will be paid).

1.3 $410,000 \text{ shares} \times \text{Tk } 10 = \text{Tk } 4,100,000$. One tenth of 410,000 is 41,000 shares.

1.4 A shareholder is a person who owns one or more shares of a company. A merchandising business buys finished goods and resells them, while a manufacturing business makes goods from materials and then sells them.

1.5 It is a service business, because it repairs and binds books for schools. Separate records let the owner see the result of each business alone and judge each one fairly.

CHAPTER II · WHAT ACCOUNTING IS

2.1 Nurul asks whether the shop earned enough profit (Tk 750,000). The bank asks whether the loan (Tk 600,000) can be repaid. The publishers ask whether the Tk 1,150,000 owed to them will be paid. The tax authority asks whether the profit return is honest. The staff ask whether wages (Tk 2,940,000) and jobs are safe.

2.t1 A household bill is not a cost of the shop. If it appears in the shop's records, the shop's profit looks too low and the bank forms a false picture of the shop's ability to repay. The business entity rule keeps the two sets of money apart.

2.p1 The book leaves the shop's inventory, so the shop's records must show that event. The rule says the shop is separate from

wages 13,000 = 28,000. Surplus = Tk 15,000. The cupboard is an asset, not an expense.

17.3 Opening owner's equity = $60,000 - 15,000 = 45,000$. Closing = $72,000 - 18,000 = 54,000$. Profit = $54,000 - 45,000 + 20,000 - 8,000 = \text{Tk } 21,000$.

17.4 In the receipts and payments account the Tk 30,000 is a payment. In the income and expenditure account it does not appear as expenditure, because it is a lasting asset. In the balance sheet it appears as an asset at its cost.

CHAPTER XVIII · PARTNERSHIP ACCOUNTS

18.p1 Interest for a year is $200,000 \times 5\% = \text{Tk } 10,000$. For six months, $10,000 \times 6 \div 12 = \text{Tk } 5,000$.

18.t1 His share of the remainder does not change, because the ratio stays 5:3:2 and he still receives two parts in ten. His interest rises by $400,000 \times 6\% = \text{Tk } 24,000$ a year, so his total grows through the first layer only.

18.t2 Outsiders such as suppliers and banks are paid first, from all the cash available. If the cash cannot cover the debts, nothing is left for the partners, whose capital is used up by the loss.

18.1 Four parts in all. The first partner receives $(3 \div 4) \times 80,000 = \text{Tk } 60,000$ and the second receives $(1 \div 4) \times 80,000 = \text{Tk } 20,000$. Check: $60,000 + 20,000 = 80,000$.

18.2 Interest: Mira $450,000 \times 8\% = 36,000$; Naveed $300,000 \times 8\% = 24,000$; total 60,000. Remainder $211,000 - 60,000 = 151,000$. Shares: Mira $(3 \div 5) \times 151,000 = 90,600$; Naveed $(2 \div 5) \times 151,000 = 60,400$. Totals: Mira Tk 126,600 and Naveed Tk 84,400, which add to 211,000.

18.3 Time is $9 \div 12 = 0.75$ year. Interest: Mira $450,000 \times 8\% \times 0.75 = 27,000$; Naveed $300,000 \times 8\% \times 0.75 = 18,000$; total 45,000. Remainder $180,000 - 45,000 = 135,000$. Shares: Mira 81,000; Naveed 54,000. Totals: Mira Tk 108,000 and Naveed Tk 72,000, which add to 180,000.

18.4 Debit Bank Tk 300,000 and credit Priya, Capital Tk 300,000, dated 1 April 2028, with a narration that the amount was paid in by the new partner on admission. Debits equal credits.

18.5 Loss = $850,000 - 700,000 = \text{Tk } 150,000$. Shares: P $(2 \div 3) \times 150,000 = \text{Tk } 100,000$; Q $(1 \div 3) \times 150,000 = \text{Tk } 50,000$. Payments: P $400,000 - 100,000 = \text{Tk } 300,000$; Q $200,000 - 50,000 = \text{Tk } 150,000$. Check: cash after debts is $700,000 - 250,000 = 450,000$, which equals $300,000 + 150,000$.

CHAPTER XIX · COMPANY ACCOUNTS: SHARES

19.p1 The premium on each share is $18 - 10 = \text{Tk } 8$. On 1,000 shares the total premium is $1,000 \times 8 = \text{Tk } 8,000$.

19.t1 The company could allot shares to every applicant in proportion, for example 2 shares for every 3 applied for, or accept some applications in full and refuse the rest. Money for shares not allotted must be returned to the applicant, or used for the later amount due if the applicant agrees.

credit Cash 4,000. Cash: $50,000 + 26,400 - 7,500 - 9,200 - 4,000 = 55,700$ debit. Accounts Payable: $18,000 - 7,500 = 10,500$ credit.

4.3 Borrowing: debit Cash 100,000, credit Bank Loan 100,000. Repaying: debit Bank Loan 40,000, credit Cash 40,000. Bank Loan has a credit balance of Tk 60,000.

4.4 An asset normally has a debit balance, so a credit balance means more cash was recorded as paid out than received, through an error or an overdraft. Check each entry against its source document.

4.5 Inventory: asset, debit, debit. Accounts Payable: liability, credit, credit. Drawings: drawings, debit, debit. Service Revenue: revenue, credit, credit. Wages: expense, debit, debit.

CHAPTER V · THE JOURNAL AND THE LEDGER

5.p1 Debit Electricity (an expense) Tk 10,000 and credit Cash Tk 10,000. Narration: Paid electricity bill in cash.

5.t1 Publisher Payable would show a credit balance of Tk 200,000 instead of Tk 120,000. The credit balances would then total Tk 740,000 against debit balances of Tk 660,000, so credits would exceed debits by Tk 80,000.

5.1 Debit Cash 100,000, credit Owner's Equity 100,000 (narration: Owner invested cash). Then debit Shelving 20,000, credit Cash 20,000 (narration: Bought shelving for cash).

5.2 Cash debit side: 100,000. Cash credit side: 20,000. Balance = $100,000 - 20,000 =$ Tk 80,000 debit.

5.3 Debit total = $125,500 + 48,250 = 173,750$. Credit total = $30,750 + 61,400 = 92,150$. Balance = $173,750 - 92,150 =$ Tk 81,600, on the debit side.

5.4 Debit side: $245,000 + 52,000 = 297,000$. Credit side: 118,500. Balance = $297,000 - 118,500 =$ Tk 178,500 debit.

5.5 Debit Equipment 30,000; credit Cash 10,000; credit Equipment Dealer Payable 20,000. Debits 30,000 equal credits 30,000. Balances: Equipment Tk 30,000 debit; Cash Tk 10,000 credit on these entries alone; Equipment Dealer Payable Tk 20,000 credit.

CHAPTER VI · THE TRIAL BALANCE

6.p1 Every entry has equal debits and credits, and posting copies both sides, so the ledger holds equal totals of debits and credits. Balancing removes the same amount from both totals, so the debit balances still equal the credit balances.

6.t1 Two examples: the 15 January rent entry left out entirely, which leaves Cash at Tk 305,000 and Rent at nil but both totals at Tk 660,000; and wages debited to Rent instead of Wages, which leaves the totals equal but Rent wrongly at Tk 95,000 and Wages at nil.

6.1 Debits: $50,000 + 30,000 =$ Tk 80,000. Credits: Tk 80,000. The totals are equal.

6.2 Debits: $187,450 + 96,300 + 150,000 + 42,000 + 78,500 =$ Tk 554,250. Credits: $64,250 + 400,000 + 90,000 =$ Tk 554,250. The

is found, it reduces Inventory to the counted figure and charges Tk 1,800 as a loss.

13.5 Holding cost is $600,000 \times 12\% =$ Tk 72,000 a year. The write-down per unit is $80 - 55 =$ Tk 25, so for 150 units it is Tk 3,750. The units are reported at $150 \times 55 =$ Tk 8,250.

CHAPTER XIV · COSTING INVENTORY: FIFO, LIFO AND WEIGHTED AVERAGE

14.p1 The oldest 60 units are 20 at Tk 300, 30 at Tk 320 and 10 at Tk 340, so the cost of goods sold is Tk 6,000 + Tk 9,600 + Tk 3,400 = Tk 19,000. The 40 units left are $40 \times$ Tk 340 = Tk 13,600. Tk 19,000 + Tk 13,600 = Tk 32,600.

14.p2 The unit cost is Tk 326, so 25 sets cost $25 \times$ Tk 326 = Tk 8,150.

14.t1 FIFO is highest at Tk 12,600, weighted average is next at Tk 12,180, and LIFO is lowest at Tk 11,600 in the case. When prices rise, FIFO charges the oldest, cheapest costs to the sale and LIFO charges the newest, dearest costs. Weighted average lies between them.

14.t2 The ranking reverses: LIFO gives the highest gross profit, FIFO the lowest, and weighted average lies between. For all three methods, cost of goods sold plus ending inventory equals the cost available.

14.1 Cost available is Tk 100 + Tk 120 = Tk 220. FIFO: ending Tk 120, cost of goods sold Tk 100, gross profit Tk 100. LIFO: ending Tk 100, cost of goods sold Tk 120, gross profit Tk 80. Weighted average: unit cost Tk 11, ending Tk 110, cost of goods sold Tk 110, gross profit Tk 90.

14.2 Cost available is Tk 1,800 + Tk 2,880 + Tk 5,200 = Tk 9,880 for 200 units; 70 remain. FIFO: ending Tk 3,640, cost of goods sold Tk 6,240, gross profit Tk 4,160. LIFO: ending Tk 3,240, cost of goods sold Tk 6,640, gross profit Tk 3,760. Weighted average: unit cost Tk 49.40, ending Tk 3,458, cost of goods sold Tk 6,422, gross profit Tk 3,978. Each pair adds to Tk 9,880.

14.3 Cost available is Tk 2,000 + Tk 1,750 = Tk 3,750 for 50 boxes; 20 remain. FIFO: ending Tk 1,400, cost of goods sold Tk 2,350, gross profit Tk 650. LIFO: ending Tk 1,600, cost of goods sold Tk 2,150, gross profit Tk 850. Weighted average: unit cost Tk 75, ending Tk 1,500, cost of goods sold Tk 2,250, gross profit Tk 750. LIFO gives the highest profit because prices fall.

14.4 FIFO: $4,160 \div 10,400 = 40.0\%$. LIFO: $3,760 \div 10,400 = 36.2\%$. Weighted average: $3,978 \div 10,400 = 38.3\%$. FIFO gives the highest ending inventory, Tk 3,640, because prices rise and FIFO leaves the newest, dearest costs in inventory.

CHAPTER XV · PLANT ASSETS

15.p1 The van price of Tk 900,000 and the painting of Tk 12,000, which is needed before use, belong in the cost. The road licence of Tk 20,000 is a cost of using the van during the first year, so it is an expense, not part of the cost.

15.t1 Profit for 2026 would be Tk 15,000 too low, and the machine

4,000, so net income is Tk 700. Balance sheet: debit 3,900; credit $3,200 + 700 = 3,900$.

8.2 Adjustments: Supplies Expense 2,700 (Supplies 4,200 less 1,500 on hand); Wages 1,300 with Wages Payable 1,300; Depreciation Expense 4,800 with Accumulated Depreciation 4,800. Adjusted totals are 131,100 each side. Expenses: $18,500 + 4,800 + 2,000 + 2,700 + 4,800 = 32,800$. Net income = $61,400 - 32,800 = \text{Tk } 28,600$. Balance sheet columns: debit 98,300; credit $69,700 + 28,600 = 98,300$.

8.3 Net loss of 3,500 thousand (21,500 - 18,000). It is entered in the credit column of the income pair ($18,000 + 3,500 = 21,500$) and the debit column of the balance sheet pair ($40,000 + 3,500 = 43,500$). In full taka, the net loss is Tk 3,500,000.

8.4 Adjusted balances: Cash 9,000; Equipment 12,000; Prepaid Insurance 900; Wages 3,400; Insurance Expense 900; Depreciation Expense 1,200 (debits, total 27,400); Accumulated Depreciation 1,200; Wages Payable 400; Unearned Revenue 900; Capital 17,400; Fees Revenue 7,500 (credits, total 27,400). Expenses 5,500; net income = $7,500 - 5,500 = \text{Tk } 2,000$. Balance sheet: debit 21,900; credit $19,900 + 2,000 = 21,900$.

8.5 A worksheet gathers all balances on one page, tests the adjustments before they are recorded, and checks the arithmetic with equal totals. It changes no account because it is working paper; the ledger changes only when the adjusting entries are journalised and posted.

CHAPTER IX · FINANCIAL STATEMENTS AND CLOSING ENTRIES

9.p1 A credit of Tk 300; credit Tk 800 less debit Tk 500. This balance is the net income.

9.t1 $\text{Tk } 788,000 + \text{Tk } 900,000 = \text{Tk } 1,688,000$. The figure mixes two years, so the 2027 income statement would overstate revenue by Tk 788,000.

9.t2 The balance sheet needs ending capital. Ending capital comes from the statement of owner's equity (starting capital plus net income minus drawings), so that statement must come first.

9.1 Income statement: what did the business earn? Statement of owner's equity: how did capital change? Balance sheet: what does the business own and owe on one date? The first two headings end *For the year ended ...*; the balance sheet heading ends *At ...*.

9.2 Net income is $\text{Tk } 12,000 - \text{Tk } 7,000 = \text{Tk } 5,000$. Ending capital is $\text{Tk } 10,000 + \text{Tk } 5,000 - \text{Tk } 1,000 = \text{Tk } 14,000$. Assets are $\text{Tk } 6,000 + \text{Tk } 2,000 + \text{Tk } 9,000 = \text{Tk } 17,000$; liabilities Tk 3,000 plus capital Tk 14,000 also give Tk 17,000.

9.3 Step 1: debit Revenue 12,000, credit Income Summary 12,000. Step 2: debit Income Summary 7,000, credit Expenses 7,000. Step 3: debit Income Summary 5,000, credit Owner's Capital 5,000. Step 4: debit Owner's Capital 1,000, credit Drawings 1,000.

9.4 Temporary: Revenue, Wages Expense, Drawings. Permanent: Prepaid Insurance, Wages Payable, Owner's Capital.

9.5 1. January 2027: debit Wages Payable 4,000, credit Wages

debit Owner's Capital 2,000, credit Income Summary 2,000. Capital falls by Tk 2,000.

CHAPTER X · ACCOUNTS OF A MERCHANDISING COMPANY

10.p1 $\text{Tk } 500 + \text{Tk } 2,000 - \text{Tk } 700 = \text{Tk } 1,800$.

10.p2 Tk 22,500, the first instalment, because it falls due within one year. The other Tk 67,500 is non-current.

10.t1 The two mistakes are equal and opposite on the same side, so total debits still equal total credits. Example: Wages debited Tk 300 too much and Rent debited Tk 300 too little.

10.t2 Cost of goods sold is Tk 2,000 too large, gross profit is Tk 2,000 too small, and inventory is missing from current assets, so the balance sheet understates assets.

10.1 $\text{Tk } 5,000 - \text{Tk } 300 - \text{Tk } 100 = \text{Tk } 4,600$.

10.2 Net sales Tk 78,000. Cost of goods sold = $12,000 + 50,000 - 1,000 - 14,000 = \text{Tk } 47,000$. Gross profit Tk 31,000. Operating expenses $6,000 + 7,000 = \text{Tk } 13,000$. Profit Tk 18,000.

10.3 Current assets Tk 22,000 ($3,000 + 4,000 + 14,000 + 1,000$); equipment Tk 20,000; total assets Tk 42,000. Current liabilities Tk 7,000 ($5,000 + 2,000$); long-term loan Tk 10,000; owner's equity Tk 25,000; total Tk 42,000.

10.4 (a) omission; (b) principle; (c) commission; (d) compensating error.

10.5 (b) Debit Vehicles Tk 40,000; credit Repairs Expense Tk 40,000. (c) Debit Electricity Expense Tk 700; credit Telephone Expense Tk 700.

10.6 Inventory: current asset. Shop fittings: non-current asset. Accounts payable: current liability. Loan due in four years: non-current liability. Loan due in six months: current liability.

CHAPTER XI · SPECIAL JOURNALS

11.p1 The subsidiary ledger holds the balance of one named customer. The accounts receivable control account holds the total owed by all customers together.

11.t1 Five credit sales make five entries and ten postings, so a special journal saves very little. The answer changes when credit sales become many, for example dozens each month, or when more than one clerk must write at the same time.

11.t2 Every sale is posted twice to the control account, so it is too high by the total of the lines. The customer accounts are correct, so the two totals disagree by exactly the amount posted twice, and the test reveals the slip.

11.t3 The agreement test reveals it. The control account, posted from the wrong total, is Tk 1,800 lower than the sum of the customer accounts, since $\text{Tk } 54,200 - \text{Tk } 52,400 = \text{Tk } 1,800$.

11.1 Lines of Tk 100, Tk 200 and Tk 300 give a total of Tk 600. Debit the accounts receivable account Tk 600 and credit Sales Tk 600, each posted once.

11.2 Total: $\text{Tk } 42,500 + \text{Tk } 18,750 + \text{Tk } 31,200 = \text{Tk } 92,450$. Closing